12066 & 12094 PG Advanced Network Engineering

Group Project: Network Design, Troubleshoot and Evaluation

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1. Design Objectives

The primary objective of this project is to architect, implement, evaluate, and troubleshoot a scalable, secure, modular, and highly available campus network infrastructure for a medium-sized educational institution using *Cisco Packet Tracer*. The design addresses the operational and technological needs of a typical Campus Area Network (CAN) across separate physical buildings, specifically the Admin Building (Admissions and Wellbeing), the IT Building (IT Staff and Server Room), and Academic Building (Teaching and Research Labs), while enabling cross-building collaboration and secure access to external Internet resources.

To achieve an enterprise-grade campus network aligned with modern engineering standards, the specific goals of this design are:

- **Hierarchical Topology & High Availability:** Establish a structured three-tier architecture (Core, Distribution, and Access layers) that isolates broadcast domains, streamlines management, and incorporates redundant Distribution-layer links managed by Rapid Spanning Tree Protocol (RSTP) to eliminate single points of failure without risking switching loops.

- **Modular Departmental Segmentation & Inter-VLAN Routing:** Implement Variable Length Subnet Masking (VLSM) and dynamic DHCP address allocation to segment departmental LANs

into dedicated Virtual Local Area Networks (VLANs), utilizing Layer 3 Switched Virtual Interfaces (SVIs) at the Distribution layer for high-speed Inter-VLAN and cross-building routing.

- **Dynamic Routing & Centralised Infrastructure Services:** Deploy Single-Area OSPF across the core and inter-building links for rapid route convergence, and host central infrastructure services (DNS and DHCP) within a dedicated server farm.

- **Multi-Layered Security Enforcement:** Enforce robust security boundary defenses by deploying an edge Firewall between the campus network and the external Internet cloud, applying Access Control Lists (ACLs) to filter inter-departmental traffic, securing device administrative access via SSH/encrypted credentials, privliledged-mode passwords and implementing Port Security alongside interface management on access switches.

- **Wireless & External Service Integration:** Provide a dual-SSID wireless access point for segregated Staff and Student mobility, as well as wireless extensions of the Admissions and Research departments, while demonstrating secure cloud service integration via external Web and FTP servers in the simulated Internet cloud.

- **Verification, Fault Tolerance, & Scalability:** Rigorously validate end-to-end ICMP/IP connectivity, DNS name resolution, dynamic addressing, and FTP file transfer capabilities, while demonstrating structured network troubleshooting under simulated hardware failure and evaluating the framework's capacity for future campus expansion.

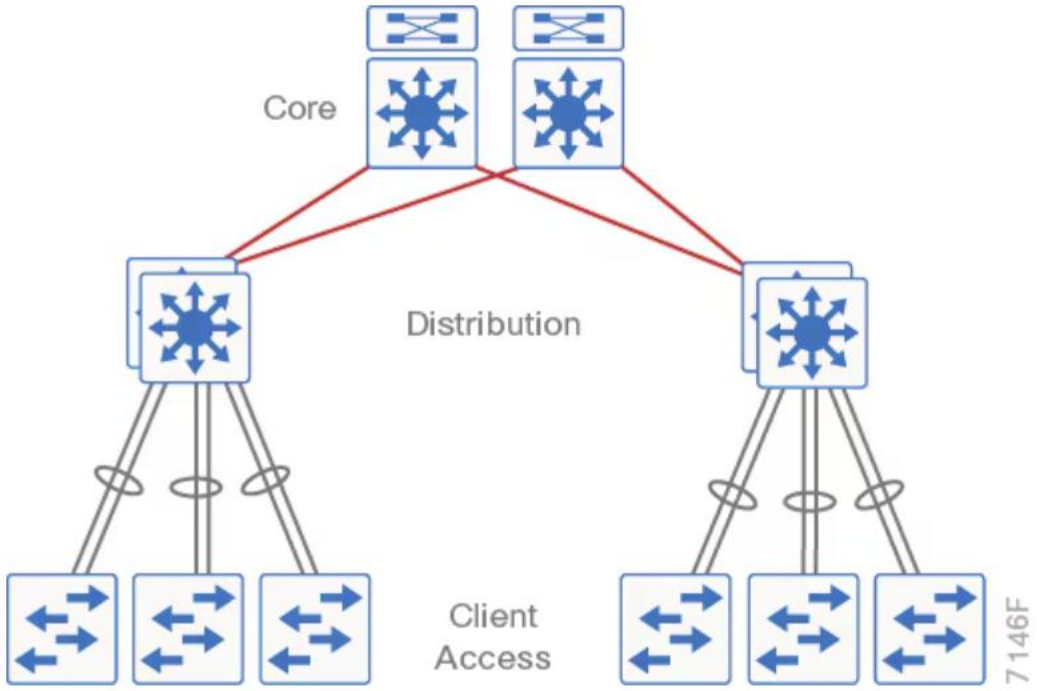
2. Design Principles

To ensure the campus network infrastructure is robust, adaptable, and easily manageable, the design adheres to established enterprise networking principles and Cisco validated design frameworks. The architecture balances performance, redundancy, security, and scalability while addressing the unique constraints and operational requirements of the educational institution.

2.1 Hierarchical Architecture

The campus network is structured around Cisco’s classic Three-Tier Hierarchical Model:

Figure 1. LAN hierarchical design



** The campus network architecture follows the Cisco Validated Design (CVD) framework for hierarchical multi-tier campus networks (Cisco Systems 2020).*

1. **Access Layer (Edge Connectivity):** Deployed at the local department level using Layer 2 switches (e.g., Cisco Catalyst 2960) and Wireless Access Points. This layer provides user-to-network connectivity, enforces VLAN separation per department, applies Port Security to prevent unauthorized physical connections, and isolates unused switchports via administrative shutdown.

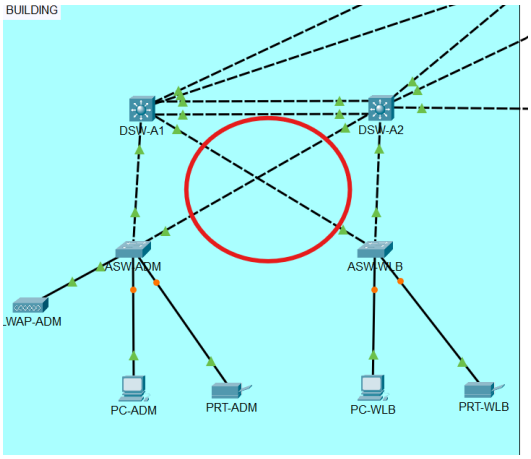
2. **Distribution Layer (Policy & Routing Boundary):** Deployed using Layer 3 switches (e.g., Cisco Catalyst 3650/3560) situated within each of the three physical campus buildings (Admin Building, IT Building, and Academic Building). This layer acts as the default gateway for local subnets using Switched Virtual Interfaces (SVIs), aggregates Access-layer links, terminates broadcast domains, and enforces Access Control Lists (ACLs) and DHCP Relay services. Lateral point-to-point links connect the three building Distribution switches directly to allow cross-building traffic flow without burdening the Core layer

3. **Core Layer (High-Speed Backbone):** Positioned at the network center to interconnect the building Distribution layers, internal Server Room, and the perimeter Firewall. Operating strictly as a high-speed routing engine running Open Shortest Path First (OSPF), the Core layer avoids CPU-intensive packet inspection to maximize throughput and minimize latency across the campus backbone.

2.2 High Availability & Redundancy

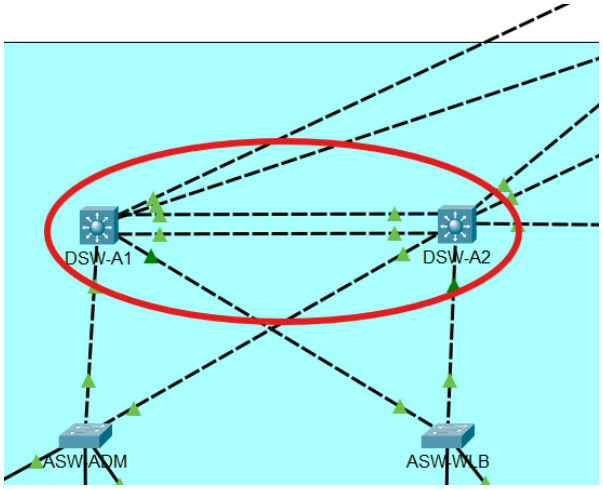
Single points of failure are systematically eliminated across the campus topology to maintain continuous operational availability:

- **Switching Redundancy:** Dual physical trunk links are provisioned between Access switches and Distribution switches. To prevent layer-2 switching loops, broadcast storms, and MAC table instability, Rapid Spanning Tree Protocol (RSTP / 802.1w) is configured. Primary and Secondary Root Bridge priorities are explicitly defined on Distribution switches to force deterministic loop-free forwarding topology.

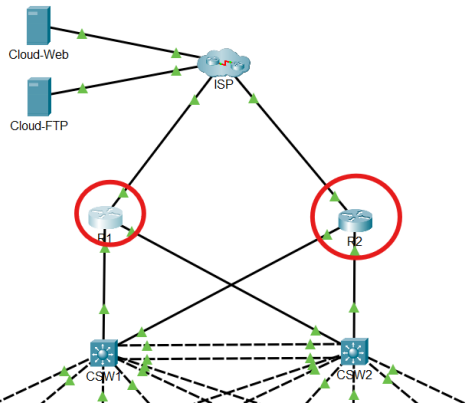


Dual trunk links between access and distribution switches

- **Dynamic Routing Resilience:** Equal-Cost Multi-Path (ECMP) routing via OSPF across core inter-building links allows traffic to dynamically reroute within seconds in the event of a physical link cut or device outage.
- **Hot Standby Router Protocol:** To protect end-user workstations against single switch or gateway failures, **Hot Standby Router Protocol (HSRP)** is deployed across the dual Layer 3 Distribution switch pair in each building (See Routing Configurations section). The downside to this is only the Active Switch is being used as the gateway, with the Standby Switch not being utilized until failover is required.
- **Load balancing at the WAN Edge:** Unlike the distribution switches, the Edge Router pairs are not configured as a single virtual router but are standalone pairs that load balance traffic. Allowing for low overhead and maximum bandwidth utilization, befitting the Core Layer.



L3 Switch pairs configured using HSRP to share

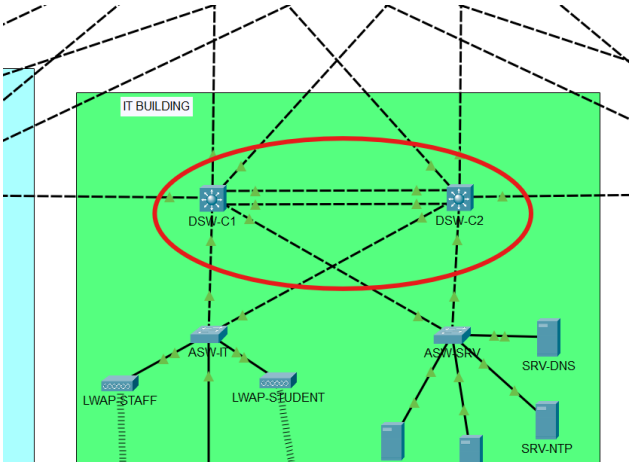


In the WAN Edge, the Router pairs are left standalone as a load-balancing pair to minimize overhead and prioritize bandwidth

one Virtual Default Gateway IP for seamless failover (See Routing Configuration section)

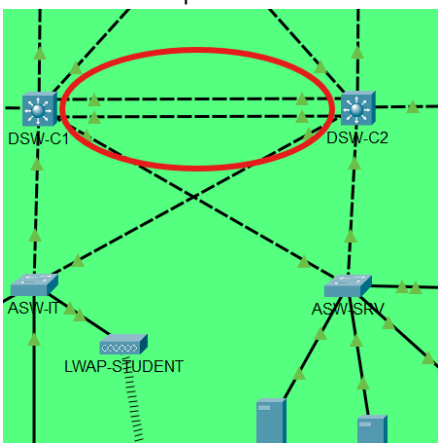
2.3 Localised Inter-VLAN Routing & Cross-Building Traffic Flow

Rather than utilizing traditional Router-on-a-Stick (RoaS) sub-interfaces—which introduce single-link trunk bottlenecks—Inter-VLAN routing is performed directly in hardware at the Distribution layer using Layer 3 SVIs. *Although the assignment specificies 'Each building has its own router, which are then connected to each other' we have permission to use L3 switches – Authors.*



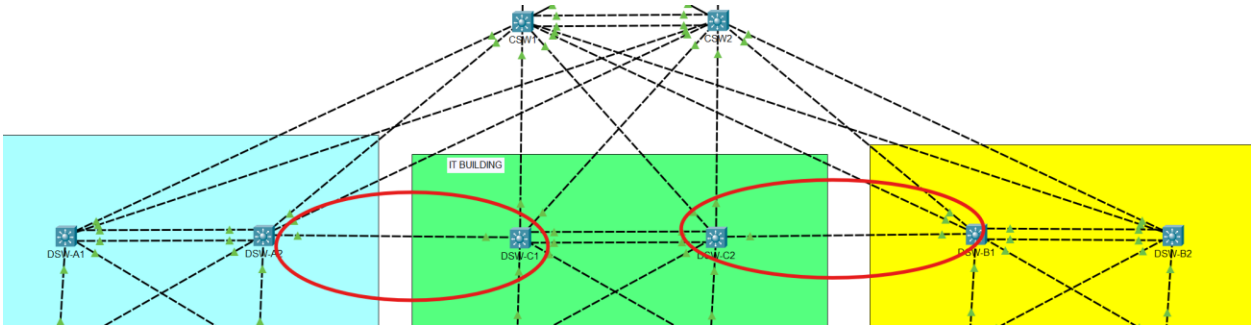
Multilayer switches at the Distribution layer

- **Building-Local Routing:** Traffic flowing between departments within the same building (e.g., Admission and Wellbeing in the Admin Building) is routed directly at the local Layer 3 Distribution switch at wire speed.



Etherchannel links between switch pairs for fault tolerance and greater throughput

- **Cross-Building Communication:** Dedicated lateral point-to-point routed links connect the Distribution entities across buildings directly. This satisfies the brief's requirement for cross-building communication without forcing inter-building traffic up to the Core layer, reserving Core bandwidth for external server and Internet-bound traffic.



Peer links between buildings

2.4 Scalability & Modular Structure

The architecture uses a block-based modular framework, allowing seamless expansion as the institution grows:

- **VLSM Address Planning:** Variable Length Subnet Masking (VLSM) optimizes IPv4 address utilization, preserving consecutive IP blocks for future departmental additions without requiring classful re-addressing.

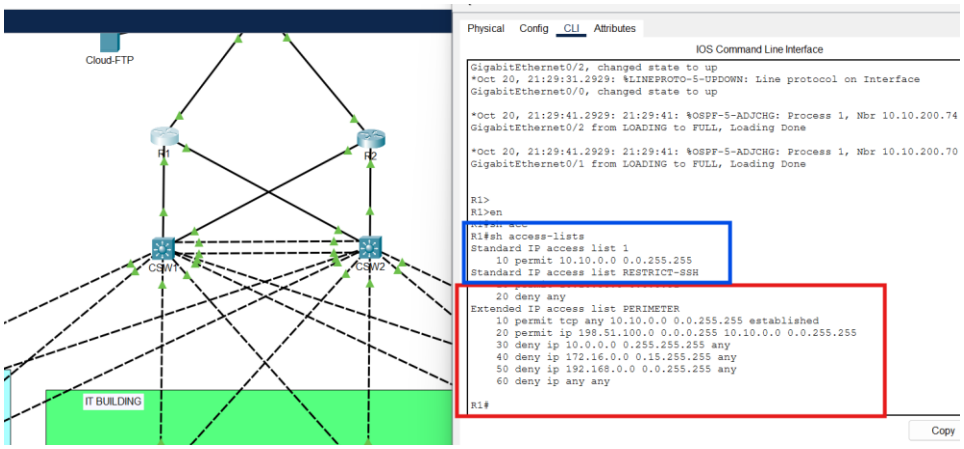
- **Modular Building Blocks:** Additional buildings or departmental LANs can be added as self-contained Distribution/Access blocks by attaching them to the core and extending OSPF routing parameters without disrupting existing services.

- **Centralised Addressing (DHCP Relay):** End-user addressing is dynamically managed via central DHCP servers in the IT Building Server Room using ip helper-address relay statements on local SVIs, avoiding decentralized IP management.

2.5 Security in Depth

Security controls are embedded across multiple network layers rather than relying solely on perimeter defenses:

- **Perimeter Defense:** The Edge Routers have ACLs only allowing return traffic and the Cloud Server Network.



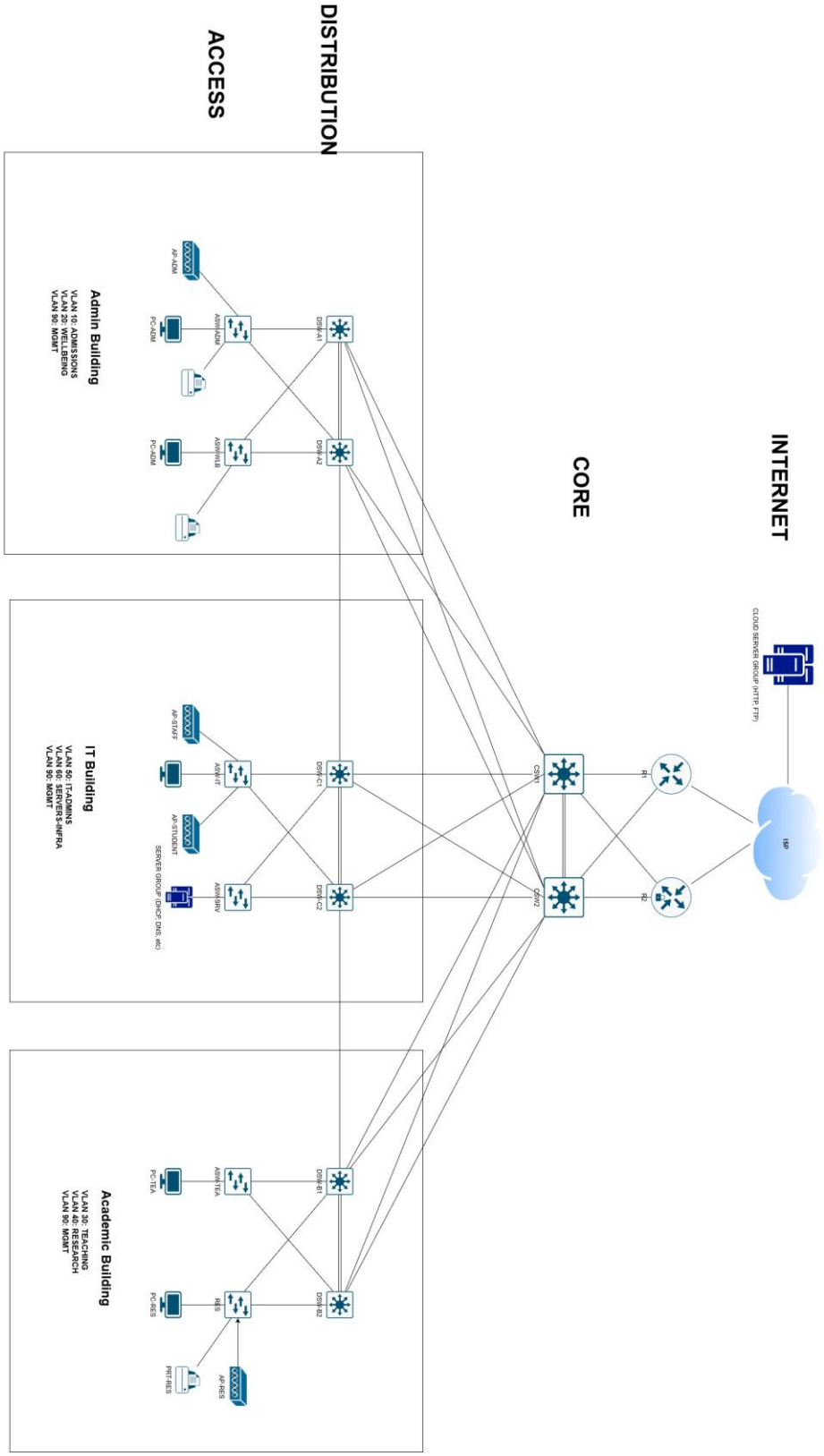
Edge Router ACLs and the SSH ACL only allowing the IT Staff access above

- **Access Control Lists (ACLs):** Standard and Extended ACLs are applied to Distribution Layer interfaces to restrict lateral movement between sensitive departmental VLANs (e.g., only granting SSH access to IT-staff PCs)).

- **Switchport & Layer 2 Hardening:** Access ports enforce Port Security (limiting MAC address entries). Unused switchports are manually placed in a shutdown state and assigned to a blackhole VLAN to mitigate unauthorized physical access attempts.

- **Administrative Protection:** Device management interfaces enforce encrypted password storage (service password-encryption), SSH access, and local console login authentication, warning banners as well as executive timeouts.

3.0 Topology Diagram (with Annotations)



4. IP Addressing Scheme

Subnet block sizes were selected using Variable Length Subnet Masking (VLSM). User subnets were provisioned with a target minimum growth buffer of 100%, while the Server Room was sized for a target 50% expansion.

4.1 Access Layer

Internal Campus Subnets / VLANs

Dept.	Bldg .	VLAN ID	VLAN NAME	Subnet CIDR	Subnet Mask	Users	Usable IP Range	Default Gateway
Admissions	Admin	10	ADMISSIONS	10.10.10.0/26	255.255.255.192	**62**	10.10.10.1 – 10.10.10.62	10.10.10.1
Wellbeing	Admin	20	WELLBEING	10.10.20.0/26	255.255.255.192	**62**	10.10.20.1 – 10.10.20.62	10.10.20.1
Teaching Lab	Academic	30	TEACHING	10.10.30.0/26	255.255.255.192	**62**	10.10.30.1 – 10.10.30.62	10.10.30.1
Research Lab	Academic	40	RESEARCH	10.10.40.0/27	255.255.255.224	**30**	10.10.40.1 – 10.10.40.30	10.10.40.1
IT Staff	IT	50	IT-ADMIN	10.10.50.0/27	255.255.255.224	**30**	10.10.50.1 – 10.10.50.30	10.10.50.1
Server Room	IT	60	SERVER-INFRA	10.10.60.0/28	255.255.255.240	**15**	10.10.60.1 – 10.10.60.14	10.10.60.1
Staff Wireless	Campus-wide	70	STAFF-WIFI	10.10.70.0/25	255.255.255.128	**126**	10.10.70.1 – 10.10.70.126	10.10.70.1
Student Wireless	Campus-wide	80	STUDENT-WIFI	10.10.80.0/25	255.255.255.128	**126**	10.10.80.1 – 10.10.80.126	10.10.80.1
Management - A	Admin	90	MGMT	10.10.91.0/26	255.255.255.192	**62**	10.10.91.1 – 10.10.91.62	10.10.91.1
Management - B	Academic	90	MGMT	10.10.92.0/26	255.255.255.192	**62**	10.10.92.1 – 10.10.92.62	10.10.92.1
Management - C	IT	90	MGMT	10.10.93.0/26	255.255.255.192	**62**	10.10.93.1 – 10.10.93.62	10.10.93.1
Blackhole	Campus-wide	99	BLACKHOLE	Unassigned	N/A	N/A	Shutdown ports only	N/A
Native	Campus-wide	100	NATIVE	Unassigned	N/A	N/A	Trunk native tagging only	N/A

**Management VLAN is divided into three subnets, one for each building*

4.2 Distribution Layer

HSRP Groups (See 'Routing Configuration' Section)

VLAN	VLAN NAME	HSRP GROUP	VIRTUAL IP	ACTIVE ROUTER	STANDBY ROUTER
VLAN 10	ADMISSIONS	STANDBY 10	10.10.10.1	DSW-A1	DSW-A2
VLAN 20	WELLBEING	STANDBY 20	10.10.20.1	DSW-A2	DSW-A1
VLAN 30	TEACHING	STANDBY 30	10.10.30.1	DSW-C1	DSW-C2

VLAN 40	RESEARCH	STANDBY 40	10.10.40.1	DSW-C2	DSW-C1
VLAN 50	IT-ADMIN	STANDBY 50	10.10.50.1	DSW-B1	DSW-B2
VLAN 60	SERVER-INFRA	STANDBY 60	10.10.60.1	DSW-B2	DSW-B1
VLAN 70	STAFF-WIFI	STANDBY 70	10.10.70.1	DSW-B1	DSW-B2
VLAN 80	STUDENT-WIFI	STANDBY 80	10.10.80.1	DSW-B2	DSW-B1
VLAN 90	MGMT	STANDBY 90	10.10.91.1	DSW-A1	DSW-A2
VLAN 90	MGMT	STANDBY 90	10.10.92.1	DSW-B1	DSW-B2
VLAN 90	MGMT	STANDBY 90	10.10.93.1	DSW-C1	DSW-C2
VLAN 99	BLACKHOLE	N/A	N/A	N/A	N/A
VLAN 100	NATIVE	N/A	N/A	N/A	N/A

Lateral Links

East Device	West Device	Network	Subnet	East IP	West IP
DSW-A2	DSW-C1	10.10.200.48/30	255.255.255.252	10.10.200.49	10.10.200.50
DSW-B2	DSW-B1	10.10.200.52/30	255.255.255.252	10.10.200.53	10.10.200.54

Uplinks to Core Layer

Distribution Device	Distribution Interface	Core Device	Core Interface	Network	Subnet	IP (Distribution)	IP (Core)
DSW-A1	G1/0/5	CSW1	G1/0/7	10.10.200.0/30	255.255.255.252	10.10.200.1	10.10.200.2
DSW-A1	G1/0/6	CSW2	G1/0/7	10.10.200.4/30	255.255.255.252	10.10.200.5	10.10.200.6
DSW-A2	G1/0/5	CSW1	G1/0/8	10.10.200.8/30	255.255.255.252	10.10.200.9	10.10.200.10
DSW-A2	G1/0/6	CSW2	G1/0/8	10.10.200.12/30	255.255.255.252	10.10.200.13	10.10.200.14
DSW-C1	G1/0/5	CSW1	G1/0/9	10.10.200.16/30	255.255.255.252	10.10.200.17	10.10.200.18
DSW-C1	G1/0/6	CSW2	G1/0/9	10.10.200.20/30	255.255.255.252	10.10.200.21	10.10.200.22
DSW-C2	G1/0/5	CSW1	G1/0/10	10.10.200.24/30	255.255.255.252	10.10.200.25	10.10.200.26
DSW-C2	G1/0/6	CSW2	G1/0/10	10.10.200.28/30	255.255.255.252	10.10.200.29	10.10.200.30
DSW-B1	G1/0/5	CSW1	G1/0/11	10.10.200.32/30	255.255.255.252	10.10.200.33	10.10.200.34
DSW-B1	G1/0/6	CSW2	G1/0/11	10.10.200.36/30	255.255.255.252	10.10.200.37	10.10.200.38
DSW-B2	G1/0/5	CSW1	G1/0/12	10.10.200.40/30	255.255.255.252	10.10.200.41	10.10.200.42
DSW-B2	G1/0/6	CSW2	G1/0/12	10.10.200.44/30	255.255.255.252	10.10.200.45	10.10.200.46

4.3 Core Layer

Uplinks to Router

Router	L3 Switch	Network	Subnet	Router IP	Interface	L3 Switch IP	L3 Switch Interface
R1	CSW1	10.10.200.60/30	255.255.255.252	10.10.200.61	G0/1	10.10.200.62	G1/0/1
R1	CSW2	10.10.200.64/30	255.255.255.252	10.10.200.65	G0/2	10.10.200.66	G1/0/1
R2	CSW1	10.10.200.68/30	255.255.255.252	10.10.200.69	G0/1	10.10.200.70	G1/0/2
R2	CSW2	10.10.200.72/30	255.255.255.252	10.10.200.73	G0/2	10.10.200.74	G1/0/2

Peer Links

East Device	West Device	Network	Subnet	East IP	West IP
CSW1	CSW2	10.10.200.56/30	255.255.255.252	10.10.200.57	10.10.200.58

Internet

Router	Network / CIDR	Local IP	R1 IP	R2 IP
ISP	203.0.113.0/29	203.0.113.1	203.0.113.2	203.0.113.3

Cloud Servers

Device	Network / CIDR	Interface	Default Gateway
Cloud - Web	198.51.100.0/24	198.51.100.10	198.51.100.1
Cloud - FTP	198.51.100.0/24	198.51.100.20	198.51.100.1

Loopback / Management IP Addressing Scheme

Device	Management IP	Loopback IP
R1	N/A	1.1.1.1
R2	N/A	2.2.2.2
CSW1	N/A	3.3.3.3
CSW2	N/A	4.4.4.4
DSW-A1	10.10.91.2	5.5.5.5
DSW-A2	10.10.91.3	6.6.6.6
DSW-C1	10.10.93.2	7.7.7.7
DSW-C2	10.10.93.3	8.8.8.8
DSW-B1	10.10.92.2	9.9.9.9
DSW-B2	10.10.92.3	10.10.10.10
ASW-ADM	10.10.91.4	N/A
ASW-WLB	10.10.91.5	N/A
ASW-IT	10.10.93.4	N/A
ASW-SRV	10.10.93.5	N/A
ASW-TEA	10.10.92.4	N/A
ASW-RES	10.10.92.5	N/A

Endpoints

Endpoint	IP Address
PC-ADM	DHCP
LP-ADM	DHCP
PRT-ADM	DHCP

PC-WLB	DHCP
PRT-WLB	DHCP
PC-TEA	DHCP
PC-RES	DHCP
PRT-RES	DHCP
LP-RES	DHCP
PC-IT	DHCP
SRV-SYSLOG	10.10.60.4
SRV-DHCP	10.10.60.5
SRV-DNS	10.10.60.6
SRV-NTP	10.10.60.7
LP-STAFF	DHCP
LP-STUDENT	DHCP

5. VLAN Scheme

VLAN ID	VLAN Name	Dept / Purpose	Trunking & Allowed Strategy
10	ADMISSIONS	Admission Dept (PCs, Printer, AP)	Allowed on Admin L2 vertical trunks & L2 peer trunk.
20	WELLBEING	Wellbeing Dept (PCs, Printer)	Allowed on Admin L2 vertical trunks & L2 peer trunk.
30	TEACHING	Teaching Lab (PCs)	Allowed on Academic L2 vertical trunks & L2 peer trunk.
40	RESEARCH	Research Lab (PCs, Printer, AP)	Allowed on Academic L2 vertical trunks & L2 peer trunk.
50	IT-ADMIN	IT Staff Workstations	Allowed on IT L2 vertical trunks & L2 peer trunk.
60	SERVER-INFRA	IT Server Room (DHCP/DNS/NTP/SYSLOG)	Allowed on IT L2 vertical trunks & L2 peer trunk.
70	STAFF-WIFI	Staff Wireless (SSID: Staff)	Tagged across all inter-switch 802.1Q trunks

80	STUDENT-WIFI	Student Wireless (SSID: Student)	Tagged across all inter-switch 802.1Q trunks
90	MGMT	Infrastructure Management (SSH/SVIs) (Divided into three subnets, see IP addressing scheme)	Tagged on all trunks
99	BLACKHOLE	Inactive / Unassigned Switchports	**Pruned / Omitted** from all trunk allowed lists.
100	NATIVE	Native 802.1Q Trunk VLAN	Set as switchport trunk native vlan 100 on trunks.

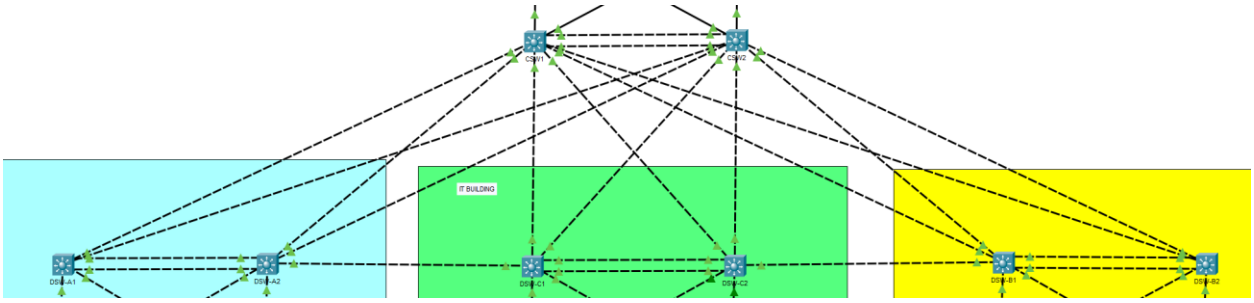
6. Routing Configurations

To ensure rapid convergence, deterministic path selection, and high availability across the 3-building campus, the Layer 3 routing framework is structured into a Single-Area OSPF (Area 0) backbone complemented by Hot Standby Router Protocol (HSRP) at the Distribution boundary (i.e two L3 switches share a Virtual IP address acting as the default gateway for the respective VLAN).

6.1 Three-Tier OSPF Area 0 Backbone Architecture

The entire Layer 3 routing domain runs **Single-Area OSPF (Process ID 1, Area 0)** across all Core Layer 3 switches, Distribution Layer 3 switches, and Perimeter Edge Gateways. As this is a lab environment, single-area OSPF was chosen for simplicity.

- **Layer 3 Point-to-Point Transit Links (**/30 **Subnets):** All inter-switch links connecting the Distribution switches up to the Core switches (12 uplinks total) and between the Core switch pair are configured as pure Layer 3 routed interfaces (Config: no switchport) using dedicated /30 point-to-point subnets (10.10.200.0/24 space).



- **OSPF Network Type Optimization (**ip ospf network point-to-point**):** To accelerate routing convergence, all /30 routed interfaces are explicitly configured with ip ospf network point-to-point. This instructs OSPF to skip the standard 40-second Designated Router (DR) and Backup

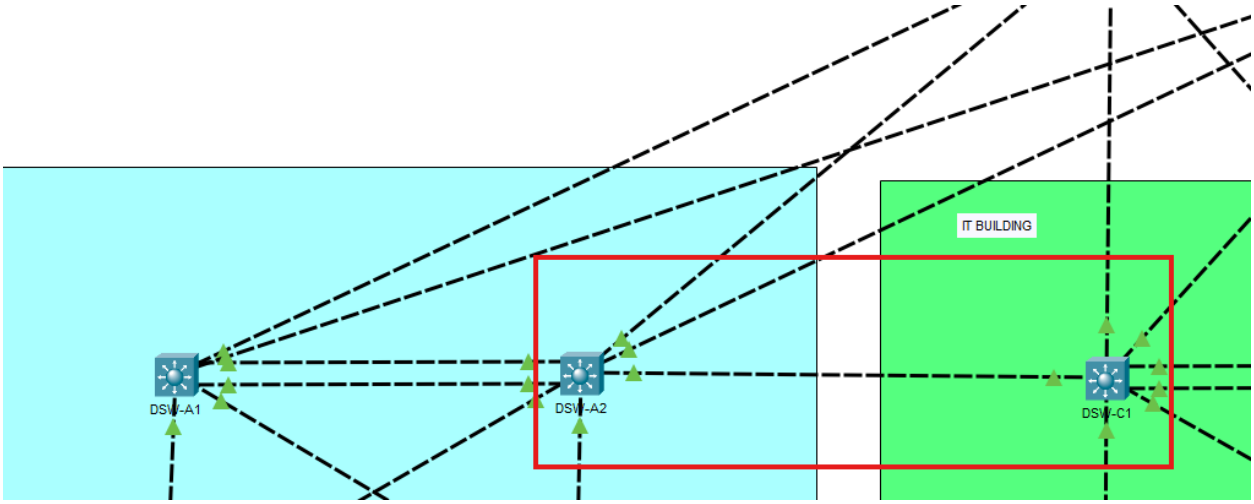
Designated Router (BDR) election process on point-to-point links, achieving sub-second adjacency formation upon link startup or failover.

- **Control-Plane Hardening (Passive Interfaces):** To mitigate security risks and reduce CPU overhead, passive-interface default is applied on all point-to-point links. End-user departmental SVIs (VLANs 10–80) remain passive; advertising their host subnets into Area 0 without broadcasting OSPF control packets to user workstations.

6.2 Localized Cross-Building Lateral Routing

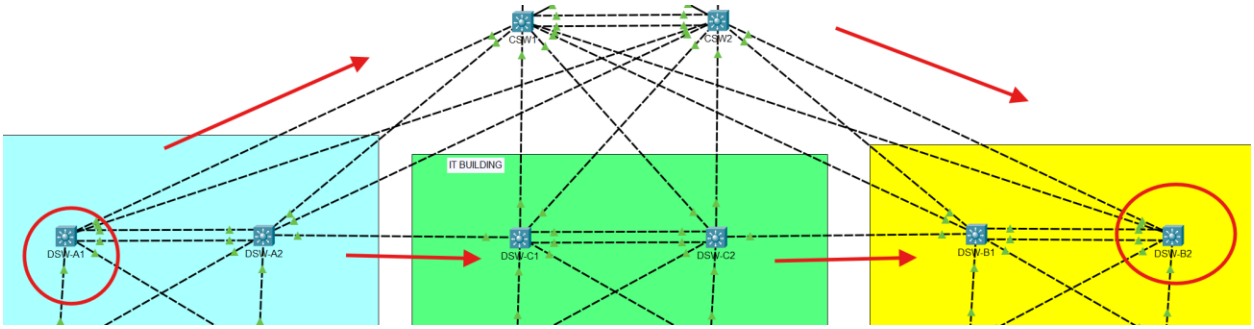
To satisfy the project's explicit requirement for **cross-building communication without forcing local traffic through the Core backbone**, dedicated **Layer 3 lateral point-to-point links** connect adjacent building Distribution switch pairs directly:

- **Direct Path Preference:** Lateral links operate natively in OSPF Area 0. When a workstation in the Admin Building (VLAN 10) communicates with a server or IT building, OSPF calculates the shortest metric cost and routes traffic **directly across the 1-hop lateral link at the Distribution layer**.



- **Core Offloading & Redundancy:** By keeping local building-to-building traffic at the Distribution layer, high-speed Core bandwidth is reserved for server farm access and Internet Cloud transit. If a physical lateral cable is severed, OSPF automatically converges within milliseconds and reroutes cross-building traffic **up through the Core switch pair as an alternative secondary path**.

Authors note: Traffic between buildings may still go through the core layer if it results in less hops than via the lateral links. For example, if traffic has to go from DSW-A1 (extreme left) to DSW-B2 (extreme right) then the packet will go through the core as it's just 2 hops, as opposed travelling laterally, which would be 5 hops. For the sake of efficient routing, we have not forced all traffic through the peer links.

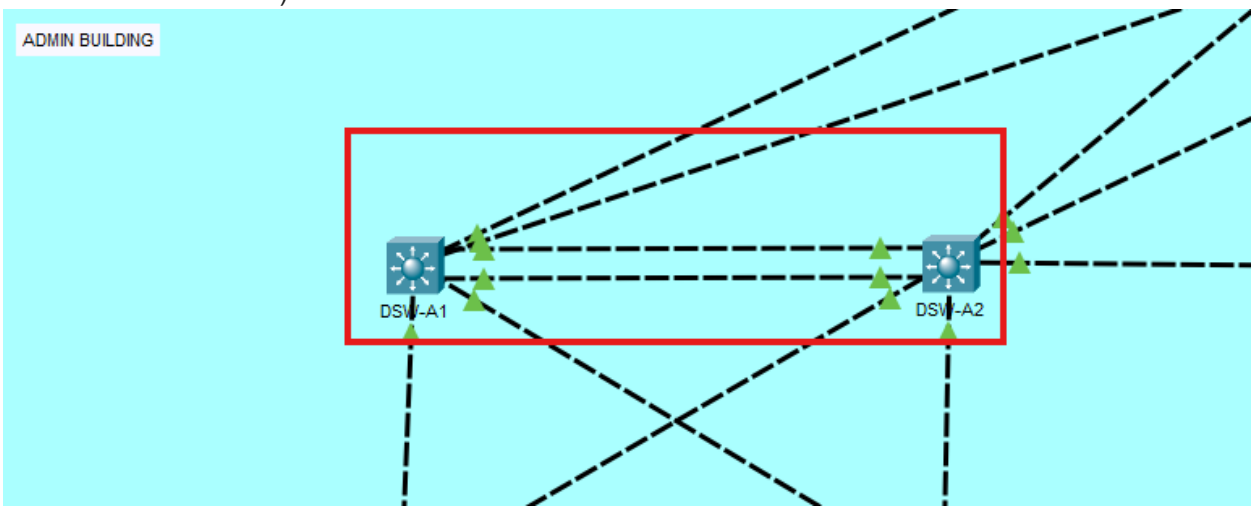


DSW-A1 takes 2 hops to DSW-B2 via the Core layer switch as opposed to 5 hops via the peer links

6.3 First-Hop Redundancy Protocol (HSRP) at Distribution

To protect end-user workstations against single switch or gateway failures, **Hot Standby Router Protocol (HSRP)** is deployed across the dual Layer 3 Distribution switch pair in each building:

- **Active/Standby Gateway Load Balancing:** Each building's Switched Virtual Interfaces (SVIs) run HSRP with preemption enabled (standby X preempt) on both switches to ensure deterministic gateway takeover.
    - **Admin Building Example:** DSW-A2 is configured with priority 110 (Active) for Admissions & Wellbeing VLANs (VLANs 10 & 20) (see HSRP Groups table in previous section).



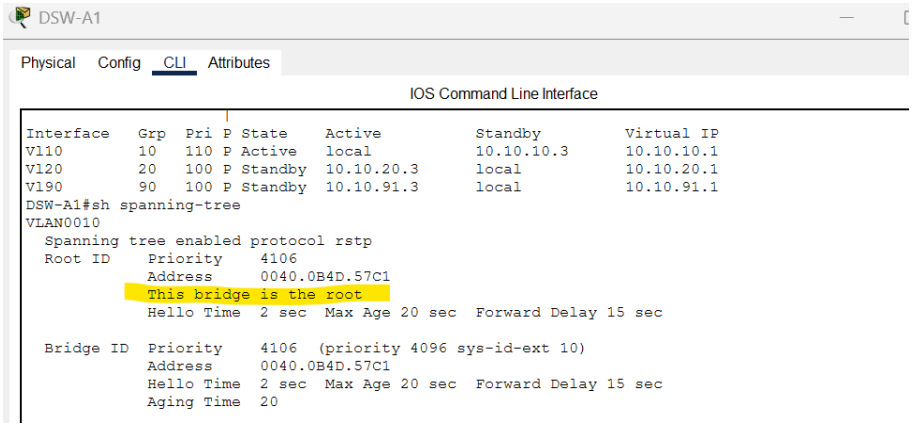
4.2 Distribution Layer

HSRP Groups (See 'Routing Configuration' Section)

VLAN	VLAN NAME	HSRP GROUP	VIRTUAL IP	ACTIVE ROUTER	STANDBY ROUTER
VLAN 10	ADMISSIONS	STANDBY 10	10.10.10.1	DSW-A1	DSW-A2
VLAN 20	WELLBEING	STANDBY 20	10.10.20.1	DSW-A2	DSW-A1
~~VLAN 30~~	~~TEACHING~~	~~STANDBY 30~~	~~10.10.30.1~~	~~DSW-C1~~	~~DSW-C2~~
VLAN 40	RESEARCH	STANDBY 40	10.10.40.1	DSW-C2	DSW-C1

Dept.	Bldg .	VLAN ID	VLAN NAME	Subnet CIDR	Subnet Mask	Users	Usable IP Range	Default Gateway
Admissions	Admin	10	ADMISSIONS	10.10.10.0/26	255.255.255.192	**62**	10.10.10.1 – 10.10.10.62	10.10.10.1
Wellbeing	Admin	20	WELLBEING	10.10.20.0/26	255.255.255.192	**62**	10.10.20.1 – 10.10.20.62	10.10.20.1

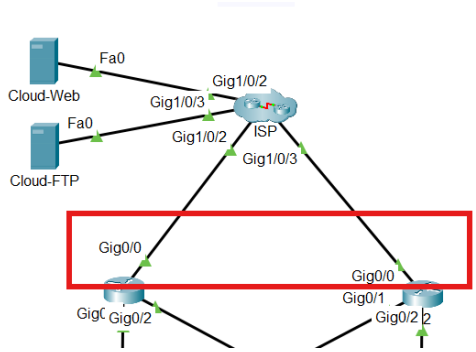
Example of HSRP being configured for two routers sharing one virtual IP, the active router in the group is always the Root bridge for the STP VLAN domain (DSW-A1 in this example below)



- **Virtual IP Alignment with STP:** For every VLAN, the **Active HSRP Gateway switch is explicitly aligned to act as the Primary Spanning Tree Root Bridge (Priority 4096)**. This ensures that Layer 2 loop-free forwarding paths lead directly to the active Layer 3 gateway, completely eliminating asymmetric routing across the intra-building peer link.

6.4 NAT/PAT Configuration

To conserve public IPv4 addresses and obscure internal network topology details from external internet threats, **Port Address Translation (PAT / NAT Overload)** is configured on the WAN-facing edge routers (R1 and R2). PAT dynamically maps all internal private IPv4 addresses (10.10.0.0/16 space, covering departmental VLANs 10–60 and Wireless VLANs 70–80) to a single public IP address assigned to the router's WAN interface (203.0.113.2 on R1 and 203.0.113.3 on R2) using unique source Layer 4 port numbers. Internal Core-facing transit interfaces (GigabitEthernet 0/1 and 0/2) are designated as ip nat inside, while the ISP-facing WAN interface (GigabitEthernet 0/0) is designated as ip nat outside. This stateful translation boundary ensures that outbound sessions to public Cloud Web (198.51.100.10) and FTP (198.51.100.20) servers establish valid translation table entries, permitting established return packets to pass through perimeter Access Control Lists (ACLs) while completely masking internal host IPs.

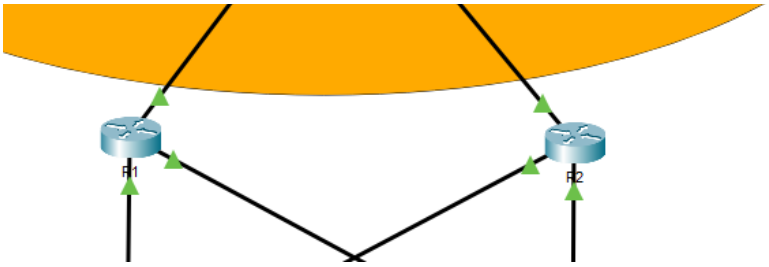


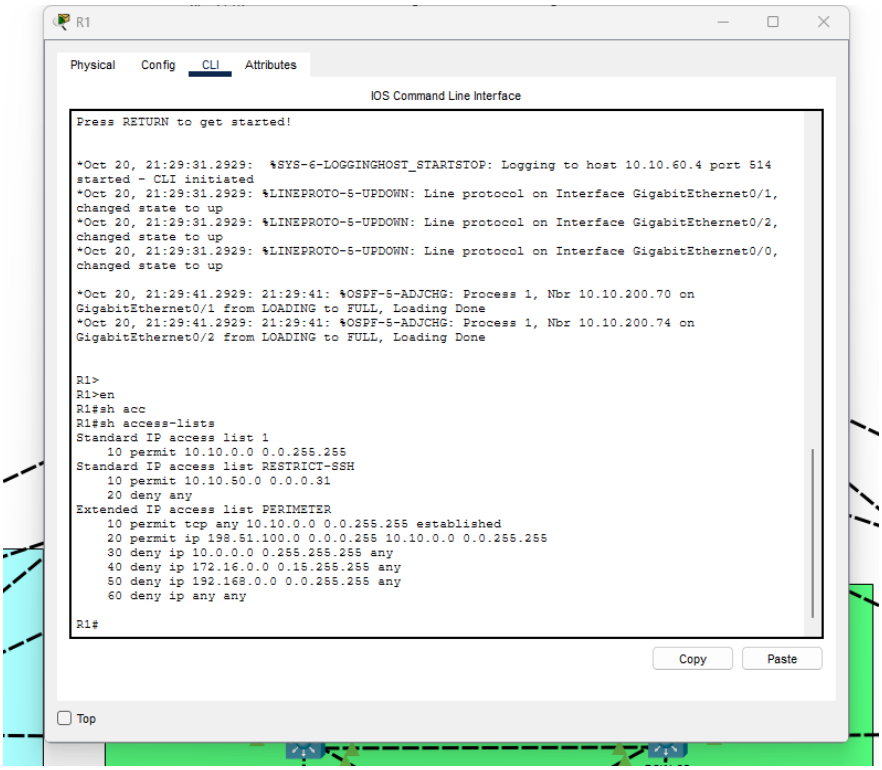
We are using PAT on a single interface IP in this lab for simplicity, in actual production, a pool would probably be configured. - Authors

Note: See Appendix for Routing Configurations (sh ip route)

7. Access Control List Details

- ACL on Edge Routers to prevent spoofed IP addresses but allow return traffic
- **All** network devices also have an ACL that only permits IT staff from remoting in via SSH





*ACL configurations (same on both routers)

```
R1#sh ip int g0/0
GigabitEthernet0/0 is up, line protocol is up (connected)
  Internet address is 203.0.113.2/29
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is not set
  Directed broadcast forwarding is disabled
  Outgoing access list is not set
  Inbound  access list is PERIMETER
  Proxy ARP is enabled
  Security level is default
```

*Proof ACLs applied on interface facing internet

```
R1#show running-config | section line vty
line vty 0 4
 access-class RESTRICT-SSH in
 exec-timeout 30 0
 logging synchronous
 login local
 transport input ssh
line vty 5 15
 access-class RESTRICT-SSH in
 exec-timeout 30 0
 logging synchronous
 login local
 transport input ssh
R1#
```

*Proof ACLS applied on virtual lines restricting access to IT Staff

Authors Note: Although the Assignment requirement specifies a Firewall to be implemented, the dedicated hardware Firewalls in Packet tracer are a relatively new feature and lack the full functionality of the physical versions. We have gone with an ACL configured on our Edge Router pair to satisfy the requirement instead.

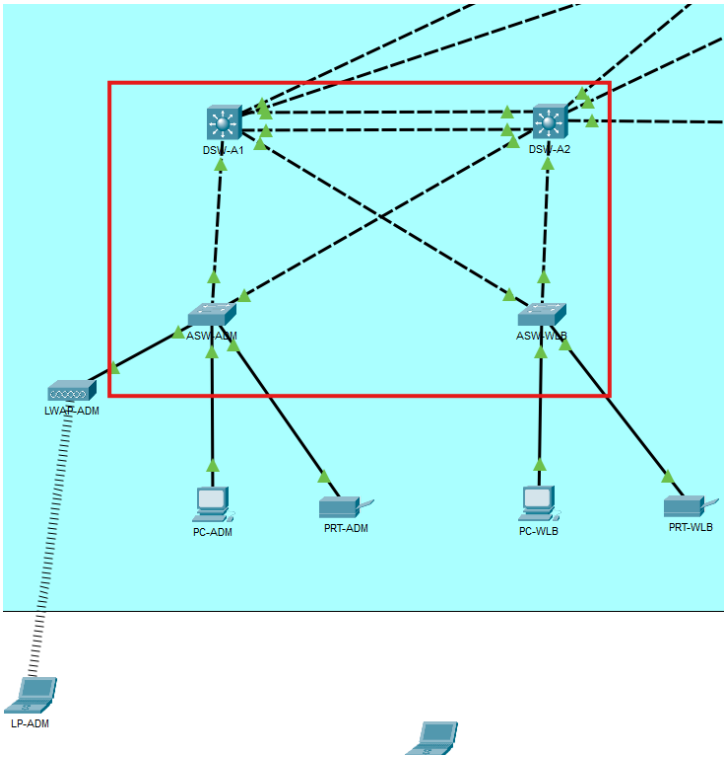
8. Security Measures

Domain	Security Mechanism	Config	Threat Mitigated
Control Plane	**Privileged EXEC Secret**	enable secret class	Unauthorized privilege escalation to administrative EXEC mode.
Control Plane	**SSH v2 & Password Encryption**	ip ssh version 2 service password-encryption	Cleartext password sniffing and packet eavesdropping.
Line Access	**Inactivity Timeout & Sync Logging**	exec-timeout 30 0 logging synchronous	Unattended terminal session hijacking & CLI buffer clutter.
Audit & Logging	**Centralized Syslog Logging**	logging 10.10.60.4 logging trap debugging	Log tampering, erased local buffers, and unmonitored security violations.
Time Sync	**Network Time Protocol (NTP)**	ntp server 10.10.60.7	Unsynchronized log timestamps preventing cross-device forensic audit correlation.
Layer 2 Edge	**Access Port Security (PG Requirement)**	switchport port-security maximum 1 switchport port-security violation shutdown	MAC address flooding attacks and rogue device connections.

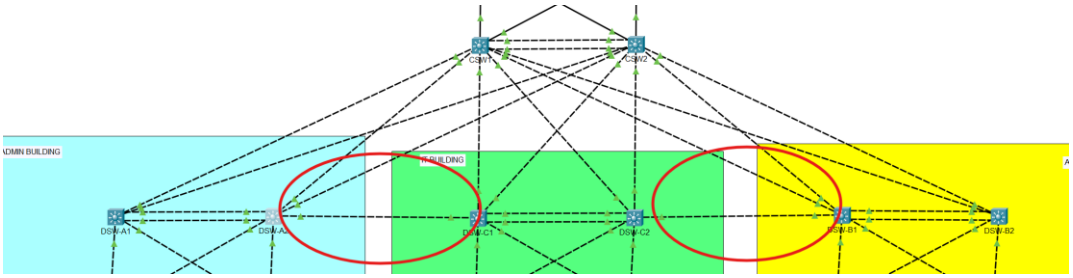
Switchports	**Port Hygiene (Blackhole VLAN)**	switchport access vlan 99 shutdown	Unauthorized physical access via unallocated wall-jacks.
Trunking	**Native VLAN Hardening**	switchport trunk native vlan 100	802.1Q Double-Tagging VLAN Hopping attacks.
Management	**VTY Management ACL**	access-class RESTRICT-SSH in	Unauthorized SSH access attempts from non-IT subnets.
Routing	**OSPF Passive Interfaces**	passive-interface default	Rogue OSPF neighbor injection and control-plane overhead.
Perimeter	**Perimeter Extended ACL & PAT**	ip access-group PERIMETER in ip nat inside source list ... overload	External IP spoofing, unsolicited inbound intrusion, & private IP exposure.

9. Switchport Status of Switches

*See appendix for the table for the STP status of ports and switches



Each building has an Access and Distribution Switch pair with Rapid Spanning Tree Protocol (RSTP / 802.1w) enabled across to ensure deterministic loop prevention and optimal traffic forwarding. Because all upstream links to the Core layer and lateral inter-building links are Layer 3 routed interfaces (**no switchport**), STP execution is confined locally within each physical building block.



Lateral links are layer 3 routed connections, segregating each building into its own STP domain

To prevent asymmetric routing, STP Root Bridge is always the HSRP Active Router. This ensures that the Layer 2 loop-free forwarding path always leads directly to the active Layer 3 default gateway without traversing unnecessary inter-switch hops.

Note: The project marking rubric says **'Switchport Status of Core Switches'** but we've changed it to Access/Distribution Layer. In this topology, Core Switches are on Layer 3 and *will not participate in STP*.

10. Connectivity Testing Screenshots & Results

**See Appendix section for screenshot proofs*

Source Device	Destination Device	Test Method	Expected Result	Actual Result	Status	Objective
IT Staff PC (10.10.50.5 / VLAN 50) *IT Building*	Admission PC (10.10.10.15 / VLAN 10) *Admin Building*	ping 10.10.10.15	0% Packet Loss (Replies received)	0% Loss (RTT < 1ms)	**PASS**	Validates inter-building Inter-VLAN routing across OSPF Area 0 backbone/lateral links.
IT Staff PC (10.10.50.5 / VLAN 50) *IT Building*	Teaching Lab PC (10.10.30.15 / VLAN 30) *Academic Building*	ping 10.10.30.15	0% Packet Loss (Replies received)	0% Loss (RTT < 1ms)	**PASS**	Confirms Layer 3 reachability between IT department and Academic student subnets.
IT Staff PC	Staff Wireless Laptop	ping 10.10.70.15	0% Packet Loss (Replies received)	0% Loss (RTT < 1ms)	**PASS**	Verifies L3 routing to wireless clients on VLAN 70.

(10.10.50.5 / VLAN 50) *IT Building*	(10.10.70.15 / VLAN 70) *Campus-wide Wi-Fi*					
IT Staff PC (10.10.50.5 / VLAN 50) *IT Building*	Central DHCP Server (10.10.60.2 / VLAN 60) *IT Server Room*	ping 10.10.60.2	0% Packet Loss (Replies received)	0% Loss (RTT < 1ms)	**PASS**	Confirms access to core server room infrastructure across local SVI boundaries.
IT Staff PC (10.10.50.5 / VLAN 50) *IT Building*	Cloud Web Server (External Public Cloud) *Internet Subnet*	ping 198.51.100.10	0% Packet Loss (Replies received)	0% Loss (RTT ~ 2ms)	**PASS**	Validates outbound perimeter routing, PAT state creation, and firewall traversal.
Admission PC (10.10.10.15 / VLAN 10) *Admin Building*	Internal DNS Server (10.10.60.3 / VLAN 60) *IT Server Room*	nslookup www.ane-labs.net	Resolves FQDN to 198.51.100.10	Resolved to 198.51.100.10	**PASS**	Demonstrates internal DNS query resolution across Inter-VLAN boundaries (UDP 53).

Student Laptop (10.10.80.15 / VLAN 80) *Campus-wide Wi-Fi*	External Web Portal (www.ane-labs.net)	Browser: www.ane-labs.net	HTML Home Page loads cleanly	Page Loaded	**PASS**	Satisfies Criteria 11 by validating end-to-end FQDN web access via PAT and Edge ACL.
IT Staff PC (10.10.50.5 / VLAN 50) *IT Building*	Cloud FTP Server (198.51.100.20)	ftp 198.51.100.20 (CLI Authentication)	230 User logged in & dir listing	Logged In (226 Complete)	**PASS**	Satisfies Postgraduate (12094) requirement for cloud FTP service demonstration.

11. DNS Resolution and Webpage Access

Source Device	**Destination Device**	**Test Method**	**Expected Result**
Staff Laptop (10.10.70.xx / VLAN 70)	Internal DNS Server (10.10.60.3 / VLAN 60) *IT Server Room*	nslookup www.campus.edu.au	Resolves FQDN to 198.51.100.10

Campus-wide Wi-Fi			
Staff Laptop (10.10.70.xx / VLAN 70) *Campus-wide Wi-Fi*	External Web Portal (www.campus.edu.au)	Browser -> http://www.campus.edu.au	HTML Home Page loads cleanly
Staff Laptop (10.10.70.xx / VLAN 70) *Campus-wide Wi-Fi*	Cloud FTP Server (198.51.100.20)	ftp 198.51.100.20 (CLI Authentication) Upload and download a file	User logged in & dir listing

12. Troubleshooting use-case scenarios

Use Case 1: Active HSRP / Spanning Tree Gateway & Root Mismatch (Traffic Asymmetry)

- **Simulated Issue:** A misconfiguration occurs on the Admin Distribution switch pair (DSW-Admin-1 and DSW-Admin-2) where DSW-A1 is configured as the **Active HSRP Gateway** for Admission (VLAN 10), but DSW-A2 is left with a lower bridge priority, making it the **Primary STP Root Bridge** for VLAN 10.
- **Symptoms Identified:** Pings between Admission PCs (10.10.10.0/26) and the gateway succeed, but simulation mode reveals frames travel up to DSW-Admin-2 via STP, cross the horizontal intra-building peer link, and then hit DSW-Admin-1 to be routed, making for inefficient routing.
- **Diagnostic Commands & Root Cause:**
    - show standby brief shows DSW-A1 as Active.
    - show spanning-tree vlan 10 reveals DSW-A2 is the Root Bridge.
- **Methodical Resolution:** Re-align STP Root Bridge priorities with HSRP gateway states by issuing spanning-tree vlan 10 root primary on DSW-A1 and spanning-tree vlan 10 priority root secondary on DSW-Admin-2.

Use Case 2: DHCP Request Dropped across Remote Buildings (ip helper-address Missing)

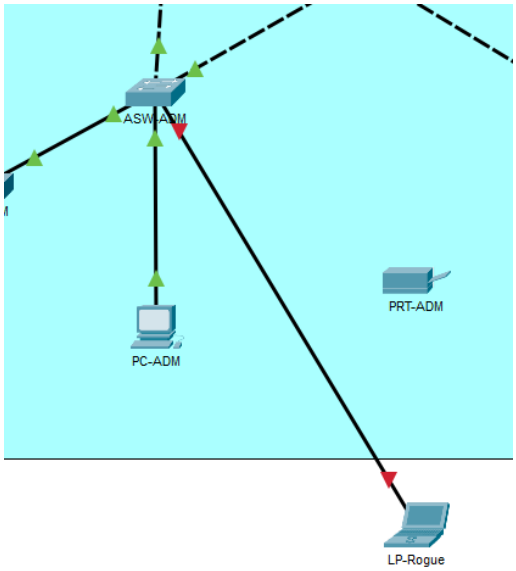
- **Simulated Issue:** End-user PCs in the Academic Building (Teaching Lab on VLAN 30) fail to obtain an IP address dynamically, assigning themselves APIPA addresses (169.254.X.X), whereas PCs in the IT Building receive DHCP leases normally.
- **Symptoms Identified:** ipconfig /renew times out on Academic PCs. Simulation mode shows the Layer 2 DHCP Discover broadcast hitting SVI 30 on DSW-A1 but dropping immediately without being converted into a Layer 3 unicast packet.
- **Diagnostic Commands & Root Cause:**
    - Executing show running-config interface Vlan 30 on DSW-Acad-1 shows that ip helper-address 10.10.60.2 was omitted from the SVI.
- **Methodical Resolution:** Navigate to interface Vlan 30 on both Academic Distribution switches and apply ip helper-address 10.10.60.2. Verify that PCs receive leases from the 10.10.30.0/26 pool.

Use Case 3: Perimeter Extended ACL Rule Order Dropping Return TCP Traffic

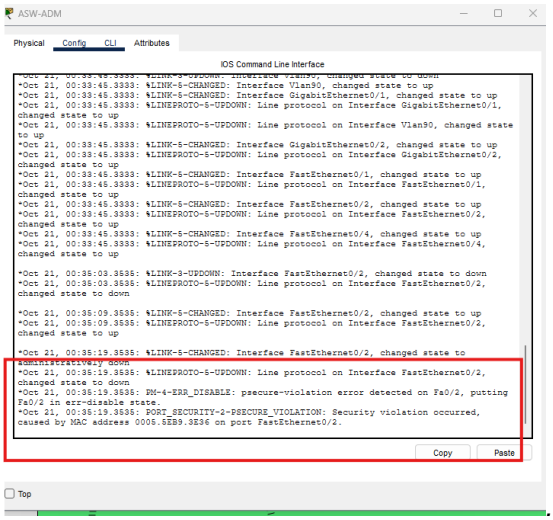
- **Simulated Issue:** Campus wireless laptops (VLAN 70 / 10.10.70.0/25) can successfully ping the external Cloud Web Server IP (198.51.100.10), but navigating to [www.ane-labs.net] in the browser returns "Server connection was reset" or times out.
- **Symptoms Identified:** Outbound TCP SYN packets reach the Web server, but return SYN-ACK packets are dropped at Edge-Router-1's WAN interface (G0/0).
- **Diagnostic Commands & Root Cause:**
    - Run show ip access-lists PERIMETER on Edge-Router-1.
    - The ACL evaluates top-down: the anti-spoofing rule deny ip 10.0.0.0 0.255.255.255 any precedes the permit tcp any 10.10.0.0 0.0.255.255 established rule. Because return packets target the translated internal IP 10.10.70.5, line 1 drops them.
- **Methodical Resolution:** Re-sequence the PERIMETER ACL so the established rule sits above all RFC 1918 anti-spoofing deny lines. Verify web page loads in the browser.

Use Case 4: Rogue Access Port Security Violation (PG Requirement, demonstrated in the)

- **Simulated Issue:** An unauthorized rogue laptop is plugged into an unassigned access port (Fa0/12) on the Admission Access Switch (Admin-ASW-1), causing the physical link to instantly drop.
- **Symptoms Identified:** Link LED turns red/off, and the port enters an err-disabled state.
- **Diagnostic Commands & Root Cause:**
    - Run show interface FastEthernet 0/12 -> displays FastEthernet0/12 is err-disabled.
    - Run show port-security interface FastEthernet 0/12 -> shows violation mode shutdown triggered by an unauthorized MAC address. Alternatively, the port was not moved to the BLACKHOLE (VLAN 666) port hygiene state.
- **Methodical Resolution:** Disconnect the unauthorized device, enforce proper port hygiene (switchport access vlan 666 and shutdown), and recover legitimate ports using shutdown followed by no shutdown.



Rogue laptop replaces the printer's connection



..and triggers PortSec

13. Scalability Discussion & Future Growth Analysis

The implemented campus network architecture is specifically engineered around a **modular, block-based design framework** aligned with Cisco Validated Designs (CVD) for Enterprise Campus Networks. The scalability and long-term flexibility of this infrastructure are analyzed across five core engineering pillars below:

1. Modular Building-Block Architecture (Horizontal & Vertical Expansion)

The network relies on a standardized, decoupled **Three-Tier Hierarchical Model** (Access, Distribution, and Core):

- **Adding Endpoints (Vertical Scaling):** Expanding user density within existing departments equires only stacking or attaching an additional Layer 2 Access switch to the building's Distribution pair via standard 802.1Q vertical trunks. The local Layer 3 Distribution Switched Virtual Interfaces (SVIs) and HSRP gateways remain unchanged.
- **Adding Campus Buildings (Horizontal Scaling):** If the institution constructs a 4th physical building (e.g., a Student Center or Library), it can be integrated as a self-contained "Distribution/Access Block". The new building's Layer 3 Distribution switch pair simply connects back to the central dual Core switches (Core-SW1 / Core-SW2) via point-to-point /30 routed links and joins the existing **OSPF Area 0 backbone**. Existing buildings and configurations require zero modification.

2. Structured VLSM Subnetting & Hierarchical Route Summarization

Subnet allocation follows Variable Length Subnet Masking (VLSM) principles built on the classless 10.10.0.0/16 internal private IP space.

- **Pre-Provisioned Host Growth Buffers:** User subnets were intentionally sized to the nearest binary power boundary (2^n), providing an **actual host growth capacity between 50% and 195%** across departments (e.g., Admission's /26 subnet accommodates up to 62 usable IPs for its 34 initial hosts).
- **Route Summarization (Supernetting):** Subnets are allocated in contiguous numerical blocks per building block. As the network expands, Core routers can summarize entire building subnets into single CIDR prefix advertisements (e.g., summarizing 10.10.10.0/26 and 10.10.20.0/26 into 10.10.0.0/23 toward the Core). This keeps OSPF Link-State Databases (LSDB) lightweight, reduces routing table memory overhead, and prevents localized link flaps from recalculating Shortest Path First (SPF) algorithms campus-wide.

3. Dynamic Routing & Non-Blocking Backbone Capacity

- **OSPF Area 0 Adaptability:** Because all backbone transit interfaces operate as pure Layer 3 routed interfaces (no switchport) using point-to-point /30 networks, adding new distribution nodes or edge connections does not introduce Spanning Tree Protocol (STP) loop vulnerabilities or broadcast storms.
- **Equal-Cost Multi-Path (ECMP) Bandwidth Scaling:** Upstream transit links from Distribution switches to the Core backbone run OSPF ECMP across redundant links. If campus bandwidth demands double in the future, additional physical interfaces can be aggregated into **Layer 3 EtherChannels** without altering the underlying routing architecture or STP topology.

4. Centralized Services & Dynamic Addressing (DHCP/DNS)

- **Centralized Management via DHCP Relay:** The end-user address allocation is decoupled from local switch hardware. New subnets or expanded IP pools are created centrally on the Server Room DHCP server (10.10.60.2), while local Distribution SVIs simply require an ip helper-address 10.10.60.5 statement. This eliminates manual IP management on client devices as host counts scale.
- **Centralized WLC & Split-MAC Wireless Architecture:** Wireless coverage scales seamlessly by deploying Lightweight Access Points (LWAPs) across new buildings. Although no WLC is used in this lab environment, one could be added if we need to scale LWAP deployment.

Section 14: Self-Reflection & SWOT Analysis

14.1 Strengths (Internal Architectural Advantages)

- **Deterministic Forwarding via HSRP/STP Integration:** Aligning Layer 2 Rapid Spanning Tree (RSTP) Root Bridge priorities (4096/8192) directly with Layer 3 HSRP gateway priorities (110/100) per VLAN completely eliminates "traffic tromboning" across intra-building distribution links, providing deterministic sub-second failover.
- **High-Speed Layer 3 Lateral Offloading:** Configuring point-to-point Layer 3 routed interfaces (no switchport with /30 subnets) directly between building Distribution switch pairs allows cross-building departmental traffic to route directly at the Distribution layer without burdening the central Core backbone.
- **Control-Plane & Switchport Hardening:** Implementing an Out-of-Band Management VLAN 99, SSH v2 with 4096-bit RSA keys, VTY management ACLs (RESTRICT-SSH-ACCESS), 30-minute

execution timeouts, Port Security, and Blackhole VLAN 99 port hygiene establishes an enterprise-grade "Defense-in-Depth" posture.

14.2 Weaknesses (Recognized Architectural Limitations)

- **Single Perimeter Gateway Boundary (WAN SPOF):** While internal campus switching features high availability, the WAN perimeter relies on a sing ISP to connect to the Cloud Architecture, creating a single point of failure for outbound Internet and external Cloud Server access.
- **No Single-Controller Wireless Architecture:** Lack of a single Wireless LAN Controller is this architecture results in management difficulty, security gaps, and poor roaming performance of LWAPs. *Author's note: The real reason why we did not configure a WLC in this lab environment is because we found it to be very buggy, especially with DHCP assignment of the LWAPs involving multiple IP Address pools.*
- **Absence of Dynamic ARP Inspection (DAI):** While static Port Security is enforced at the Access layer, advanced First-Hop Security features like DAI and DHCP Snooping are disabled due to Packet Tracer simulator command constraints, leaving local subnets partially vulnerable to malicious ARP poisoning.

14.3 Opportunities (Future Architectural Enhancements)

- **High-Availability Edge Firewall Pair & Dual-Homing:** The WAN edge can be upgraded to an Active/Standby Firewall Cluster paired with dual multihomed ISP connections running External BGP (eBGP). This would provide seamless perimeter failover and load balancing if a primary ISP link drops.
- **WLC High Availability Stateful Switchover (HA-SSO):** Deploying a redundant standby WLC in an HA pair would allow CAPWAP tunnels and client AP databases to sync in real-time, enabling hitless wireless failover without client re-authentication.
- **Full Dual-Stack IPv6 Backbone Deployment:** While the current IPv4 plan is fully operational, expanding the parallel IPv6 addressing scheme across all departmental SVIs and OSPFv3 processes would ensure long-term compatibility with global IPv6 internet standards.

14.4 Threats (External & Environmental Security Risks)

- **Perimeter DDoS & TCP SYN Flood Vulnerability:** Because the perimeter router uses standard stateful PAT overload without dedicated Distributed Denial of Service (DDoS) scrubbing hardware, external botnets could exhaust the router's NAT translation table memory.
- **Physical Rogue AP & Wireless Eavesdropping:** If an unauthorized user physically connects a rogue consumer Wi-Fi router into an active wall jack, it could broadcast an unencrypted SSID, exposing departmental traffic to rogue interception.
- **Simulator State-Engine Inconsistencies:** Packet Tracer's simplified state engine occasionally drops valid TCP three-way handshakes (SYN/SYN-ACK) during rapid topology changes or when evaluating complex ACL rule orders. Recognizing these simulator anomalies was critical during testing to differentiate between software glitches and true network protocol failures.

- END OF REPORT -

References

Cisco Systems 2020, *Campus LAN and Wireless LAN Design Guide*, Cisco Validated Design (CVD), Cisco Systems, San Jose, CA, viewed 12 October 2026, https://www.cisco.com/c/en/us/td/docs/solutions/CVD/Campus/cisco-campus-lan-wlan-design-guide.html.

APPENDIX

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Packet Tracer Information
6. Routing Configurations
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11. DNS Resolution and Webpage Access Screenshots
12. Troubleshooting Use-case Scenarios
IOS Configurations

Packet Tracer Information

Access Domain / Service	Target Device / Interface	Username / Parameter	Password / Secret
Console Line Access	All Routers & Switches	admin	cisco
SSH v2 Remote Management	All Routers & Switches	admin	cisco
Privileged EXEC Mode	All Routers & Switches	N/A	class

Cloud FTP Service	Cloud-FTP-SRV (198.51.100.20)	anelabs	cisco
Campus Wireless Networks (ALL Wifis)	All LWAPs / WLC SSIDs	N/A	Cisco123
External Web Services	Cloud-Web-SRV (198.51.100.10)	URL: www.ane-labs.net	Public Access

Routing Configuration (IPv4 Routing Tables)

DSW-A1

```
DSW-A1(config)#do sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 10.10.200.2 to network 0.0.0.0

1.0.0.0/32 is subnetted, 1 subnets
O 1.1.1.1 [110/3] via 10.10.200.2, 00:52:30, GigabitEthernet1/0/5
[110/3] via 10.10.200.6, 00:52:30, GigabitEthernet1/0/6
2.0.0.0/32 is subnetted, 1 subnets
O 2.2.2.2 [110/3] via 10.10.200.2, 00:52:30, GigabitEthernet1/0/5
[110/3] via 10.10.200.6, 00:52:30, GigabitEthernet1/0/6
5.0.0.0/32 is subnetted, 1 subnets
C 5.5.5.5 is directly connected, Loopback0
6.0.0.0/32 is subnetted, 1 subnets
O 6.6.6.6 [110/2] via 10.10.91.3, 00:52:15, Vlan90
7.0.0.0/32 is subnetted, 1 subnets
O 7.7.7.7 [110/3] via 10.10.200.2, 00:52:30, GigabitEthernet1/0/5
[110/3] via 10.10.200.6, 00:52:30, GigabitEthernet1/0/6
8.0.0.0/32 is subnetted, 1 subnets
O 8.8.8.8 [110/3] via 10.10.200.2, 00:52:30, GigabitEthernet1/0/5
[110/3] via 10.10.200.6, 00:52:30, GigabitEthernet1/0/6
9.0.0.0/32 is subnetted, 1 subnets
O 9.9.9.9 [110/3] via 10.10.200.2, 00:52:15, GigabitEthernet1/0/5
[110/3] via 10.10.200.6, 00:52:15, GigabitEthernet1/0/6
[110/3] via 10.10.91.3, 00:52:15, Vlan90
10.0.0.0/8 is variably subnetted, 35 subnets, 6 masks
C 10.10.10.0/26 is directly connected, Vlan10
L 10.10.10.2/32 is directly connected, Vlan10
O 10.10.10.10/32 [110/3] via 10.10.200.2, 00:52:30, GigabitEthernet1/0/5
[110/3] via 10.10.200.6, 00:52:30, GigabitEthernet1/0/6
C 10.10.20.0/26 is directly connected, Vlan20
L 10.10.20.2/32 is directly connected, Vlan20
O 10.10.30.0/26 [110/3] via 10.10.200.2, 00:52:30, GigabitEthernet1/0/5
```

```
[110/3] via 10.10.200.6, 00:52:30, GigabitEthernet1/0/6
O 10.10.40.0/27 [110/3] via 10.10.200.2, 00:52:30, GigabitEthernet1/0/5
[110/3] via 10.10.200.6, 00:52:30, GigabitEthernet1/0/6
O 10.10.50.0/27 [110/3] via 10.10.200.2, 00:52:15, GigabitEthernet1/0/5
[110/3] via 10.10.200.6, 00:52:15, GigabitEthernet1/0/6
[110/3] via 10.10.91.3, 00:52:15, Vlan90
O 10.10.60.0/28 [110/3] via 10.10.200.2, 00:52:15, GigabitEthernet1/0/5
[110/3] via 10.10.200.6, 00:52:15, GigabitEthernet1/0/6
[110/3] via 10.10.91.3, 00:52:15, Vlan90
O 10.10.70.0/25 [110/3] via 10.10.200.2, 00:52:15, GigabitEthernet1/0/5
[110/3] via 10.10.200.6, 00:52:15, GigabitEthernet1/0/6
[110/3] via 10.10.91.3, 00:52:15, Vlan90
O 10.10.80.0/25 [110/3] via 10.10.200.2, 00:52:15, GigabitEthernet1/0/5
[110/3] via 10.10.200.6, 00:52:15, GigabitEthernet1/0/6
[110/3] via 10.10.91.3, 00:52:15, Vlan90
C 10.10.91.0/26 is directly connected, Vlan90
L 10.10.91.2/32 is directly connected, Vlan90
O 10.10.92.0/26 [110/3] via 10.10.200.2, 00:52:05, GigabitEthernet1/0/5
[110/3] via 10.10.200.6, 00:52:05, GigabitEthernet1/0/6
O 10.10.93.0/26 [110/3] via 10.10.200.2, 00:52:15, GigabitEthernet1/0/5
[110/3] via 10.10.200.6, 00:52:15, GigabitEthernet1/0/6
[110/3] via 10.10.91.3, 00:52:15, Vlan90
C 10.10.200.0/30 is directly connected, GigabitEthernet1/0/5
L 10.10.200.1/32 is directly connected, GigabitEthernet1/0/5
C 10.10.200.4/30 is directly connected, GigabitEthernet1/0/6
L 10.10.200.5/32 is directly connected, GigabitEthernet1/0/6
O 10.10.200.8/30 [110/2] via 10.10.200.2, 00:52:15, GigabitEthernet1/0/5
[110/2] via 10.10.91.3, 00:52:15, Vlan90
O 10.10.200.12/30 [110/2] via 10.10.200.6, 00:52:15, GigabitEthernet1/0/6
[110/2] via 10.10.91.3, 00:52:15, Vlan90
O 10.10.200.16/30 [110/2] via 10.10.200.2, 00:52:30, GigabitEthernet1/0/5
O 10.10.200.20/30 [110/2] via 10.10.200.6, 00:52:30, GigabitEthernet1/0/6
O 10.10.200.24/30 [110/2] via 10.10.200.2, 00:52:30, GigabitEthernet1/0/5
O 10.10.200.28/30 [110/2] via 10.10.200.6, 00:52:30, GigabitEthernet1/0/6
O 10.10.200.32/30 [110/2] via 10.10.200.2, 00:52:30, GigabitEthernet1/0/5
O 10.10.200.36/30 [110/2] via 10.10.200.6, 00:52:30, GigabitEthernet1/0/6
O 10.10.200.40/30 [110/2] via 10.10.200.2, 00:52:30, GigabitEthernet1/0/5
O 10.10.200.44/30 [110/2] via 10.10.200.6, 00:52:30, GigabitEthernet1/0/6
O 10.10.200.48/30 [110/2] via 10.10.91.3, 00:52:15, Vlan90
O 10.10.200.52/30 [110/3] via 10.10.200.2, 00:52:30, GigabitEthernet1/0/5
[110/3] via 10.10.200.6, 00:52:30, GigabitEthernet1/0/6
O 10.10.200.60/30 [110/2] via 10.10.200.2, 00:52:30, GigabitEthernet1/0/5
O 10.10.200.64/30 [110/2] via 10.10.200.6, 00:52:30, GigabitEthernet1/0/6
O 10.10.200.68/30 [110/2] via 10.10.200.2, 00:52:30, GigabitEthernet1/0/5
O 10.10.200.72/30 [110/2] via 10.10.200.6, 00:52:30, GigabitEthernet1/0/6
O*E2 0.0.0.0/0 [110/1] via 10.10.200.2, 00:52:30, GigabitEthernet1/0/5
[110/1] via 10.10.200.6, 00:52:30, GigabitEthernet1/0/6
```

DSW-A2

```
DSW-A2(config)#do sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
```

```
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 10.10.200.10 to network 0.0.0.0

1.0.0.0/32 is subnetted, 1 subnets
O 1.1.1.1 [110/3] via 10.10.200.10, 00:53:39, GigabitEthernet1/0/5
[110/3] via 10.10.200.14, 00:53:39, GigabitEthernet1/0/6
2.0.0.0/32 is subnetted, 1 subnets
O 2.2.2.2 [110/3] via 10.10.200.10, 00:53:39, GigabitEthernet1/0/5
[110/3] via 10.10.200.14, 00:53:39, GigabitEthernet1/0/6
5.0.0.0/32 is subnetted, 1 subnets
O 5.5.5.5 [110/2] via 10.10.91.2, 00:53:14, Vlan90
6.0.0.0/32 is subnetted, 1 subnets
C 6.6.6.6 is directly connected, Loopback0
7.0.0.0/32 is subnetted, 1 subnets
O 7.7.7.7 [110/3] via 10.10.200.10, 00:53:39, GigabitEthernet1/0/5
[110/3] via 10.10.200.14, 00:53:39, GigabitEthernet1/0/6
8.0.0.0/32 is subnetted, 1 subnets
O 8.8.8.8 [110/3] via 10.10.200.10, 00:53:39, GigabitEthernet1/0/5
[110/3] via 10.10.200.14, 00:53:39, GigabitEthernet1/0/6
9.0.0.0/32 is subnetted, 1 subnets
O 9.9.9.9 [110/2] via 10.10.200.50, 00:53:39, GigabitEthernet1/0/24
10.0.0.0/8 is variably subnetted, 36 subnets, 6 masks
C 10.10.10.0/26 is directly connected, Vlan10
L 10.10.10.3/32 is directly connected, Vlan10
O 10.10.10.10/32 [110/3] via 10.10.200.10, 00:53:39, GigabitEthernet1/0/5
[110/3] via 10.10.200.14, 00:53:39, GigabitEthernet1/0/6
C 10.10.20.0/26 is directly connected, Vlan20
L 10.10.20.3/32 is directly connected, Vlan20
O 10.10.30.0/26 [110/3] via 10.10.200.10, 00:53:39, GigabitEthernet1/0/5
[110/3] via 10.10.200.14, 00:53:39, GigabitEthernet1/0/6
O 10.10.40.0/27 [110/3] via 10.10.200.10, 00:53:39, GigabitEthernet1/0/5
[110/3] via 10.10.200.14, 00:53:39, GigabitEthernet1/0/6
O 10.10.50.0/27 [110/2] via 10.10.200.50, 00:53:39, GigabitEthernet1/0/24
O 10.10.60.0/28 [110/2] via 10.10.200.50, 00:53:39, GigabitEthernet1/0/24
O 10.10.70.0/25 [110/2] via 10.10.200.50, 00:53:39, GigabitEthernet1/0/24
O 10.10.80.0/25 [110/2] via 10.10.200.50, 00:53:39, GigabitEthernet1/0/24
C 10.10.91.0/26 is directly connected, Vlan90
L 10.10.91.3/32 is directly connected, Vlan90
O 10.10.92.0/26 [110/3] via 10.10.200.10, 00:53:14, GigabitEthernet1/0/5
[110/3] via 10.10.200.14, 00:53:14, GigabitEthernet1/0/6
O 10.10.93.0/26 [110/2] via 10.10.200.50, 00:53:39, GigabitEthernet1/0/24
O 10.10.200.0/30 [110/2] via 10.10.200.10, 00:53:14, GigabitEthernet1/0/5
[110/2] via 10.10.91.2, 00:53:14, Vlan90
O 10.10.200.4/30 [110/2] via 10.10.200.14, 00:53:14, GigabitEthernet1/0/6
[110/2] via 10.10.91.2, 00:53:14, Vlan90
C 10.10.200.8/30 is directly connected, GigabitEthernet1/0/5
L 10.10.200.9/32 is directly connected, GigabitEthernet1/0/5
C 10.10.200.12/30 is directly connected, GigabitEthernet1/0/6
L 10.10.200.13/32 is directly connected, GigabitEthernet1/0/6
O 10.10.200.16/30 [110/2] via 10.10.200.10, 00:53:39, GigabitEthernet1/0/5
[110/2] via 10.10.200.50, 00:53:39, GigabitEthernet1/0/24
O 10.10.200.20/30 [110/2] via 10.10.200.14, 00:53:39, GigabitEthernet1/0/6
[110/2] via 10.10.200.50, 00:53:39, GigabitEthernet1/0/24
```

```
O 10.10.200.24/30 [110/2] via 10.10.200.10, 00:53:39, GigabitEthernet1/0/5
O 10.10.200.28/30 [110/2] via 10.10.200.14, 00:53:39, GigabitEthernet1/0/6
O 10.10.200.32/30 [110/2] via 10.10.200.10, 00:53:39, GigabitEthernet1/0/5
O 10.10.200.36/30 [110/2] via 10.10.200.14, 00:53:39, GigabitEthernet1/0/6
O 10.10.200.40/30 [110/2] via 10.10.200.10, 00:53:39, GigabitEthernet1/0/5
O 10.10.200.44/30 [110/2] via 10.10.200.14, 00:53:39, GigabitEthernet1/0/6
C 10.10.200.48/30 is directly connected, GigabitEthernet1/0/24
L 10.10.200.49/32 is directly connected, GigabitEthernet1/0/24
O 10.10.200.52/30 [110/3] via 10.10.200.10, 00:53:39, GigabitEthernet1/0/5
[110/3] via 10.10.200.14, 00:53:39, GigabitEthernet1/0/6
O 10.10.200.60/30 [110/2] via 10.10.200.10, 00:53:39, GigabitEthernet1/0/5
O 10.10.200.64/30 [110/2] via 10.10.200.14, 00:53:39, GigabitEthernet1/0/6
O 10.10.200.68/30 [110/2] via 10.10.200.10, 00:53:39, GigabitEthernet1/0/5
O 10.10.200.72/30 [110/2] via 10.10.200.14, 00:53:39, GigabitEthernet1/0/6
O*E2 0.0.0.0/0 [110/1] via 10.10.200.10, 00:53:39, GigabitEthernet1/0/5
[110/1] via 10.10.200.14, 00:53:39, GigabitEthernet1/0/6
```

DSW- B1

```
DSW-B1(config)#do sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 10.10.200.34 to network 0.0.0.0

1.0.0.0/32 is subnetted, 1 subnets
O 1.1.1.1 [110/3] via 10.10.200.34, 00:54:14, GigabitEthernet1/0/5
[110/3] via 10.10.200.38, 00:54:14, GigabitEthernet1/0/6
2.0.0.0/32 is subnetted, 1 subnets
O 2.2.2.2 [110/3] via 10.10.200.34, 00:54:14, GigabitEthernet1/0/5
[110/3] via 10.10.200.38, 00:54:14, GigabitEthernet1/0/6
5.0.0.0/32 is subnetted, 1 subnets
O 5.5.5.5 [110/3] via 10.10.200.34, 00:54:14, GigabitEthernet1/0/5
[110/3] via 10.10.200.38, 00:54:14, GigabitEthernet1/0/6
6.0.0.0/32 is subnetted, 1 subnets
O 6.6.6.6 [110/3] via 10.10.200.34, 00:54:14, GigabitEthernet1/0/5
[110/3] via 10.10.200.38, 00:54:14, GigabitEthernet1/0/6
7.0.0.0/32 is subnetted, 1 subnets
C 7.7.7.7 is directly connected, Loopback0
8.0.0.0/32 is subnetted, 1 subnets
O 8.8.8.8 [110/2] via 10.10.92.3, 00:53:49, Vlan90
9.0.0.0/32 is subnetted, 1 subnets
O 9.9.9.9 [110/3] via 10.10.200.34, 00:54:14, GigabitEthernet1/0/5
[110/3] via 10.10.200.38, 00:54:14, GigabitEthernet1/0/6
10.0.0.0/8 is variably subnetted, 36 subnets, 6 masks
O 10.10.10.0/26 [110/3] via 10.10.200.34, 00:54:14, GigabitEthernet1/0/5
[110/3] via 10.10.200.38, 00:54:14, GigabitEthernet1/0/6
O 10.10.10.10/32 [110/2] via 10.10.200.53, 00:54:29, GigabitEthernet1/0/24
O 10.10.20.0/26 [110/3] via 10.10.200.34, 00:54:14, GigabitEthernet1/0/5
[110/3] via 10.10.200.38, 00:54:14, GigabitEthernet1/0/6
```

```
C 10.10.30.0/26 is directly connected, Vlan30
L 10.10.30.2/32 is directly connected, Vlan30
C 10.10.40.0/27 is directly connected, Vlan40
L 10.10.40.2/32 is directly connected, Vlan40
O 10.10.50.0/27 [110/2] via 10.10.200.53, 00:54:29, GigabitEthernet1/0/24
O 10.10.60.0/28 [110/2] via 10.10.200.53, 00:54:29, GigabitEthernet1/0/24
O 10.10.70.0/25 [110/2] via 10.10.200.53, 00:54:29, GigabitEthernet1/0/24
O 10.10.80.0/25 [110/2] via 10.10.200.53, 00:54:29, GigabitEthernet1/0/24
O 10.10.91.0/26 [110/3] via 10.10.200.34, 00:53:59, GigabitEthernet1/0/5
[110/3] via 10.10.200.38, 00:53:59, GigabitEthernet1/0/6
C 10.10.92.0/26 is directly connected, Vlan90
L 10.10.92.2/32 is directly connected, Vlan90
O 10.10.93.0/26 [110/2] via 10.10.200.53, 00:54:29, GigabitEthernet1/0/24
O 10.10.200.0/30 [110/2] via 10.10.200.34, 00:54:14, GigabitEthernet1/0/5
O 10.10.200.4/30 [110/2] via 10.10.200.38, 00:54:14, GigabitEthernet1/0/6
O 10.10.200.8/30 [110/2] via 10.10.200.34, 00:54:14, GigabitEthernet1/0/5
O 10.10.200.12/30 [110/2] via 10.10.200.38, 00:54:14, GigabitEthernet1/0/6
O 10.10.200.16/30 [110/2] via 10.10.200.34, 00:54:14, GigabitEthernet1/0/5
O 10.10.200.20/30 [110/2] via 10.10.200.38, 00:54:14, GigabitEthernet1/0/6
O 10.10.200.24/30 [110/2] via 10.10.200.53, 00:54:14, GigabitEthernet1/0/24
[110/2] via 10.10.200.34, 00:54:14, GigabitEthernet1/0/5
O 10.10.200.28/30 [110/2] via 10.10.200.53, 00:54:14, GigabitEthernet1/0/24
[110/2] via 10.10.200.38, 00:54:14, GigabitEthernet1/0/6
C 10.10.200.32/30 is directly connected, GigabitEthernet1/0/5
L 10.10.200.33/32 is directly connected, GigabitEthernet1/0/5
C 10.10.200.36/30 is directly connected, GigabitEthernet1/0/6
L 10.10.200.37/32 is directly connected, GigabitEthernet1/0/6
O 10.10.200.40/30 [110/2] via 10.10.200.34, 00:53:49, GigabitEthernet1/0/5
[110/2] via 10.10.92.3, 00:53:49, Vlan90
O 10.10.200.44/30 [110/2] via 10.10.200.38, 00:53:49, GigabitEthernet1/0/6
[110/2] via 10.10.92.3, 00:53:49, Vlan90
O 10.10.200.48/30 [110/3] via 10.10.200.34, 00:54:14, GigabitEthernet1/0/5
[110/3] via 10.10.200.38, 00:54:14, GigabitEthernet1/0/6
C 10.10.200.52/30 is directly connected, GigabitEthernet1/0/24
L 10.10.200.54/32 is directly connected, GigabitEthernet1/0/24
O 10.10.200.60/30 [110/2] via 10.10.200.34, 00:54:14, GigabitEthernet1/0/5
O 10.10.200.64/30 [110/2] via 10.10.200.38, 00:54:14, GigabitEthernet1/0/6
O 10.10.200.68/30 [110/2] via 10.10.200.34, 00:54:14, GigabitEthernet1/0/5
O 10.10.200.72/30 [110/2] via 10.10.200.38, 00:54:14, GigabitEthernet1/0/6
O*E2 0.0.0.0/0 [110/1] via 10.10.200.34, 00:54:14, GigabitEthernet1/0/5
[110/1] via 10.10.200.38, 00:54:14, GigabitEthernet1/0/6
```

DSW-B2

```
DSW-B2(config)#do sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 10.10.200.42 to network 0.0.0.0
```

1.0.0.0/32 is subnetted, 1 subnets
O 1.1.1.1 [110/3] via 10.10.200.42, 00:55:10, GigabitEthernet1/0/5
[110/3] via 10.10.200.46, 00:55:10, GigabitEthernet1/0/6
2.0.0.0/32 is subnetted, 1 subnets
O 2.2.2.2 [110/3] via 10.10.200.42, 00:55:10, GigabitEthernet1/0/5
[110/3] via 10.10.200.46, 00:55:10, GigabitEthernet1/0/6
5.0.0.0/32 is subnetted, 1 subnets
O 5.5.5.5 [110/3] via 10.10.200.42, 00:55:10, GigabitEthernet1/0/5
[110/3] via 10.10.200.46, 00:55:10, GigabitEthernet1/0/6
6.0.0.0/32 is subnetted, 1 subnets
O 6.6.6.6 [110/3] via 10.10.200.42, 00:55:10, GigabitEthernet1/0/5
[110/3] via 10.10.200.46, 00:55:10, GigabitEthernet1/0/6
7.0.0.0/32 is subnetted, 1 subnets
O 7.7.7.7 [110/2] via 10.10.92.2, 00:54:45, Vlan90
8.0.0.0/32 is subnetted, 1 subnets
C 8.8.8.8 is directly connected, Loopback0
9.0.0.0/32 is subnetted, 1 subnets
O 9.9.9.9 [110/3] via 10.10.200.42, 00:55:00, GigabitEthernet1/0/5
[110/3] via 10.10.200.46, 00:55:00, GigabitEthernet1/0/6
10.0.0.0/8 is variably subnetted, 35 subnets, 6 masks
O 10.10.10.0/26 [110/3] via 10.10.200.42, 00:55:10, GigabitEthernet1/0/5
[110/3] via 10.10.200.46, 00:55:10, GigabitEthernet1/0/6
O 10.10.10.10/32 [110/3] via 10.10.200.42, 00:54:45, GigabitEthernet1/0/5
[110/3] via 10.10.200.46, 00:54:45, GigabitEthernet1/0/6
[110/3] via 10.10.92.2, 00:54:45, Vlan90
O 10.10.20.0/26 [110/3] via 10.10.200.42, 00:55:10, GigabitEthernet1/0/5
[110/3] via 10.10.200.46, 00:55:10, GigabitEthernet1/0/6
C 10.10.30.0/26 is directly connected, Vlan30
L 10.10.30.3/32 is directly connected, Vlan30
C 10.10.40.0/27 is directly connected, Vlan40
L 10.10.40.3/32 is directly connected, Vlan40
O 10.10.50.0/27 [110/3] via 10.10.200.42, 00:54:45, GigabitEthernet1/0/5
[110/3] via 10.10.200.46, 00:54:45, GigabitEthernet1/0/6
[110/3] via 10.10.92.2, 00:54:45, Vlan90
O 10.10.60.0/28 [110/3] via 10.10.200.42, 00:54:45, GigabitEthernet1/0/5
[110/3] via 10.10.200.46, 00:54:45, GigabitEthernet1/0/6
[110/3] via 10.10.92.2, 00:54:45, Vlan90
O 10.10.70.0/25 [110/3] via 10.10.200.42, 00:54:45, GigabitEthernet1/0/5
[110/3] via 10.10.200.46, 00:54:45, GigabitEthernet1/0/6
[110/3] via 10.10.92.2, 00:54:45, Vlan90
O 10.10.80.0/25 [110/3] via 10.10.200.42, 00:54:45, GigabitEthernet1/0/5
[110/3] via 10.10.200.46, 00:54:45, GigabitEthernet1/0/6
[110/3] via 10.10.92.2, 00:54:45, Vlan90
O 10.10.91.0/26 [110/3] via 10.10.200.42, 00:54:45, GigabitEthernet1/0/5
[110/3] via 10.10.200.46, 00:54:45, GigabitEthernet1/0/6
C 10.10.92.0/26 is directly connected, Vlan90
L 10.10.92.3/32 is directly connected, Vlan90
O 10.10.93.0/26 [110/3] via 10.10.200.42, 00:54:45, GigabitEthernet1/0/5
[110/3] via 10.10.200.46, 00:54:45, GigabitEthernet1/0/6
[110/3] via 10.10.92.2, 00:54:45, Vlan90
O 10.10.200.0/30 [110/2] via 10.10.200.42, 00:55:10, GigabitEthernet1/0/5
O 10.10.200.4/30 [110/2] via 10.10.200.46, 00:55:10, GigabitEthernet1/0/6
O 10.10.200.8/30 [110/2] via 10.10.200.42, 00:55:10, GigabitEthernet1/0/5
O 10.10.200.12/30 [110/2] via 10.10.200.46, 00:55:10, GigabitEthernet1/0/6

```
O 10.10.200.16/30 [110/2] via 10.10.200.42, 00:55:10, GigabitEthernet1/0/5
O 10.10.200.20/30 [110/2] via 10.10.200.46, 00:55:10, GigabitEthernet1/0/6
O 10.10.200.24/30 [110/2] via 10.10.200.42, 00:55:10, GigabitEthernet1/0/5
O 10.10.200.28/30 [110/2] via 10.10.200.46, 00:55:10, GigabitEthernet1/0/6
O 10.10.200.32/30 [110/2] via 10.10.200.42, 00:54:45, GigabitEthernet1/0/5
[110/2] via 10.10.92.2, 00:54:45, Vlan90
O 10.10.200.36/30 [110/2] via 10.10.200.46, 00:54:45, GigabitEthernet1/0/6
[110/2] via 10.10.92.2, 00:54:45, Vlan90
C 10.10.200.40/30 is directly connected, GigabitEthernet1/0/5
L 10.10.200.41/32 is directly connected, GigabitEthernet1/0/5
C 10.10.200.44/30 is directly connected, GigabitEthernet1/0/6
L 10.10.200.45/32 is directly connected, GigabitEthernet1/0/6
O 10.10.200.48/30 [110/3] via 10.10.200.42, 00:55:10, GigabitEthernet1/0/5
[110/3] via 10.10.200.46, 00:55:10, GigabitEthernet1/0/6
O 10.10.200.52/30 [110/2] via 10.10.92.2, 00:54:45, Vlan90
O 10.10.200.60/30 [110/2] via 10.10.200.42, 00:55:10, GigabitEthernet1/0/5
O 10.10.200.64/30 [110/2] via 10.10.200.46, 00:55:10, GigabitEthernet1/0/6
O 10.10.200.68/30 [110/2] via 10.10.200.42, 00:55:10, GigabitEthernet1/0/5
O 10.10.200.72/30 [110/2] via 10.10.200.46, 00:55:10, GigabitEthernet1/0/6
O*E2 0.0.0.0/0 [110/1] via 10.10.200.42, 00:55:10, GigabitEthernet1/0/5
[110/1] via 10.10.200.46, 00:55:10, GigabitEthernet1/0/6
```

DSW-C1

```
DSW-C1(config)#do sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 10.10.200.18 to network 0.0.0.0

1.0.0.0/32 is subnetted, 1 subnets
O 1.1.1.1 [110/3] via 10.10.200.18, 00:55:42, GigabitEthernet1/0/5
[110/3] via 10.10.200.22, 00:55:42, GigabitEthernet1/0/6
2.0.0.0/32 is subnetted, 1 subnets
O 2.2.2.2 [110/3] via 10.10.200.18, 00:55:42, GigabitEthernet1/0/5
[110/3] via 10.10.200.22, 00:55:42, GigabitEthernet1/0/6
5.0.0.0/32 is subnetted, 1 subnets
O 5.5.5.5 [110/3] via 10.10.200.18, 00:55:22, GigabitEthernet1/0/5
[110/3] via 10.10.200.22, 00:55:22, GigabitEthernet1/0/6
[110/3] via 10.10.200.49, 00:55:22, GigabitEthernet1/0/24
6.0.0.0/32 is subnetted, 1 subnets
O 6.6.6.6 [110/2] via 10.10.200.49, 00:55:42, GigabitEthernet1/0/24
7.0.0.0/32 is subnetted, 1 subnets
O 7.7.7.7 [110/3] via 10.10.200.18, 00:55:42, GigabitEthernet1/0/5
[110/3] via 10.10.200.22, 00:55:42, GigabitEthernet1/0/6
8.0.0.0/32 is subnetted, 1 subnets
O 8.8.8.8 [110/3] via 10.10.200.18, 00:55:42, GigabitEthernet1/0/5
[110/3] via 10.10.200.22, 00:55:42, GigabitEthernet1/0/6
9.0.0.0/32 is subnetted, 1 subnets
```

```
C 9.9.9.9 is directly connected, Loopback0
10.0.0.0/8 is variably subnetted, 38 subnets, 6 masks
O 10.10.10.0/26 [110/2] via 10.10.200.49, 00:55:42, GigabitEthernet1/0/24
O 10.10.10.10/32 [110/3] via 10.10.200.18, 00:55:42, GigabitEthernet1/0/5
[110/3] via 10.10.200.22, 00:55:42, GigabitEthernet1/0/6
O 10.10.20.0/26 [110/2] via 10.10.200.49, 00:55:42, GigabitEthernet1/0/24
O 10.10.30.0/26 [110/3] via 10.10.200.18, 00:55:42, GigabitEthernet1/0/5
[110/3] via 10.10.200.22, 00:55:42, GigabitEthernet1/0/6
O 10.10.40.0/27 [110/3] via 10.10.200.18, 00:55:42, GigabitEthernet1/0/5
[110/3] via 10.10.200.22, 00:55:42, GigabitEthernet1/0/6
C 10.10.50.0/27 is directly connected, Vlan50
L 10.10.50.2/32 is directly connected, Vlan50
C 10.10.60.0/28 is directly connected, Vlan60
L 10.10.60.2/32 is directly connected, Vlan60
C 10.10.70.0/25 is directly connected, Vlan70
L 10.10.70.2/32 is directly connected, Vlan70
C 10.10.80.0/25 is directly connected, Vlan80
L 10.10.80.2/32 is directly connected, Vlan80
O 10.10.91.0/26 [110/2] via 10.10.200.49, 00:55:22, GigabitEthernet1/0/24
O 10.10.92.0/26 [110/3] via 10.10.200.18, 00:55:12, GigabitEthernet1/0/5
[110/3] via 10.10.200.22, 00:55:12, GigabitEthernet1/0/6
C 10.10.93.0/26 is directly connected, Vlan90
L 10.10.93.2/32 is directly connected, Vlan90
O 10.10.200.0/30 [110/2] via 10.10.200.18, 00:55:42, GigabitEthernet1/0/5
O 10.10.200.4/30 [110/2] via 10.10.200.22, 00:55:42, GigabitEthernet1/0/6
O 10.10.200.8/30 [110/2] via 10.10.200.49, 00:55:42, GigabitEthernet1/0/24
[110/2] via 10.10.200.18, 00:55:42, GigabitEthernet1/0/5
O 10.10.200.12/30 [110/2] via 10.10.200.49, 00:55:42, GigabitEthernet1/0/24
[110/2] via 10.10.200.22, 00:55:42, GigabitEthernet1/0/6
C 10.10.200.16/30 is directly connected, GigabitEthernet1/0/5
L 10.10.200.17/32 is directly connected, GigabitEthernet1/0/5
C 10.10.200.20/30 is directly connected, GigabitEthernet1/0/6
L 10.10.200.21/32 is directly connected, GigabitEthernet1/0/6
O 10.10.200.24/30 [110/2] via 10.10.200.18, 00:55:42, GigabitEthernet1/0/5
O 10.10.200.28/30 [110/2] via 10.10.200.22, 00:55:42, GigabitEthernet1/0/6
O 10.10.200.32/30 [110/2] via 10.10.200.18, 00:55:42, GigabitEthernet1/0/5
O 10.10.200.36/30 [110/2] via 10.10.200.22, 00:55:42, GigabitEthernet1/0/6
O 10.10.200.40/30 [110/2] via 10.10.200.18, 00:55:42, GigabitEthernet1/0/5
O 10.10.200.44/30 [110/2] via 10.10.200.22, 00:55:42, GigabitEthernet1/0/6
C 10.10.200.48/30 is directly connected, GigabitEthernet1/0/24
L 10.10.200.50/32 is directly connected, GigabitEthernet1/0/24
O 10.10.200.52/30 [110/3] via 10.10.200.18, 00:55:42, GigabitEthernet1/0/5
[110/3] via 10.10.200.22, 00:55:42, GigabitEthernet1/0/6
O 10.10.200.60/30 [110/2] via 10.10.200.18, 00:55:42, GigabitEthernet1/0/5
O 10.10.200.64/30 [110/2] via 10.10.200.22, 00:55:42, GigabitEthernet1/0/6
O 10.10.200.68/30 [110/2] via 10.10.200.18, 00:55:42, GigabitEthernet1/0/5
O 10.10.200.72/30 [110/2] via 10.10.200.22, 00:55:42, GigabitEthernet1/0/6
O*E2 0.0.0.0/0 [110/1] via 10.10.200.18, 00:55:42, GigabitEthernet1/0/5
[110/1] via 10.10.200.22, 00:55:42, GigabitEthernet1/0/6
```

DSW-C2

```
DSW-C2(config)#do sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
```

```
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 10.10.200.26 to network 0.0.0.0

1.0.0.0/32 is subnetted, 1 subnets
O 1.1.1.1 [110/3] via 10.10.200.26, 00:56:08, GigabitEthernet1/0/5
[110/3] via 10.10.200.30, 00:56:08, GigabitEthernet1/0/6
2.0.0.0/32 is subnetted, 1 subnets
O 2.2.2.2 [110/3] via 10.10.200.26, 00:56:08, GigabitEthernet1/0/5
[110/3] via 10.10.200.30, 00:56:08, GigabitEthernet1/0/6
5.0.0.0/32 is subnetted, 1 subnets
O 5.5.5.5 [110/3] via 10.10.200.26, 00:56:08, GigabitEthernet1/0/5
[110/3] via 10.10.200.30, 00:56:08, GigabitEthernet1/0/6
6.0.0.0/32 is subnetted, 1 subnets
O 6.6.6.6 [110/3] via 10.10.200.26, 00:56:08, GigabitEthernet1/0/5
[110/3] via 10.10.200.30, 00:56:08, GigabitEthernet1/0/6
7.0.0.0/32 is subnetted, 1 subnets
O 7.7.7.7 [110/2] via 10.10.200.54, 00:56:08, GigabitEthernet1/0/24
8.0.0.0/32 is subnetted, 1 subnets
O 8.8.8.8 [110/3] via 10.10.200.26, 00:55:43, GigabitEthernet1/0/5
[110/3] via 10.10.200.30, 00:55:43, GigabitEthernet1/0/6
[110/3] via 10.10.200.54, 00:55:43, GigabitEthernet1/0/24
9.0.0.0/32 is subnetted, 1 subnets
O 9.9.9.9 [110/3] via 10.10.200.26, 00:55:58, GigabitEthernet1/0/5
[110/3] via 10.10.200.30, 00:55:58, GigabitEthernet1/0/6
10.0.0.0/8 is variably subnetted, 38 subnets, 6 masks
O 10.10.10.0/26 [110/3] via 10.10.200.26, 00:56:08, GigabitEthernet1/0/5
[110/3] via 10.10.200.30, 00:56:08, GigabitEthernet1/0/6
C 10.10.10.10/32 is directly connected, Loopback0
O 10.10.20.0/26 [110/3] via 10.10.200.26, 00:56:08, GigabitEthernet1/0/5
[110/3] via 10.10.200.30, 00:56:08, GigabitEthernet1/0/6
O 10.10.30.0/26 [110/2] via 10.10.200.54, 00:56:08, GigabitEthernet1/0/24
O 10.10.40.0/27 [110/2] via 10.10.200.54, 00:56:08, GigabitEthernet1/0/24
C 10.10.50.0/27 is directly connected, Vlan50
L 10.10.50.3/32 is directly connected, Vlan50
C 10.10.60.0/28 is directly connected, Vlan60
L 10.10.60.3/32 is directly connected, Vlan60
C 10.10.70.0/25 is directly connected, Vlan70
L 10.10.70.3/32 is directly connected, Vlan70
C 10.10.80.0/25 is directly connected, Vlan80
L 10.10.80.3/32 is directly connected, Vlan80
O 10.10.91.0/26 [110/3] via 10.10.200.26, 00:55:43, GigabitEthernet1/0/5
[110/3] via 10.10.200.30, 00:55:43, GigabitEthernet1/0/6
O 10.10.92.0/26 [110/2] via 10.10.200.54, 00:55:43, GigabitEthernet1/0/24
C 10.10.93.0/26 is directly connected, Vlan90
L 10.10.93.3/32 is directly connected, Vlan90
O 10.10.200.0/30 [110/2] via 10.10.200.26, 00:56:08, GigabitEthernet1/0/5
O 10.10.200.4/30 [110/2] via 10.10.200.30, 00:56:08, GigabitEthernet1/0/6
O 10.10.200.8/30 [110/2] via 10.10.200.26, 00:56:08, GigabitEthernet1/0/5
O 10.10.200.12/30 [110/2] via 10.10.200.30, 00:56:08, GigabitEthernet1/0/6
O 10.10.200.16/30 [110/2] via 10.10.200.26, 00:56:08, GigabitEthernet1/0/5
```

```
O 10.10.200.20/30 [110/2] via 10.10.200.30, 00:56:08, GigabitEthernet1/0/6
C 10.10.200.24/30 is directly connected, GigabitEthernet1/0/5
L 10.10.200.25/32 is directly connected, GigabitEthernet1/0/5
C 10.10.200.28/30 is directly connected, GigabitEthernet1/0/6
L 10.10.200.29/32 is directly connected, GigabitEthernet1/0/6
O 10.10.200.32/30 [110/2] via 10.10.200.54, 00:56:08, GigabitEthernet1/0/24
[110/2] via 10.10.200.26, 00:56:08, GigabitEthernet1/0/5
O 10.10.200.36/30 [110/2] via 10.10.200.54, 00:56:08, GigabitEthernet1/0/24
[110/2] via 10.10.200.30, 00:56:08, GigabitEthernet1/0/6
O 10.10.200.40/30 [110/2] via 10.10.200.26, 00:56:08, GigabitEthernet1/0/5
O 10.10.200.44/30 [110/2] via 10.10.200.30, 00:56:08, GigabitEthernet1/0/6
O 10.10.200.48/30 [110/3] via 10.10.200.26, 00:56:08, GigabitEthernet1/0/5
[110/3] via 10.10.200.30, 00:56:08, GigabitEthernet1/0/6
C 10.10.200.52/30 is directly connected, GigabitEthernet1/0/24
L 10.10.200.53/32 is directly connected, GigabitEthernet1/0/24
O 10.10.200.60/30 [110/2] via 10.10.200.26, 00:56:08, GigabitEthernet1/0/5
O 10.10.200.64/30 [110/2] via 10.10.200.30, 00:56:08, GigabitEthernet1/0/6
O 10.10.200.68/30 [110/2] via 10.10.200.26, 00:56:08, GigabitEthernet1/0/5
O 10.10.200.72/30 [110/2] via 10.10.200.30, 00:56:08, GigabitEthernet1/0/6
O*E2 0.0.0.0/0 [110/1] via 10.10.200.26, 00:56:08, GigabitEthernet1/0/5
[110/1] via 10.10.200.30, 00:56:08, GigabitEthernet1/0/6
```

CSW1

```
CSW1(config)#do sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 10.10.200.61 to network 0.0.0.0

1.0.0.0/32 is subnetted, 1 subnets
O 1.1.1.1 [110/2] via 10.10.200.61, 00:56:35, GigabitEthernet1/0/1
2.0.0.0/32 is subnetted, 1 subnets
O 2.2.2.2 [110/2] via 10.10.200.69, 00:56:35, GigabitEthernet1/0/2
3.0.0.0/32 is subnetted, 1 subnets
C 3.3.3.3 is directly connected, Loopback0
5.0.0.0/32 is subnetted, 1 subnets
O 5.5.5.5 [110/2] via 10.10.200.1, 00:56:35, GigabitEthernet1/0/7
6.0.0.0/32 is subnetted, 1 subnets
O 6.6.6.6 [110/2] via 10.10.200.9, 00:56:35, GigabitEthernet1/0/8
7.0.0.0/32 is subnetted, 1 subnets
O 7.7.7.7 [110/2] via 10.10.200.33, 00:56:35, GigabitEthernet1/0/11
8.0.0.0/32 is subnetted, 1 subnets
O 8.8.8.8 [110/2] via 10.10.200.41, 00:56:35, GigabitEthernet1/0/12
9.0.0.0/32 is subnetted, 1 subnets
O 9.9.9.9 [110/2] via 10.10.200.17, 00:56:35, GigabitEthernet1/0/9
10.0.0.0/8 is variably subnetted, 38 subnets, 6 masks
O 10.10.10.0/26 [110/2] via 10.10.200.1, 00:56:35, GigabitEthernet1/0/7
[110/2] via 10.10.200.9, 00:56:35, GigabitEthernet1/0/8
O 10.10.10.10/32 [110/2] via 10.10.200.25, 00:56:35, GigabitEthernet1/0/10
```

```
O 10.10.20.0/26 [110/2] via 10.10.200.1, 00:56:35, GigabitEthernet1/0/7
[110/2] via 10.10.200.9, 00:56:35, GigabitEthernet1/0/8
O 10.10.30.0/26 [110/2] via 10.10.200.33, 00:56:35, GigabitEthernet1/0/11
[110/2] via 10.10.200.41, 00:56:35, GigabitEthernet1/0/12
O 10.10.40.0/27 [110/2] via 10.10.200.33, 00:56:35, GigabitEthernet1/0/11
[110/2] via 10.10.200.41, 00:56:35, GigabitEthernet1/0/12
O 10.10.50.0/27 [110/2] via 10.10.200.17, 00:56:35, GigabitEthernet1/0/9
[110/2] via 10.10.200.25, 00:56:35, GigabitEthernet1/0/10
O 10.10.60.0/28 [110/2] via 10.10.200.17, 00:56:35, GigabitEthernet1/0/9
[110/2] via 10.10.200.25, 00:56:35, GigabitEthernet1/0/10
O 10.10.70.0/25 [110/2] via 10.10.200.17, 00:56:35, GigabitEthernet1/0/9
[110/2] via 10.10.200.25, 00:56:35, GigabitEthernet1/0/10
O 10.10.80.0/25 [110/2] via 10.10.200.17, 00:56:35, GigabitEthernet1/0/9
[110/2] via 10.10.200.25, 00:56:35, GigabitEthernet1/0/10
O 10.10.91.0/26 [110/2] via 10.10.200.1, 00:56:10, GigabitEthernet1/0/7
[110/2] via 10.10.200.9, 00:56:10, GigabitEthernet1/0/8
O 10.10.92.0/26 [110/2] via 10.10.200.33, 00:56:10, GigabitEthernet1/0/11
[110/2] via 10.10.200.41, 00:56:10, GigabitEthernet1/0/12
O 10.10.93.0/26 [110/2] via 10.10.200.17, 00:56:35, GigabitEthernet1/0/9
[110/2] via 10.10.200.25, 00:56:35, GigabitEthernet1/0/10
C 10.10.200.0/30 is directly connected, GigabitEthernet1/0/7
L 10.10.200.2/32 is directly connected, GigabitEthernet1/0/7
O 10.10.200.4/30 [110/2] via 10.10.200.1, 00:56:35, GigabitEthernet1/0/7
C 10.10.200.8/30 is directly connected, GigabitEthernet1/0/8
L 10.10.200.10/32 is directly connected, GigabitEthernet1/0/8
O 10.10.200.12/30 [110/2] via 10.10.200.9, 00:56:35, GigabitEthernet1/0/8
C 10.10.200.16/30 is directly connected, GigabitEthernet1/0/9
L 10.10.200.18/32 is directly connected, GigabitEthernet1/0/9
O 10.10.200.20/30 [110/2] via 10.10.200.17, 00:56:35, GigabitEthernet1/0/9
C 10.10.200.24/30 is directly connected, GigabitEthernet1/0/10
L 10.10.200.26/32 is directly connected, GigabitEthernet1/0/10
O 10.10.200.28/30 [110/2] via 10.10.200.25, 00:56:35, GigabitEthernet1/0/10
C 10.10.200.32/30 is directly connected, GigabitEthernet1/0/11
L 10.10.200.34/32 is directly connected, GigabitEthernet1/0/11
O 10.10.200.36/30 [110/2] via 10.10.200.33, 00:56:35, GigabitEthernet1/0/11
C 10.10.200.40/30 is directly connected, GigabitEthernet1/0/12
L 10.10.200.42/32 is directly connected, GigabitEthernet1/0/12
O 10.10.200.44/30 [110/2] via 10.10.200.41, 00:56:35, GigabitEthernet1/0/12
O 10.10.200.48/30 [110/2] via 10.10.200.9, 00:56:35, GigabitEthernet1/0/8
[110/2] via 10.10.200.17, 00:56:35, GigabitEthernet1/0/9
O 10.10.200.52/30 [110/2] via 10.10.200.25, 00:56:35, GigabitEthernet1/0/10
[110/2] via 10.10.200.33, 00:56:35, GigabitEthernet1/0/11
C 10.10.200.60/30 is directly connected, GigabitEthernet1/0/1
L 10.10.200.62/32 is directly connected, GigabitEthernet1/0/1
O 10.10.200.64/30 [110/2] via 10.10.200.61, 00:56:35, GigabitEthernet1/0/1
C 10.10.200.68/30 is directly connected, GigabitEthernet1/0/2
L 10.10.200.70/32 is directly connected, GigabitEthernet1/0/2
O 10.10.200.72/30 [110/2] via 10.10.200.69, 00:56:35, GigabitEthernet1/0/2
O*E2 0.0.0.0/0 [110/1] via 10.10.200.61, 00:56:35, GigabitEthernet1/0/1
[110/1] via 10.10.200.69, 00:56:35, GigabitEthernet1/0/2
```

CSW2

```
CSW2(config)#do sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
```

```
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 10.10.200.65 to network 0.0.0.0

1.0.0.0/32 is subnetted, 1 subnets
O 1.1.1.1 [110/2] via 10.10.200.65, 00:58:03, GigabitEthernet1/0/1
2.0.0.0/32 is subnetted, 1 subnets
O 2.2.2.2 [110/2] via 10.10.200.73, 00:58:03, GigabitEthernet1/0/2
4.0.0.0/32 is subnetted, 1 subnets
C 4.4.4.4 is directly connected, Loopback0
5.0.0.0/32 is subnetted, 1 subnets
O 5.5.5.5 [110/2] via 10.10.200.5, 00:58:03, GigabitEthernet1/0/7
6.0.0.0/32 is subnetted, 1 subnets
O 6.6.6.6 [110/2] via 10.10.200.13, 00:58:03, GigabitEthernet1/0/8
7.0.0.0/32 is subnetted, 1 subnets
O 7.7.7.7 [110/2] via 10.10.200.37, 00:58:03, GigabitEthernet1/0/11
8.0.0.0/32 is subnetted, 1 subnets
O 8.8.8.8 [110/2] via 10.10.200.45, 00:58:03, GigabitEthernet1/0/12
9.0.0.0/32 is subnetted, 1 subnets
O 9.9.9.9 [110/2] via 10.10.200.21, 00:58:03, GigabitEthernet1/0/9
10.0.0.0/8 is variably subnetted, 38 subnets, 6 masks
O 10.10.10.0/26 [110/2] via 10.10.200.5, 00:58:03, GigabitEthernet1/0/7
[110/2] via 10.10.200.13, 00:58:03, GigabitEthernet1/0/8
O 10.10.10.10/32 [110/2] via 10.10.200.29, 00:58:03, GigabitEthernet1/0/10
O 10.10.20.0/26 [110/2] via 10.10.200.5, 00:58:03, GigabitEthernet1/0/7
[110/2] via 10.10.200.13, 00:58:03, GigabitEthernet1/0/8
O 10.10.30.0/26 [110/2] via 10.10.200.37, 00:58:03, GigabitEthernet1/0/11
[110/2] via 10.10.200.45, 00:58:03, GigabitEthernet1/0/12
O 10.10.40.0/27 [110/2] via 10.10.200.37, 00:58:03, GigabitEthernet1/0/11
[110/2] via 10.10.200.45, 00:58:03, GigabitEthernet1/0/12
O 10.10.50.0/27 [110/2] via 10.10.200.21, 00:58:03, GigabitEthernet1/0/9
[110/2] via 10.10.200.29, 00:58:03, GigabitEthernet1/0/10
O 10.10.60.0/28 [110/2] via 10.10.200.21, 00:58:03, GigabitEthernet1/0/9
[110/2] via 10.10.200.29, 00:58:03, GigabitEthernet1/0/10
O 10.10.70.0/25 [110/2] via 10.10.200.21, 00:58:03, GigabitEthernet1/0/9
[110/2] via 10.10.200.29, 00:58:03, GigabitEthernet1/0/10
O 10.10.80.0/25 [110/2] via 10.10.200.21, 00:58:03, GigabitEthernet1/0/9
[110/2] via 10.10.200.29, 00:58:03, GigabitEthernet1/0/10
O 10.10.91.0/26 [110/2] via 10.10.200.5, 00:57:48, GigabitEthernet1/0/7
[110/2] via 10.10.200.13, 00:57:48, GigabitEthernet1/0/8
O 10.10.92.0/26 [110/2] via 10.10.200.37, 00:57:38, GigabitEthernet1/0/11
[110/2] via 10.10.200.45, 00:57:38, GigabitEthernet1/0/12
O 10.10.93.0/26 [110/2] via 10.10.200.21, 00:58:03, GigabitEthernet1/0/9
[110/2] via 10.10.200.29, 00:58:03, GigabitEthernet1/0/10
O 10.10.200.0/30 [110/2] via 10.10.200.5, 00:58:03, GigabitEthernet1/0/7
C 10.10.200.4/30 is directly connected, GigabitEthernet1/0/7
L 10.10.200.6/32 is directly connected, GigabitEthernet1/0/7
O 10.10.200.8/30 [110/2] via 10.10.200.13, 00:58:03, GigabitEthernet1/0/8
C 10.10.200.12/30 is directly connected, GigabitEthernet1/0/8
L 10.10.200.14/32 is directly connected, GigabitEthernet1/0/8
```

```
O 10.10.200.16/30 [110/2] via 10.10.200.21, 00:58:03, GigabitEthernet1/0/9
C 10.10.200.20/30 is directly connected, GigabitEthernet1/0/9
L 10.10.200.22/32 is directly connected, GigabitEthernet1/0/9
O 10.10.200.24/30 [110/2] via 10.10.200.29, 00:58:03, GigabitEthernet1/0/10
C 10.10.200.28/30 is directly connected, GigabitEthernet1/0/10
L 10.10.200.30/32 is directly connected, GigabitEthernet1/0/10
O 10.10.200.32/30 [110/2] via 10.10.200.37, 00:58:03, GigabitEthernet1/0/11
C 10.10.200.36/30 is directly connected, GigabitEthernet1/0/11
L 10.10.200.38/32 is directly connected, GigabitEthernet1/0/11
O 10.10.200.40/30 [110/2] via 10.10.200.45, 00:58:03, GigabitEthernet1/0/12
C 10.10.200.44/30 is directly connected, GigabitEthernet1/0/12
L 10.10.200.46/32 is directly connected, GigabitEthernet1/0/12
O 10.10.200.48/30 [110/2] via 10.10.200.13, 00:58:03, GigabitEthernet1/0/8
[110/2] via 10.10.200.21, 00:58:03, GigabitEthernet1/0/9
O 10.10.200.52/30 [110/2] via 10.10.200.29, 00:58:03, GigabitEthernet1/0/10
[110/2] via 10.10.200.37, 00:58:03, GigabitEthernet1/0/11
O 10.10.200.60/30 [110/2] via 10.10.200.65, 00:58:03, GigabitEthernet1/0/1
C 10.10.200.64/30 is directly connected, GigabitEthernet1/0/1
L 10.10.200.66/32 is directly connected, GigabitEthernet1/0/1
O 10.10.200.68/30 [110/2] via 10.10.200.73, 00:58:03, GigabitEthernet1/0/2
C 10.10.200.72/30 is directly connected, GigabitEthernet1/0/2
L 10.10.200.74/32 is directly connected, GigabitEthernet1/0/2
O*E2 0.0.0.0/0 [110/1] via 10.10.200.65, 00:58:03, GigabitEthernet1/0/1
[110/1] via 10.10.200.73, 00:58:03, GigabitEthernet1/0/2
```

R1

```
R1(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 203.0.113.1 to network 0.0.0.0

1.0.0.0/32 is subnetted, 1 subnets
C 1.1.1.1/32 is directly connected, Loopback0
2.0.0.0/32 is subnetted, 1 subnets
O 2.2.2.2/32 [110/3] via 10.10.200.62, 00:58:21, GigabitEthernet0/1
[110/3] via 10.10.200.66, 00:58:21, GigabitEthernet0/2
5.0.0.0/32 is subnetted, 1 subnets
O 5.5.5.5/32 [110/3] via 10.10.200.62, 00:58:21, GigabitEthernet0/1
[110/3] via 10.10.200.66, 00:58:21, GigabitEthernet0/2
6.0.0.0/32 is subnetted, 1 subnets
O 6.6.6.6/32 [110/3] via 10.10.200.62, 00:58:21, GigabitEthernet0/1
[110/3] via 10.10.200.66, 00:58:21, GigabitEthernet0/2
7.0.0.0/32 is subnetted, 1 subnets
O 7.7.7.7/32 [110/3] via 10.10.200.62, 00:58:21, GigabitEthernet0/1
[110/3] via 10.10.200.66, 00:58:21, GigabitEthernet0/2
8.0.0.0/32 is subnetted, 1 subnets
O 8.8.8.8/32 [110/3] via 10.10.200.62, 00:58:21, GigabitEthernet0/1
[110/3] via 10.10.200.66, 00:58:21, GigabitEthernet0/2
```

```
9.0.0.0/32 is subnetted, 1 subnets
O 9.9.9.9/32 [110/3] via 10.10.200.62, 00:58:21, GigabitEthernet0/1
[110/3] via 10.10.200.66, 00:58:21, GigabitEthernet0/2
10.0.0.0/8 is variably subnetted, 32 subnets, 6 masks
O 10.10.10.0/26 [110/3] via 10.10.200.62, 00:58:21, GigabitEthernet0/1
[110/3] via 10.10.200.66, 00:58:21, GigabitEthernet0/2
O 10.10.10.10/32 [110/3] via 10.10.200.62, 00:58:21, GigabitEthernet0/1
[110/3] via 10.10.200.66, 00:58:21, GigabitEthernet0/2
O 10.10.20.0/26 [110/3] via 10.10.200.62, 00:58:21, GigabitEthernet0/1
[110/3] via 10.10.200.66, 00:58:21, GigabitEthernet0/2
O 10.10.30.0/26 [110/3] via 10.10.200.62, 00:58:21, GigabitEthernet0/1
[110/3] via 10.10.200.66, 00:58:21, GigabitEthernet0/2
O 10.10.40.0/27 [110/3] via 10.10.200.62, 00:58:21, GigabitEthernet0/1
[110/3] via 10.10.200.66, 00:58:21, GigabitEthernet0/2
O 10.10.50.0/27 [110/3] via 10.10.200.62, 00:58:21, GigabitEthernet0/1
[110/3] via 10.10.200.66, 00:58:21, GigabitEthernet0/2
O 10.10.60.0/28 [110/3] via 10.10.200.62, 00:58:21, GigabitEthernet0/1
[110/3] via 10.10.200.66, 00:58:21, GigabitEthernet0/2
O 10.10.70.0/25 [110/3] via 10.10.200.62, 00:58:21, GigabitEthernet0/1
[110/3] via 10.10.200.66, 00:58:21, GigabitEthernet0/2
O 10.10.80.0/25 [110/3] via 10.10.200.62, 00:58:21, GigabitEthernet0/1
[110/3] via 10.10.200.66, 00:58:21, GigabitEthernet0/2
O 10.10.91.0/26 [110/3] via 10.10.200.62, 00:58:06, GigabitEthernet0/1
[110/3] via 10.10.200.66, 00:58:06, GigabitEthernet0/2
O 10.10.92.0/26 [110/3] via 10.10.200.62, 00:57:56, GigabitEthernet0/1
[110/3] via 10.10.200.66, 00:57:56, GigabitEthernet0/2
O 10.10.93.0/26 [110/3] via 10.10.200.62, 00:58:21, GigabitEthernet0/1
[110/3] via 10.10.200.66, 00:58:21, GigabitEthernet0/2
O 10.10.200.0/30 [110/2] via 10.10.200.62, 00:58:21, GigabitEthernet0/1
O 10.10.200.4/30 [110/2] via 10.10.200.66, 00:58:21, GigabitEthernet0/2
O 10.10.200.8/30 [110/2] via 10.10.200.62, 00:58:21, GigabitEthernet0/1
O 10.10.200.12/30 [110/2] via 10.10.200.66, 00:58:21, GigabitEthernet0/2
O 10.10.200.16/30 [110/2] via 10.10.200.62, 00:58:21, GigabitEthernet0/1
O 10.10.200.20/30 [110/2] via 10.10.200.66, 00:58:21, GigabitEthernet0/2
O 10.10.200.24/30 [110/2] via 10.10.200.62, 00:58:21, GigabitEthernet0/1
O 10.10.200.28/30 [110/2] via 10.10.200.66, 00:58:21, GigabitEthernet0/2
O 10.10.200.32/30 [110/2] via 10.10.200.62, 00:58:21, GigabitEthernet0/1
O 10.10.200.36/30 [110/2] via 10.10.200.66, 00:58:21, GigabitEthernet0/2
O 10.10.200.40/30 [110/2] via 10.10.200.62, 00:58:21, GigabitEthernet0/1
O 10.10.200.44/30 [110/2] via 10.10.200.66, 00:58:21, GigabitEthernet0/2
O 10.10.200.48/30 [110/3] via 10.10.200.62, 00:58:21, GigabitEthernet0/1
[110/3] via 10.10.200.66, 00:58:21, GigabitEthernet0/2
O 10.10.200.52/30 [110/3] via 10.10.200.62, 00:58:21, GigabitEthernet0/1
[110/3] via 10.10.200.66, 00:58:21, GigabitEthernet0/2
C 10.10.200.60/30 is directly connected, GigabitEthernet0/1
L 10.10.200.61/32 is directly connected, GigabitEthernet0/1
C 10.10.200.64/30 is directly connected, GigabitEthernet0/2
L 10.10.200.65/32 is directly connected, GigabitEthernet0/2
O 10.10.200.68/30 [110/2] via 10.10.200.62, 00:58:21, GigabitEthernet0/1
O 10.10.200.72/30 [110/2] via 10.10.200.66, 00:58:21, GigabitEthernet0/2
203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C 203.0.113.0/29 is directly connected, GigabitEthernet0/0
L 203.0.113.2/32 is directly connected, GigabitEthernet0/0
S* 0.0.0.0/0 [1/0] via 203.0.113.1
```

R2

```
R1(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

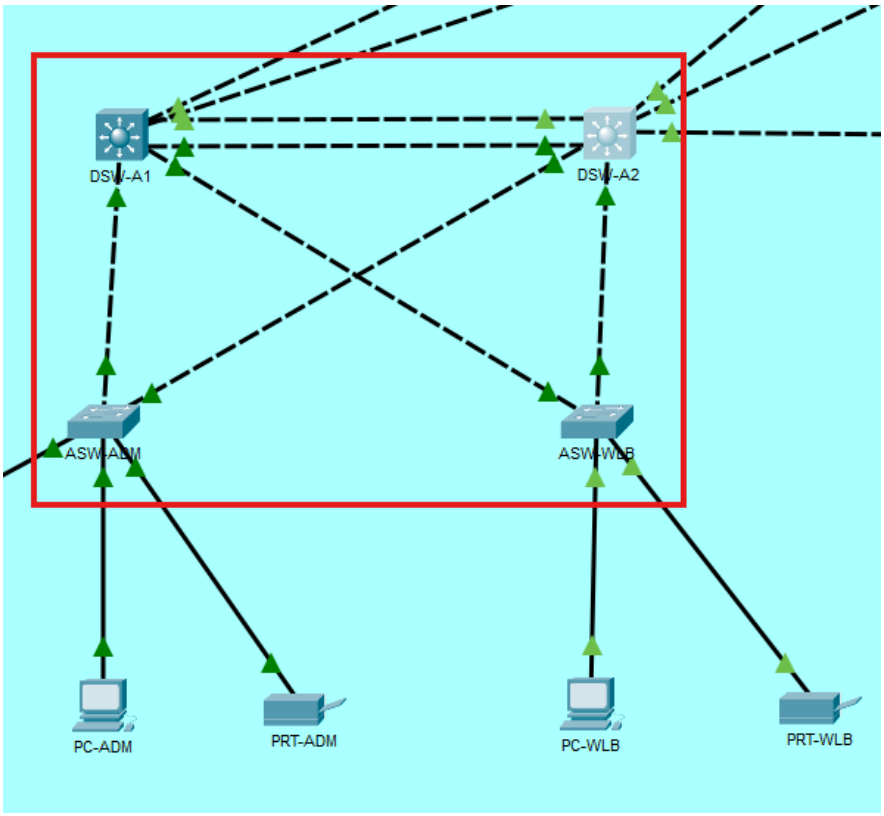
Gateway of last resort is 203.0.113.1 to network 0.0.0.0

1.0.0.0/32 is subnetted, 1 subnets
O 1.1.1.1/32 [110/3] via 10.10.200.74, 00:58:45, GigabitEthernet0/2
[110/3] via 10.10.200.70, 00:58:45, GigabitEthernet0/1
2.0.0.0/32 is subnetted, 1 subnets
C 2.2.2.2/32 is directly connected, Loopback0
5.0.0.0/32 is subnetted, 1 subnets
O 5.5.5.5/32 [110/3] via 10.10.200.74, 00:58:45, GigabitEthernet0/2
[110/3] via 10.10.200.70, 00:58:45, GigabitEthernet0/1
6.0.0.0/32 is subnetted, 1 subnets
O 6.6.6.6/32 [110/3] via 10.10.200.74, 00:58:45, GigabitEthernet0/2
[110/3] via 10.10.200.70, 00:58:45, GigabitEthernet0/1
7.0.0.0/32 is subnetted, 1 subnets
O 7.7.7.7/32 [110/3] via 10.10.200.74, 00:58:45, GigabitEthernet0/2
[110/3] via 10.10.200.70, 00:58:45, GigabitEthernet0/1
8.0.0.0/32 is subnetted, 1 subnets
O 8.8.8.8/32 [110/3] via 10.10.200.74, 00:58:45, GigabitEthernet0/2
[110/3] via 10.10.200.70, 00:58:45, GigabitEthernet0/1
9.0.0.0/32 is subnetted, 1 subnets
O 9.9.9.9/32 [110/3] via 10.10.200.74, 00:58:35, GigabitEthernet0/2
[110/3] via 10.10.200.70, 00:58:35, GigabitEthernet0/1
10.0.0.0/8 is variably subnetted, 32 subnets, 6 masks
O 10.10.10.0/26 [110/3] via 10.10.200.74, 00:58:45, GigabitEthernet0/2
[110/3] via 10.10.200.70, 00:58:45, GigabitEthernet0/1
O 10.10.10.10/32 [110/3] via 10.10.200.74, 00:58:45, GigabitEthernet0/2
[110/3] via 10.10.200.70, 00:58:45, GigabitEthernet0/1
O 10.10.20.0/26 [110/3] via 10.10.200.74, 00:58:45, GigabitEthernet0/2
[110/3] via 10.10.200.70, 00:58:45, GigabitEthernet0/1
O 10.10.30.0/26 [110/3] via 10.10.200.74, 00:58:45, GigabitEthernet0/2
[110/3] via 10.10.200.70, 00:58:45, GigabitEthernet0/1
O 10.10.40.0/27 [110/3] via 10.10.200.74, 00:58:45, GigabitEthernet0/2
[110/3] via 10.10.200.70, 00:58:45, GigabitEthernet0/1
O 10.10.50.0/27 [110/3] via 10.10.200.74, 00:58:45, GigabitEthernet0/2
[110/3] via 10.10.200.70, 00:58:45, GigabitEthernet0/1
O 10.10.60.0/28 [110/3] via 10.10.200.74, 00:58:45, GigabitEthernet0/2
[110/3] via 10.10.200.70, 00:58:45, GigabitEthernet0/1
O 10.10.70.0/25 [110/3] via 10.10.200.74, 00:58:45, GigabitEthernet0/2
[110/3] via 10.10.200.70, 00:58:45, GigabitEthernet0/1
O 10.10.80.0/25 [110/3] via 10.10.200.74, 00:58:45, GigabitEthernet0/2
[110/3] via 10.10.200.70, 00:58:45, GigabitEthernet0/1
O 10.10.91.0/26 [110/3] via 10.10.200.74, 00:58:20, GigabitEthernet0/2
[110/3] via 10.10.200.70, 00:58:20, GigabitEthernet0/1
O 10.10.92.0/26 [110/3] via 10.10.200.74, 00:58:20, GigabitEthernet0/2
[110/3] via 10.10.200.70, 00:58:20, GigabitEthernet0/1
```

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O 10.10.93.0/26 [110/3] via 10.10.200.74, 00:58:45, GigabitEthernet0/2
[110/3] via 10.10.200.70, 00:58:45, GigabitEthernet0/1
O 10.10.200.0/30 [110/2] via 10.10.200.70, 00:58:45, GigabitEthernet0/1
O 10.10.200.4/30 [110/2] via 10.10.200.74, 00:58:45, GigabitEthernet0/2
O 10.10.200.8/30 [110/2] via 10.10.200.70, 00:58:45, GigabitEthernet0/1
O 10.10.200.12/30 [110/2] via 10.10.200.74, 00:58:45, GigabitEthernet0/2
O 10.10.200.16/30 [110/2] via 10.10.200.70, 00:58:45, GigabitEthernet0/1
O 10.10.200.20/30 [110/2] via 10.10.200.74, 00:58:45, GigabitEthernet0/2
O 10.10.200.24/30 [110/2] via 10.10.200.70, 00:58:45, GigabitEthernet0/1
O 10.10.200.28/30 [110/2] via 10.10.200.74, 00:58:45, GigabitEthernet0/2
O 10.10.200.32/30 [110/2] via 10.10.200.70, 00:58:45, GigabitEthernet0/1
O 10.10.200.36/30 [110/2] via 10.10.200.74, 00:58:45, GigabitEthernet0/2
O 10.10.200.40/30 [110/2] via 10.10.200.70, 00:58:45, GigabitEthernet0/1
O 10.10.200.44/30 [110/2] via 10.10.200.74, 00:58:45, GigabitEthernet0/2
O 10.10.200.48/30 [110/3] via 10.10.200.74, 00:58:45, GigabitEthernet0/2
[110/3] via 10.10.200.70, 00:58:45, GigabitEthernet0/1
O 10.10.200.52/30 [110/3] via 10.10.200.74, 00:58:45, GigabitEthernet0/2
[110/3] via 10.10.200.70, 00:58:45, GigabitEthernet0/1
O 10.10.200.60/30 [110/2] via 10.10.200.70, 00:58:45, GigabitEthernet0/1
O 10.10.200.64/30 [110/2] via 10.10.200.74, 00:58:45, GigabitEthernet0/2
C 10.10.200.68/30 is directly connected, GigabitEthernet0/1
L 10.10.200.69/32 is directly connected, GigabitEthernet0/1
C 10.10.200.72/30 is directly connected, GigabitEthernet0/2
L 10.10.200.73/32 is directly connected, GigabitEthernet0/2
203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C 203.0.113.0/29 is directly connected, GigabitEthernet0/0
L 203.0.113.3/32 is directly connected, GigabitEthernet0/0
S* 0.0.0.0/0 [1/0] via 203.0.113.1
```

9. Switchport Status (STP Status of all Switches participating in STP)

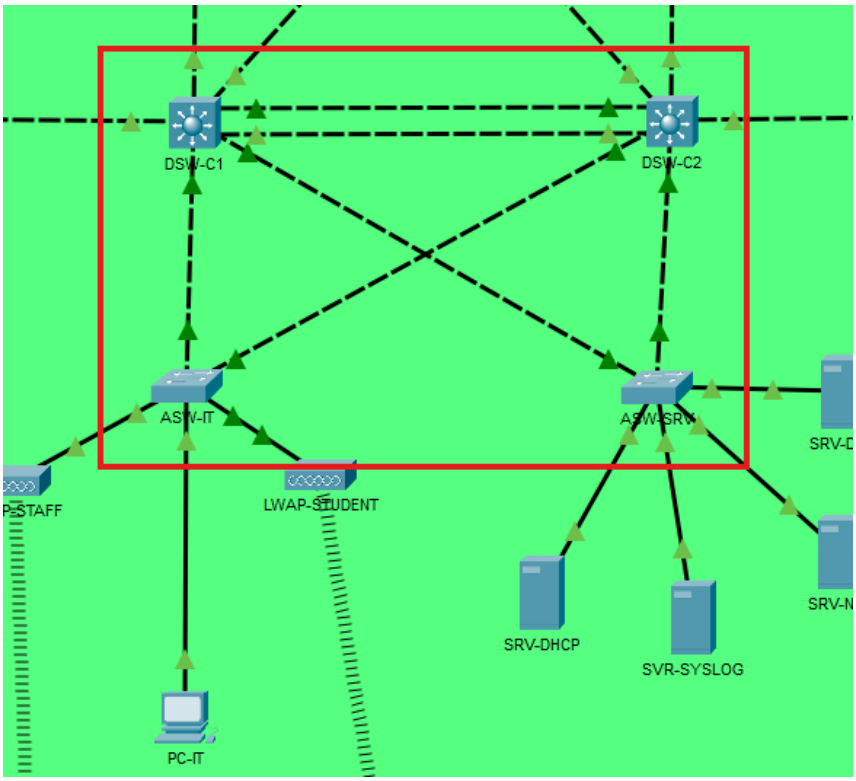
ADMIN BUILDING



SWITCH	VLAN (NAME)	ROOT	PORT	STATUS
ASW-ADM	VLAN10 (ADMISSIONS)	NO	F0/1	Designated
			F0/2	Designated
			F0/4	Designated
			G0/1	Root
			G0/2	Alternate
	VLAN20 (WELLBEING)	NO	G0/1	Alternate
			G0/2	Root
	VLAN90 (MGMT)	NO	G0/1	Alternate
			G0/2	Root
ASW-WLB	VLAN20 (WELLBEING)	NO	F0/1	Designated
			F0/2	Designated
			G0/1	Alternate
			G0/2	Root
	VLAN10 (ADMISSIONS)	NO	G0/1	Root

			G0/2	Alternate
	VLAN90 (MGMT)	NO	G0/1	Alternate
			G0/2	Root
DSW-A1	VLAN10 (ADMISSIONS)	YES	G1/0/1	Designated
			G1/0/2	Designated
			Po1	Designated
	VLAN20 (WELLBEING)	NO	G1/0/1	Designated
			G1/0/2	Designated
			Po1	Root
	VLAN90 (MGMT)	NO	G1/0/1	Designated
			G1/0/2	Designated
			Po1	Root
DSW-A2	VLAN10 (ADMISSIONS)	NO	G1/0/1	Designated
			G1/0/2	Designated
			Po1	Root
	VLAN20 (WELLBEING)	YES	G1/0/1	Designated
			G1/0/2	Designated
			Po1	Designated
	VLAN90 (MGMT)	YES	G1/0/1	Designated
			G1/0/2	Designated
			Po1	Designated

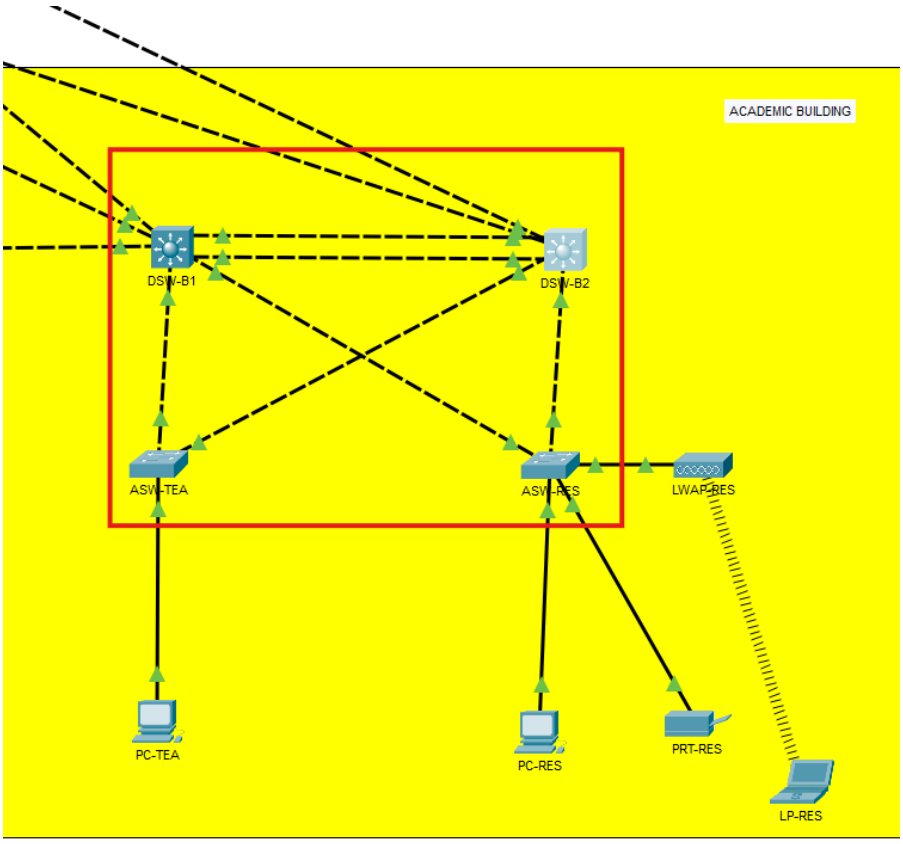
IT BUILDING



SWITCH	VLAN (NAME)	ROOT	PORT	STATUS
ASW-IT	VLAN50 (IT-ADMIN)	NO	F0/1	Designated
			G0/1	Root
			G0/2	Alternate
	VLAN60 (SERVERS-INFRA)	NO	G0/1	Alternate
			G0/2	Root
	VLAN90 (MGMT)	NO	G0/1	Root
			G0/2	Alternate
ASW-SRV	VLAN60 (SERVERS-INFRA)	NO	F0/1	Designated
			F0/2	Designated
			F0/3	Designated
			F0/4	Designated
			G0/1	Alternate
			G0/2	Root
	VLAN50 (IT-ADMIN)	NO	G0/1	Root

			G0/2	Alternate
	VLAN90 (MGMT)	NO	G0/1	Root
			G0/2	Alternate
DSW-C1	VLAN50 (IT-ADMIN)	YES	G1/0/1	Designated
			G1/0/2	Designated
			Po1	Designated
	VLAN60 (SERVERS-INFRA)	NO	G1/0/1	Designated
			G1/0/2	Designated
			Po1	Root
	VLAN90 (MGMT)	YES	G1/0/1	Designated
			G1/0/2	Designated
			Po1	Designated
DSW-C2	VLAN50 (IT-ADMIN)	NO	G1/0/1	Designated
			G1/0/2	Designated
			Po1	Root
	VLAN60 (SERVERS-INFRA)	YES	G1/0/1	Designated
			G1/0/2	Designated
			Po1	Designated
	VLAN90 (MGMT)	YES	G1/0/1	Designated
			G1/0/2	Designated
			Po1	Root

ACADEMIC BUILDING

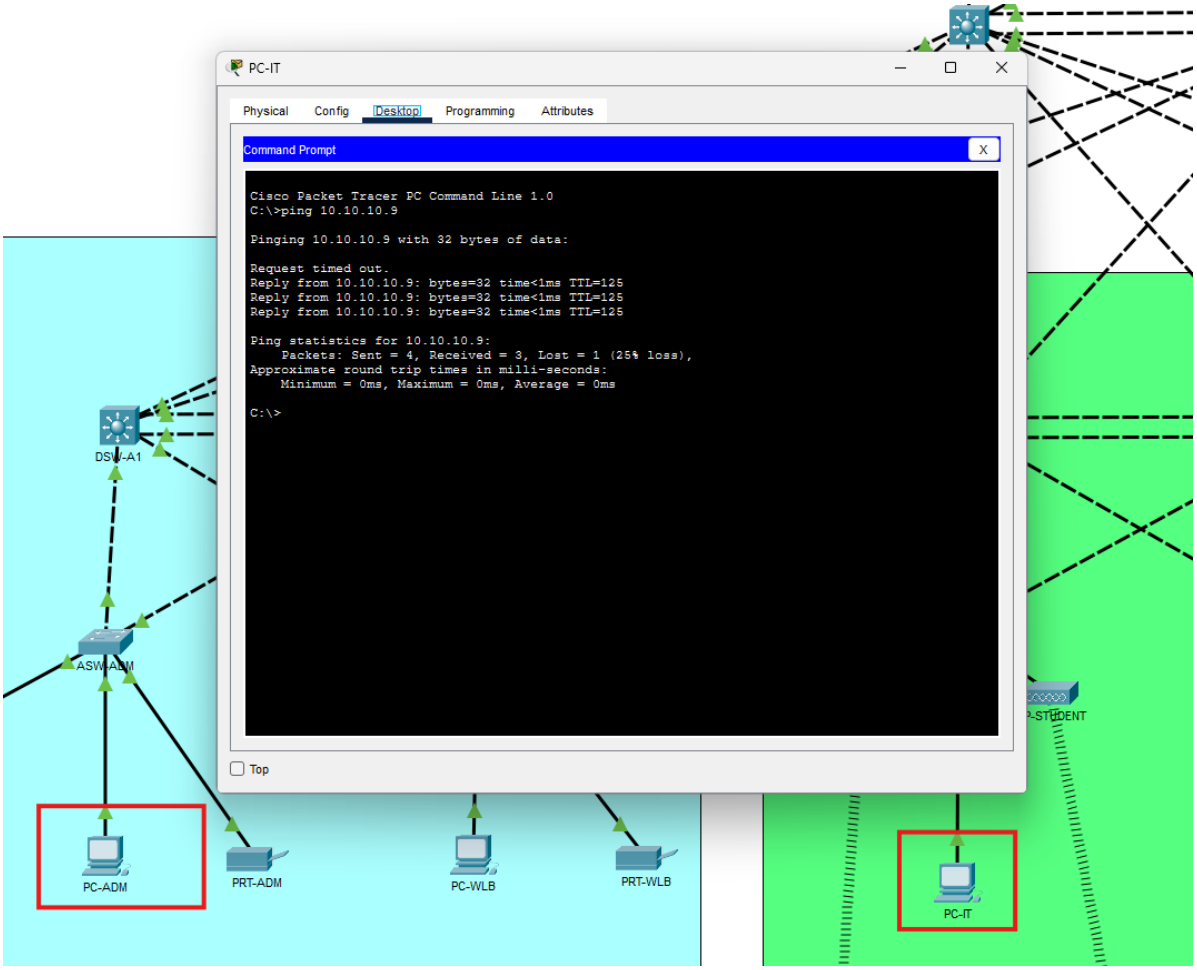


SWITCH	VLAN (NAME)	ROOT	PORT	STATUS
ASW-TEA	VLAN30 (TEACHING)	NO	F0/1	Designated
			G0/1	Root
			G0/2	Alternate
	VLAN40 (RESEARCH)	NO	G0/1	Alternate
			G0/2	Root
	VLAN90 (MGMT)	NO	G0/1	Root
			G0/2	Alternate
ASW-RES	VLAN40 (RESEARCH)	NO	F0/1	Designated
			F0/2	Designated
			F0/3	Designated
			G0/1	Alternate
			G0/2	Root
	VLAN30 (TEACHING)	NO	G0/1	Root

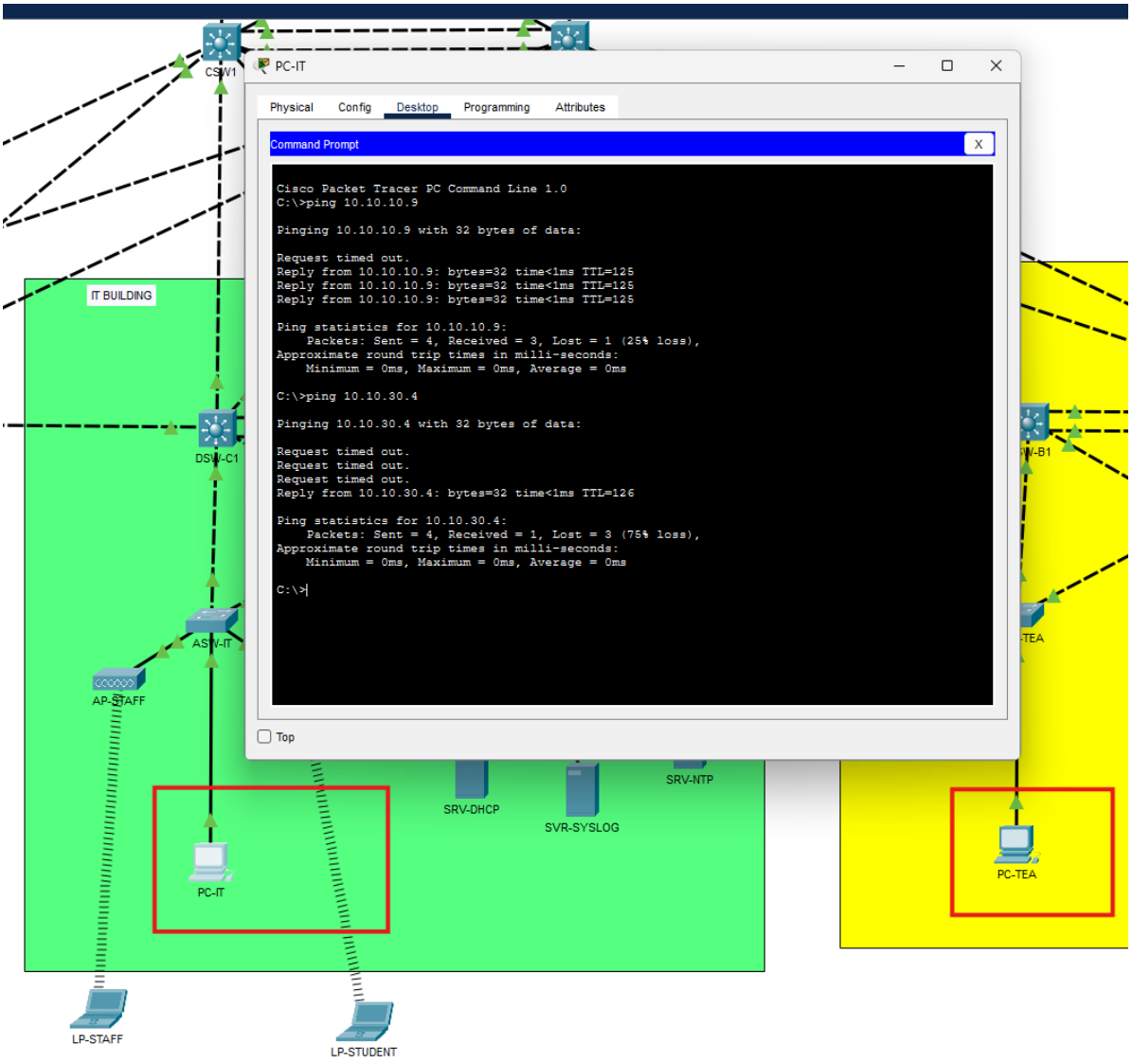
			G0/2	Alternate
	VLAN90 (MGMT)	NO	G0/1	Alternate
			G0/2	Root
DSW-B1	VLAN30 (TEACHING)	YES	G1/0/1	Designated
			G1/0/2	Designated
			Po1	Designated
	VLAN40 (RESEARCH)	NO	G1/0/1	Designated
			G1/0/2	Designated
			Po1	Root
	VLAN90 (MGMT)	YES	G1/0/1	Designated
			G1/0/2	Designated
			Po1	Designated
DSW-B2	VLAN30 (TEACHING)	NO	G1/0/1	Designated
			G1/0/2	Designated
			Po1	Root
	VLAN40 (RESEARCH)	YES	G1/0/1	Designated
			G1/0/2	Designated
			Po1	Designated
	VLAN90 (MGMT)	YES	G1/0/1	Designated
			G1/0/2	Designated
			Po1	Root

10. Connectivity Testing Screenshots & Results

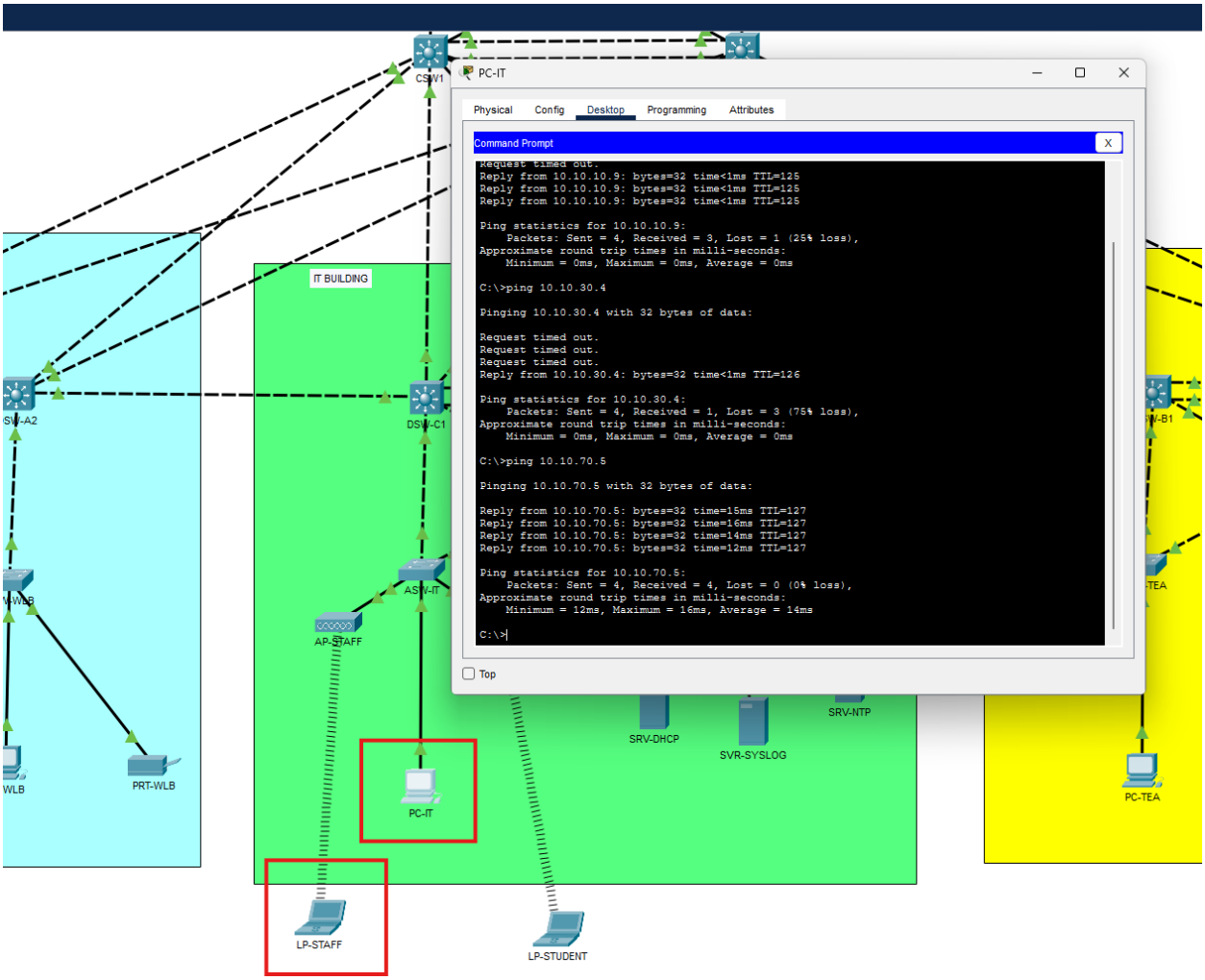
10.1 Pinging from IT Staff PC to Admissions PC


PC-IT
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.10.9
Pinging 10.10.10.9 with 32 bytes of data:
Request timed out.
Reply from 10.10.10.9: bytes=32 time<1ms TTL=125
Reply from 10.10.10.9: bytes=32 time<1ms TTL=125
Reply from 10.10.10.9: bytes=32 time<1ms TTL=125
Ping statistics for 10.10.10.9:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
Top
DSW-A1
ASW-ADM
PC-ADM
PRT-ADM
PC-WLB
PRT-WLB
PC-IT

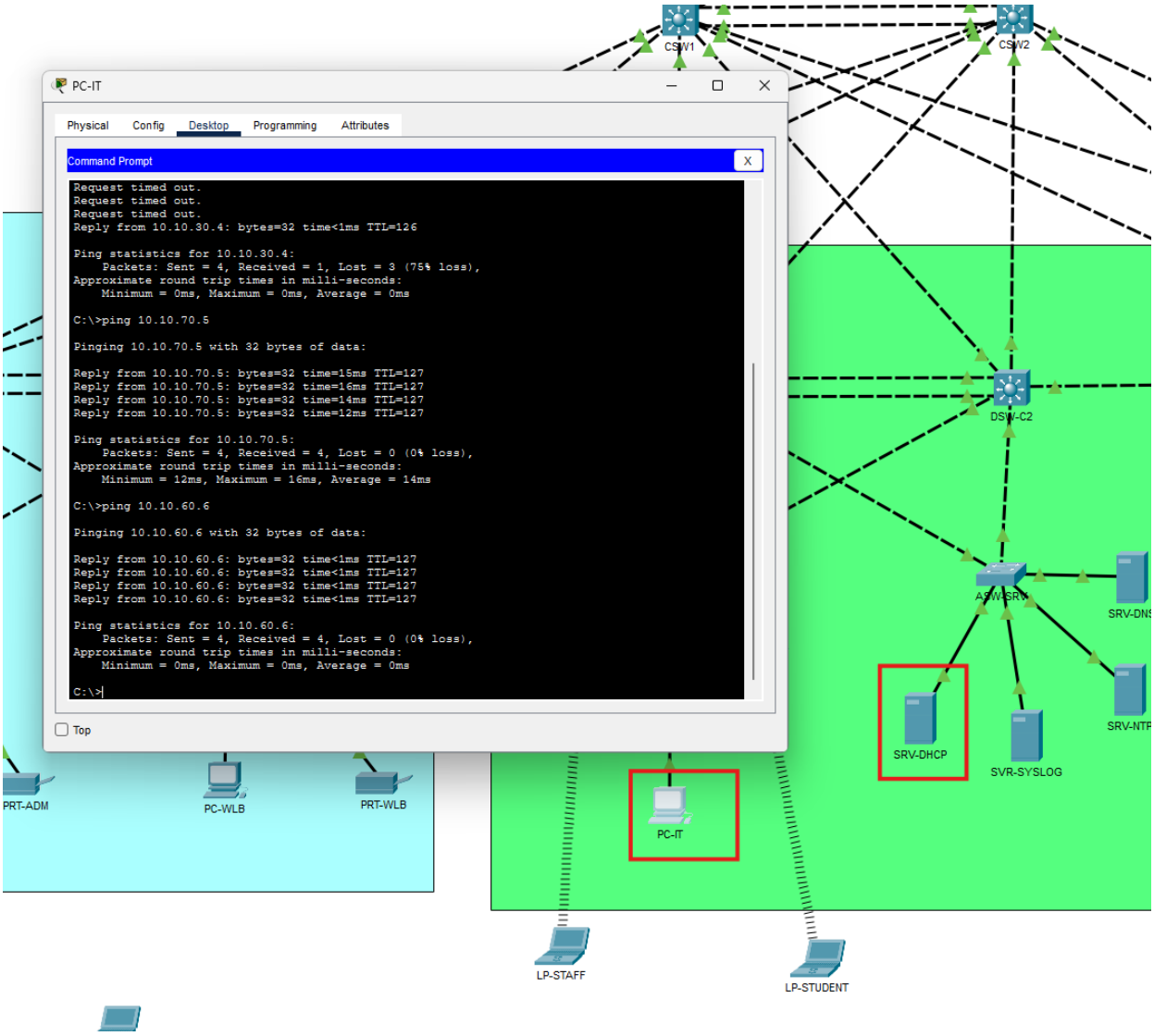
10.2 Pinging from IT Staff PC to a Teaching PC



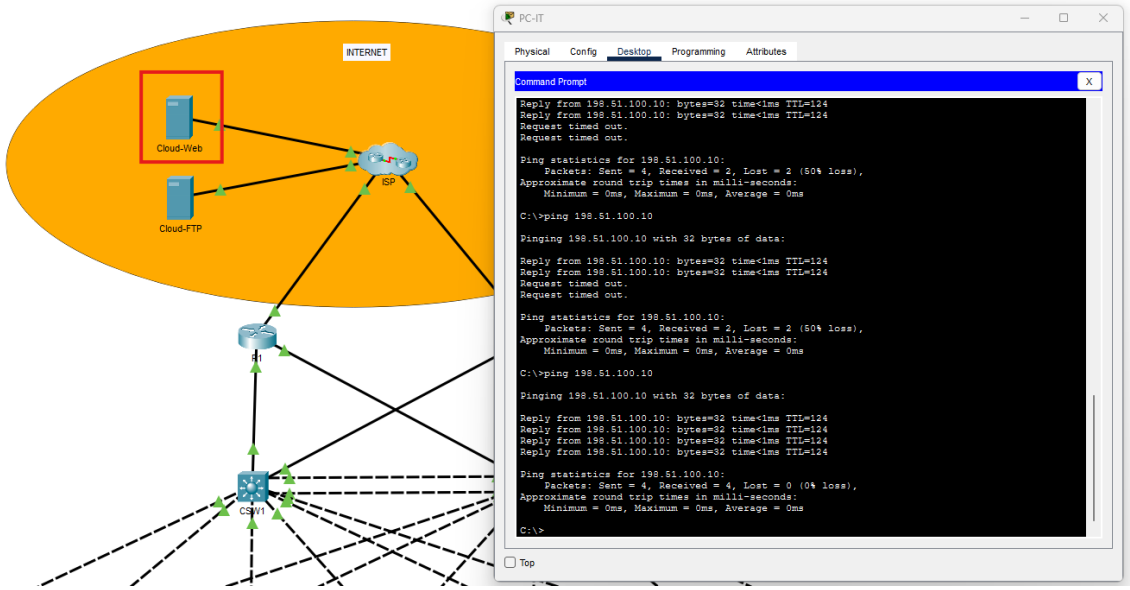
10.3 Pinging from IT Staff PC to Staff Laptop



10.4 Pinging from IT Staff PC to DHCP Server (10.10.60.5)



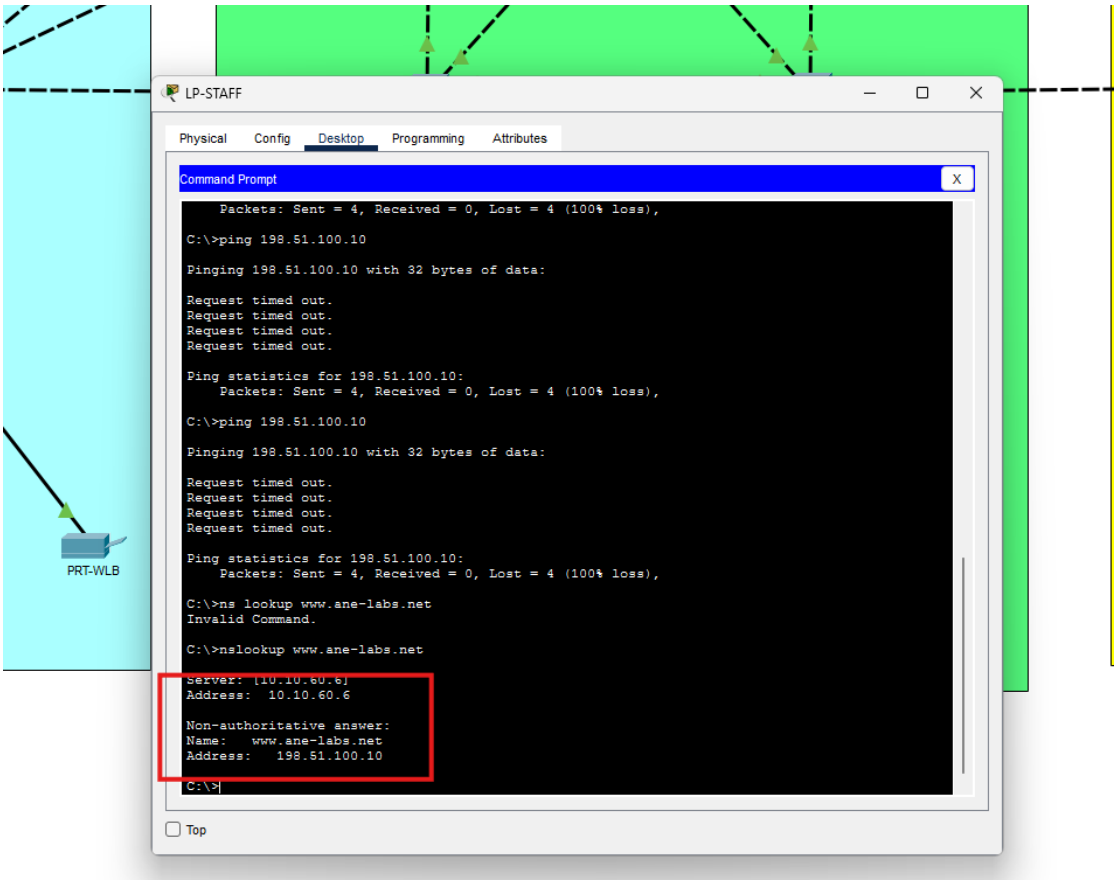
10.5 Pinging from IT-Staff PC to Cloud Web Server (198.51.100.10)



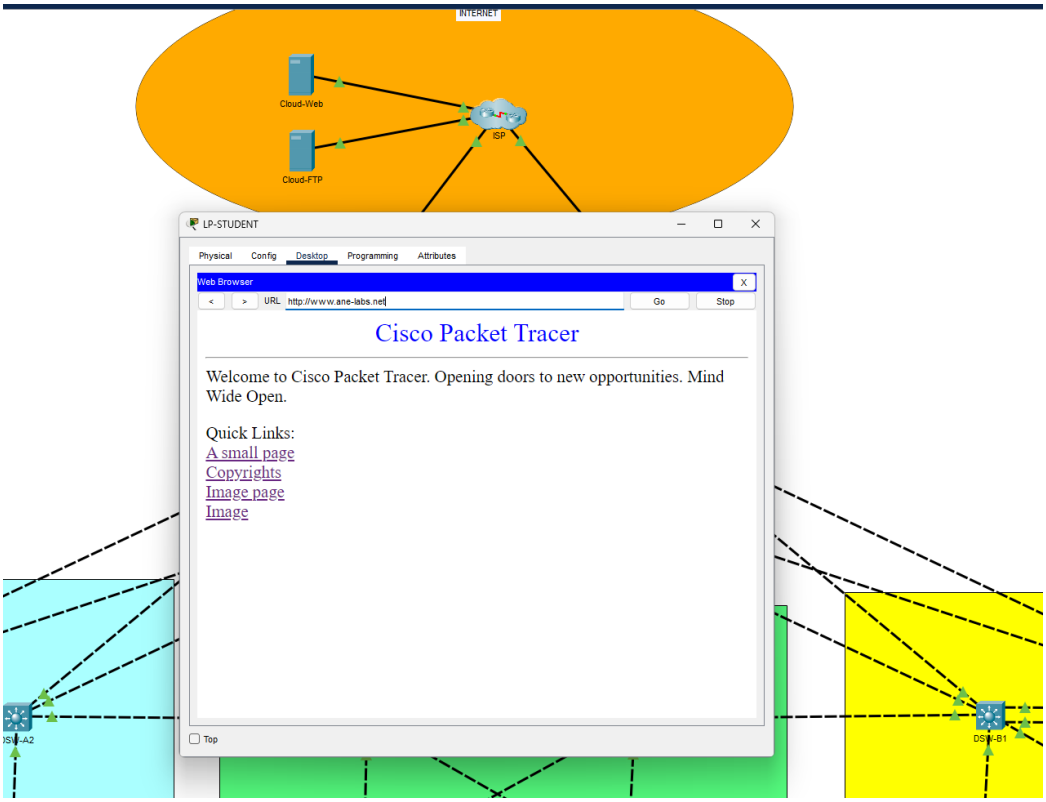
11. DNS Resolution, Webpage and Cloud Access Screenshots

Test Laptop: LP-STAFF (10.10.70.xxx)

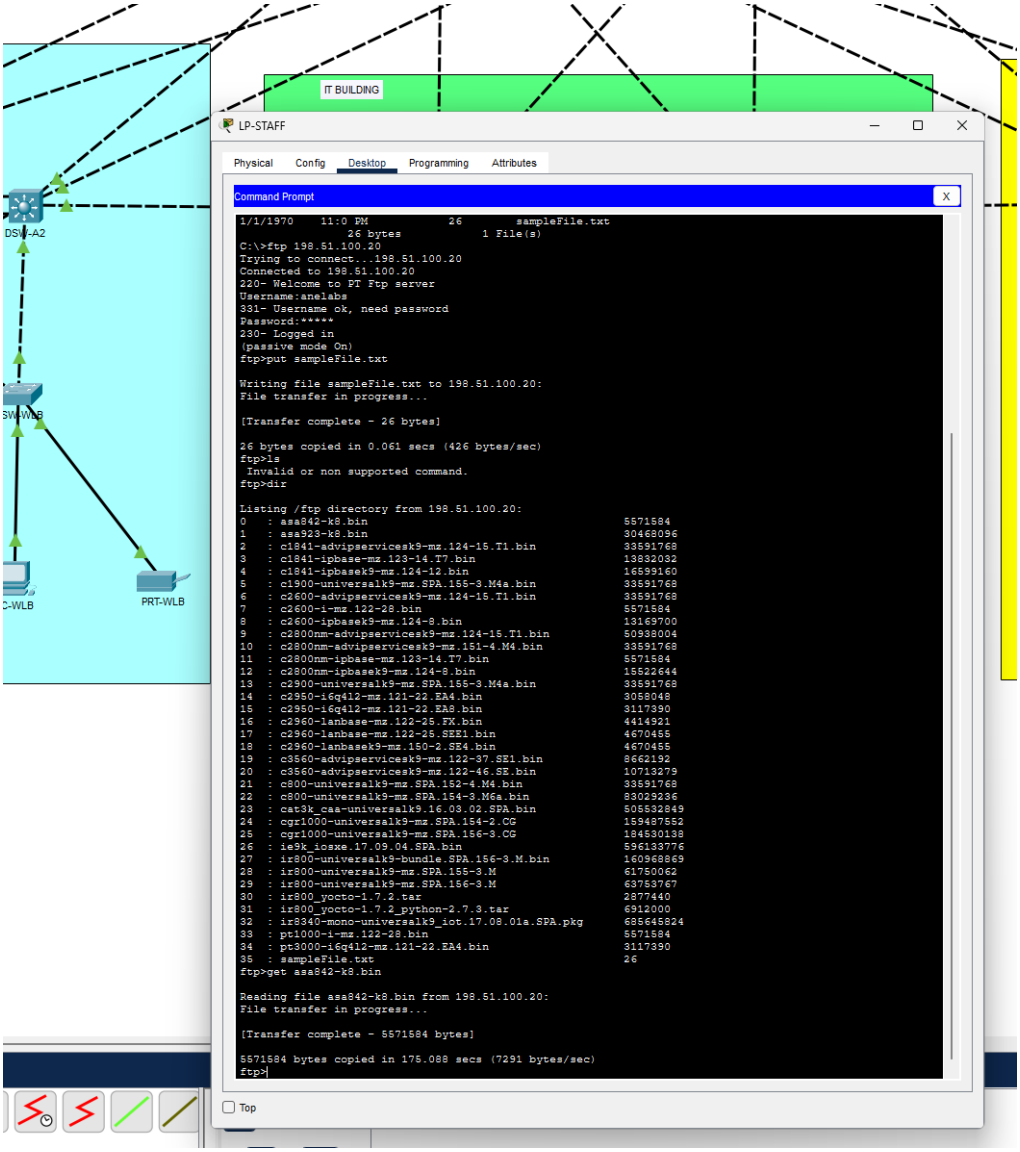
11.1 Internal DNS Name Resolution



11.2 External Web Portal Access



11.3 Cloud FTP File Transfer (Upload and Download)



IOS Configurations

Access Layer

ASW-ADM:

```
ASW-ADM(config)#do sh run
Building configuration...

Current configuration : 4152 bytes
!
version 15.0
```

```
service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname ASW-ADM
!
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
!
!
ip ssh version 2
ip domain-name ane-labs.net
ip name-server 10.10.60.6
!
username admin secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
description **Access port to an Admissions Dept PC**
switchport access vlan 10
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 0001.4366.2805
spanning-tree portfast
spanning-tree bpduguard enable
!
interface FastEthernet0/2
description **Access port to an Admissions Dept Printer**
switchport access vlan 10
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 0002.4A2A.67D2
spanning-tree portfast
spanning-tree bpduguard enable
!
interface FastEthernet0/3
switchport access vlan 99
switchport trunk native vlan 90
switchport trunk allowed vlan 10,90
switchport mode access
switchport port-security
shutdown
!
interface FastEthernet0/4
switchport access vlan 10
switchport trunk native vlan 90
switchport trunk allowed vlan 10,90
switchport mode access
!
interface FastEthernet0/5
```

```
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/6
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/7
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/8
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/9
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/10
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/11
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/12
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/13
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/14
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/15
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/16
switchport access vlan 99
```

```
switchport mode access
shutdown
!
interface FastEthernet0/17
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/18
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/19
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/20
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/21
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/22
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/23
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/24
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet0/1
switchport trunk native vlan 100
switchport trunk allowed vlan 10,20,70,80,90,100
switchport mode trunk
!
interface GigabitEthernet0/2
switchport trunk native vlan 100
switchport trunk allowed vlan 10,20,70,80,90,100
switchport mode trunk
!
interface Vlan1
no ip address
shutdown
```

```
!
interface Vlan90
description **Management IP Address**
ip address 10.10.91.4 255.255.255.192
!
ip default-gateway 10.10.91.1
!
banner motd ^C^C
logging trap debugging
logging 10.10.60.4
!
!
!
ip access-list standard RESTRICT-SSH
remark *** Permit SSH access strictly from IT Staff Subnet ***
permit 10.10.50.0 0.0.0.31
deny any
line con 0
logging synchronous
login local
exec-timeout 30 0
!
line vty 0 4
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
login local
transport input ssh
line vty 5 15
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
login local
transport input ssh
!
!
!
ntp server 5.5.5.5
!
end
```

ASW-WLB:

```
ASW-WLB(config)#do sh run
Building configuration...

Current configuration : 3982 bytes
!
version 15.0
service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname ASW-WLB
```

```
!
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
!
!
ip ssh version 2
ip domain-name ane-labs.net
ip name-server 10.10.60.6
!
username admin secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
description **Access port to a Wellbeing PC**
switchport access vlan 20
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 00D0.973B.0C9E
spanning-tree portfast
spanning-tree bpduguard enable
!
interface FastEthernet0/2
description **Access port to a Wellbeing Printer**
switchport access vlan 20
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 0001.C722.E925
spanning-tree portfast
spanning-tree bpduguard enable
!
interface FastEthernet0/3
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/4
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/5
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/6
switchport access vlan 99
switchport mode access
shutdown
!
```

```
interface FastEthernet0/7
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/8
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/9
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/10
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/11
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/12
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/13
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/14
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/15
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/16
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/17
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/18
```

```
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/19
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/20
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/21
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/22
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/23
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/24
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet0/1
switchport trunk native vlan 100
switchport trunk allowed vlan 10,20,70,80,90,100
switchport mode trunk
!
interface GigabitEthernet0/2
switchport trunk native vlan 100
switchport trunk allowed vlan 10,20,70,80,90,100
switchport mode trunk
!
interface Vlan1
no ip address
shutdown
!
interface Vlan90
description **Management IP Address**
ip address 10.10.91.5 255.255.255.192
!
ip default-gateway 10.10.91.1
!
banner motd ^C^C
logging trap debugging
```

```
logging 10.10.60.4
!
!
!
ip access-list standard RESTRICT-SSH
remark *** Permit SSH access strictly from IT Staff Subnet ***
permit 10.10.50.0 0.0.0.31
deny any
line con 0
logging synchronous
login local
exec-timeout 30 0
!
line vty 0 4
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
login local
transport input ssh
line vty 5 15
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
login local
transport input ssh
!
!
!
ntp server 6.6.6.6
!
end
```

ASW-TEA

```
ASW-TEA(config)#do sh run
Building configuration...

Current configuration : 3666 bytes
!
version 15.0
service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname ASW-TEA
!
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
!
!
ip ssh version 2
ip domain-name ane-labs.net
ip name-server 10.10.60.6
!
username admin secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
```

```
!
!
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
switchport access vlan 30
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 00D0.FF48.4125
spanning-tree portfast
spanning-tree bpduguard enable
!
interface FastEthernet0/2
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/3
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/4
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/5
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/6
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/7
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/8
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/9
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/10
```

```
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/11
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/12
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/13
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/14
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/15
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/16
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/17
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/18
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/19
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/20
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/21
switchport access vlan 99
```

```
switchport mode access
shutdown
!
interface FastEthernet0/22
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/23
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/24
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet0/1
switchport trunk native vlan 100
switchport trunk allowed vlan 30,40,70,80,90,100
switchport mode trunk
!
interface GigabitEthernet0/2
switchport trunk native vlan 100
switchport trunk allowed vlan 30,40,70,80,90,100
switchport mode trunk
!
interface Vlan1
no ip address
shutdown
!
interface Vlan90
ip address 10.10.92.4 255.255.255.192
!
ip default-gateway 10.10.92.1
!
banner motd ^C^C
logging trap debugging
logging 10.10.60.4
!
!
!
ip access-list standard RESTRICT-SSH
remark *** Permit SSH access strictly from IT Staff Subnet ***
permit 10.10.50.0 0.0.0.31
deny any
line con 0
logging synchronous
login local
exec-timeout 30 0
!
line vty 0 4
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
```

```
login local
transport input ssh
line vty 5 15
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
login local
transport input ssh
!
!
!
ntp server 7.7.7.7
!
end
```

ASW-RES

```
ASW-RES(config)#do sh run
Building configuration...

Current configuration : 4214 bytes
!
version 15.0
service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname ASW-RES
!
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
!
!
ip ssh version 2
ip domain-name ane-labs.net
ip name-server 10.10.60.6
!
username admin secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
switchport access vlan 40
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 0010.1159.7D48
spanning-tree portfast
spanning-tree bpduguard enable
!
interface FastEthernet0/2
switchport access vlan 40
```

```
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 0090.212A.32BA
spanning-tree portfast
spanning-tree bpduguard enable
!
interface FastEthernet0/3
switchport access vlan 40
switchport trunk native vlan 90
switchport trunk allowed vlan 40,90
switchport mode access
spanning-tree portfast trunk
spanning-tree bpduguard enable
!
interface FastEthernet0/4
switchport access vlan 90
switchport trunk native vlan 90
switchport trunk allowed vlan 40,90
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 00D0.5881.0701
spanning-tree portfast
spanning-tree bpduguard enable
!
interface FastEthernet0/5
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/6
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/7
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/8
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/9
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/10
switchport access vlan 99
switchport mode access
shutdown
!
```

```
interface FastEthernet0/11
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/12
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/13
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/14
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/15
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/16
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/17
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/18
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/19
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/20
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/21
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/22
```

```
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/23
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/24
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet0/1
switchport trunk native vlan 100
switchport trunk allowed vlan 30,40,70,80,90,100
switchport mode trunk
!
interface GigabitEthernet0/2
switchport trunk native vlan 100
switchport trunk allowed vlan 30,40,70,80,90,100
switchport mode trunk
!
interface Vlan1
no ip address
shutdown
!
interface Vlan90
ip address 10.10.92.5 255.255.255.192
!
ip default-gateway 10.10.92.1
!
banner motd ^C^C
logging trap debugging
logging 10.10.60.4
!
!
!
ip access-list standard RESTRICT-SSH
remark *** Permit SSH access strictly from IT Staff Subnet ***
permit 10.10.50.0 0.0.0.31
deny any
line con 0
logging synchronous
login local
exec-timeout 30 0
!
line vty 0 4
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
login local
transport input ssh
line vty 5 15
access-class RESTRICT-SSH in
```

```
exec-timeout 30 0
logging synchronous
login local
transport input ssh
!
!
!
ntp server 8.8.8.8
!
end
```

ASW-IT

```
ASW-IT(config)#do sh run
Building configuration...

Current configuration : 3975 bytes
!
version 15.0
service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname ASW-IT
!
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
!
!
ip ssh version 2
ip domain-name ane-labs.net
ip name-server 10.10.60.6
!
username admin secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
switchport access vlan 50
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 0001.64C4.9ABD
spanning-tree portfast
spanning-tree bpduguard enable
!
interface FastEthernet0/2
switchport access vlan 70
switchport trunk native vlan 90
switchport trunk allowed vlan 70,90
switchport mode access
```

```
spanning-tree portfast trunk
spanning-tree bpduguard enable
!
interface FastEthernet0/3
switchport access vlan 80
switchport trunk native vlan 90
switchport trunk allowed vlan 80,90
switchport mode access
switchport port-security mac-address sticky
switchport port-security mac-address sticky 0001.C741.1201
spanning-tree bpduguard enable
!
interface FastEthernet0/4
switchport access vlan 90
switchport mode access
!
interface FastEthernet0/5
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/6
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/7
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/8
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/9
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/10
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/11
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/12
switchport access vlan 99
switchport mode access
shutdown
!
```

```
interface FastEthernet0/13
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/14
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/15
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/16
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/17
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/18
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/19
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/20
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/21
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/22
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/23
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/24
```

```
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet0/1
switchport trunk native vlan 100
switchport trunk allowed vlan 50,60,70,80,90,100
switchport mode trunk
!
interface GigabitEthernet0/2
switchport trunk native vlan 100
switchport trunk allowed vlan 50,60,70,80,90,100
switchport mode trunk
!
interface Vlan1
no ip address
shutdown
!
interface Vlan90
ip address 10.10.93.4 255.255.255.192
!
ip default-gateway 10.10.93.1
!
banner motd ^C^C
logging trap debugging
logging 10.10.60.4
!
!
!
ip access-list standard RESTRICT-SSH
remark *** Permit SSH access strictly from IT Staff Subnet ***
permit 10.10.50.0 0.0.0.31
deny any
line con 0
logging synchronous
login local
exec-timeout 30 0
!
line vty 0 4
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
login local
transport input ssh
line vty 5 15
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
login local
transport input ssh
!
!
!
ntp server 9.9.9.9
!
end
```

ASW-SRV

```
ASW-SRV(config)#do sh run
Building configuration...

Current configuration : 4204 bytes
!
version 15.0
service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname ASW-SRV
!
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
!
!
ip ssh version 2
ip domain-name ane-labs.net
ip name-server 10.10.60.6
!
username admin secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
switchport access vlan 60
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 000B.BE67.7A99
spanning-tree portfast
spanning-tree bpduguard enable
!
interface FastEthernet0/2
switchport access vlan 60
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 0001.C931.4495
spanning-tree portfast
spanning-tree bpduguard enable
!
interface FastEthernet0/3
switchport access vlan 60
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 00E0.F792.C509
spanning-tree portfast
```

```
spanning-tree bpduguard enable
!
interface FastEthernet0/4
switchport access vlan 60
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 0060.702D.A22A
spanning-tree portfast
spanning-tree bpduguard enable
!
interface FastEthernet0/5
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/6
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/7
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/8
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/9
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/10
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/11
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/12
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/13
switchport access vlan 99
switchport mode access
shutdown
!
```

```
interface FastEthernet0/14
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/15
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/16
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/17
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/18
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/19
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/20
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/21
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/22
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/23
switchport access vlan 99
switchport mode access
shutdown
!
interface FastEthernet0/24
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet0/1
```

```
switchport trunk native vlan 100
switchport trunk allowed vlan 50,60,70,80,90,100
switchport mode trunk
!
interface GigabitEthernet0/2
switchport trunk native vlan 100
switchport trunk allowed vlan 50,60,70,80,90,100
switchport mode trunk
!
interface Vlan1
no ip address
shutdown
!
interface Vlan90
ip address 10.10.93.5 255.255.255.192
!
ip default-gateway 10.10.93.1
!
banner motd ^C^C
logging trap debugging
logging 10.10.60.4
!
!
!
ip access-list standard RESTRICT-SSH
remark *** Permit SSH access strictly from IT Staff Subnet ***
permit 10.10.50.0 0.0.0.31
deny any
line con 0
logging synchronous
login local
exec-timeout 30 0
!
line vty 0 4
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
login local
transport input ssh
line vty 5 15
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
login local
transport input ssh
!
!
!
ntp server 10.10.10.10
!
end
```

Distribution Layer

DSW-A1

```
DSW-A1(config)#do sh run
Building configuration...

Current configuration : 5286 bytes
!
version 16.3.2
service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname DSW-A1
!
!
no profinet
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
!
!
!
!
!
no ip cef
ip routing
!
no ipv6 cef
!
!
!
username admin privilege 15 secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
!
!
!
!
!
!
!
ip ssh version 2
ip domain-name ane-labs.net
ip name-server 10.10.60.6
!
!
spanning-tree mode rapid-pvst
spanning-tree vlan 10 priority 4096
spanning-tree vlan 20,90 priority 28672
!
!
!
!
!
!
interface Loopback0
ip address 5.5.5.5 255.255.255.255
```

```
ip ospf 1 area 0
!
interface Port-channel1
switchport trunk native vlan 100
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100
switchport mode trunk
switchport nonegotiate
!
interface GigabitEthernet1/0/1
switchport trunk native vlan 100
switchport trunk allowed vlan 10,20,70,80,90,100
switchport mode trunk
!
interface GigabitEthernet1/0/2
switchport trunk native vlan 100
switchport trunk allowed vlan 10,20,70,80,90,100
switchport mode trunk
!
interface GigabitEthernet1/0/3
switchport trunk native vlan 100
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100
switchport mode trunk
switchport nonegotiate
channel-group 1 mode desirable
!
interface GigabitEthernet1/0/4
switchport trunk native vlan 100
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100
switchport mode trunk
switchport nonegotiate
channel-group 1 mode desirable
!
interface GigabitEthernet1/0/5
no switchport
ip address 10.10.200.1 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
!
interface GigabitEthernet1/0/6
no switchport
ip address 10.10.200.5 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
!
interface GigabitEthernet1/0/7
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/8
```

```
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/9
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/10
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/11
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/12
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/13
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/14
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/15
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/16
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/17
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/18
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/19
switchport access vlan 99
```

```
switchport mode access
shutdown
!
interface GigabitEthernet1/0/20
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/21
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/22
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/23
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/24
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/1/1
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/1/2
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/1/3
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/1/4
switchport access vlan 99
switchport mode access
shutdown
!
interface Vlan1
no ip address
shutdown
!
interface Vlan10
mac-address 0040.0b4d.5701
ip address 10.10.10.2 255.255.255.192
ip helper-address 10.10.60.5
```

```
ip ospf 1 area 0
standby 10 ip 10.10.10.1
standby 10 priority 110
standby 10 preempt
!
interface Vlan20
mac-address 0040.0b4d.5702
ip address 10.10.20.2 255.255.255.192
ip helper-address 10.10.60.5
ip ospf 1 area 0
standby 20 ip 10.10.20.1
standby 20 preempt
!
interface Vlan90
mac-address 0040.0b4d.5703
ip address 10.10.91.2 255.255.255.192
ip helper-address 10.10.60.5
ip ospf 1 area 0
standby 90 ip 10.10.91.1
standby 90 preempt
!
router ospf 1
log-adjacency-changes
passive-interface Loopback0
passive-interface Vlan10
passive-interface Vlan20
!
ip default-gateway 10.10.90.1
ip classless
!
ip flow-export version 9
!
!
ip access-list standard RESTRICT-SSH
remark *** Permit SSH access strictly from IT Staff Subnet ***
permit 10.10.50.0 0.0.0.31
deny any
!
banner motd ^C^C
!
!
!
!
logging trap debugging
logging 10.10.60.4
line con 0
exec-timeout 30 0
logging synchronous
login local
!
line aux 0
!
line vty 0 4
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
```

```
login local
transport input ssh
line vty 5 15
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
login local
transport input ssh
!
!
!
ntp server 10.10.60.7
!
end
```

DSW-A2

```
DSW-A2(config)#do sh run
Building configuration...

Current configuration : 5579 bytes
!
version 16.3.2
service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname DSW-A2
!
!
no profinet
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
!
!
!
!
!
no ip cef
ip routing
!
no ipv6 cef
!
!
!
username admin privilege 15 secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
!
!
!
!
```

```
!
!
!
ip ssh version 2
ip domain-name ane-labs.net
ip name-server 10.10.60.6
!
!
spanning-tree mode rapid-pvst
spanning-tree vlan 20,90 priority 4096
spanning-tree vlan 10 priority 24576
!
!
!
!
!
!
interface Loopback0
ip address 6.6.6.6 255.255.255.255
ip ospf 1 area 0
!
interface Port-channel1
switchport trunk native vlan 100
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100
switchport mode trunk
switchport nonegotiate
!
interface GigabitEthernet1/0/1
switchport trunk native vlan 100
switchport trunk allowed vlan 10,20,70,80,90,100
switchport mode trunk
!
interface GigabitEthernet1/0/2
switchport trunk native vlan 100
switchport trunk allowed vlan 10,20,70,80,90,100
switchport mode trunk
!
interface GigabitEthernet1/0/3
switchport trunk native vlan 100
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100
switchport mode trunk
switchport nonegotiate
channel-group 1 mode desirable
!
interface GigabitEthernet1/0/4
switchport trunk native vlan 100
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100
switchport mode trunk
switchport nonegotiate
channel-group 1 mode desirable
!
interface GigabitEthernet1/0/5
no switchport
ip address 10.10.200.9 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
```

```
ip ospf 1 area 0
duplex auto
speed auto
!
interface GigabitEthernet1/0/6
no switchport
ip address 10.10.200.13 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
!
interface GigabitEthernet1/0/7
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/8
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/9
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/10
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/11
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/12
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/13
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/14
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/15
switchport access vlan 99
switchport mode access
```

```
shutdown
!
interface GigabitEthernet1/0/16
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/17
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/18
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/19
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/20
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/21
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/22
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/23
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/24
no switchport
ip address 10.10.200.49 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
spanning-tree bpdufilter enable
!
interface GigabitEthernet1/1/1
switchport access vlan 99
switchport mode access
shutdown
```

```
!
interface GigabitEthernet1/1/2
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/1/3
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/1/4
switchport access vlan 99
switchport mode access
shutdown
!
interface Vlan1
no ip address
shutdown
!
interface Vlan10
mac-address 0090.2b7b.7a01
ip address 10.10.10.3 255.255.255.192
ip helper-address 10.10.60.5
ip ospf 1 area 0
standby 10 ip 10.10.10.1
standby 10 preempt
!
interface Vlan20
mac-address 0090.2b7b.7a02
ip address 10.10.20.3 255.255.255.192
ip helper-address 10.10.60.5
ip ospf 1 area 0
standby 20 ip 10.10.20.1
standby 20 priority 110
standby 20 preempt
!
interface Vlan70
mac-address 0090.2b7b.7a03
no ip address
shutdown
!
interface Vlan80
mac-address 0090.2b7b.7a04
no ip address
shutdown
!
interface Vlan90
mac-address 0090.2b7b.7a05
ip address 10.10.91.3 255.255.255.192
ip helper-address 10.10.60.5
ip ospf 1 area 0
standby 90 ip 10.10.91.1
standby 90 priority 110
standby 90 preempt
!
```

```
router ospf 1
log-adjacency-changes
passive-interface Loopback0
passive-interface Vlan10
passive-interface Vlan20
!
ip default-gateway 10.10.90.1
ip classless
!
ip flow-export version 9
!
!
ip access-list standard RESTRICT-SSH
remark *** Permit SSH access strictly from IT Staff Subnet ***
permit 10.10.50.0 0.0.0.31
deny any
!
banner motd ^C^C
!
!
!
!
logging trap debugging
logging 10.10.60.4
line con 0
exec-timeout 30 0
logging synchronous
login local
!
line aux 0
!
line vty 0 4
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
login local
transport input ssh
line vty 5 15
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
login local
transport input ssh
!
!
!
ntp server 10.10.60.7
!
end
```

DSW-B1

```
DSW-B1(config)#do sh run
Building configuration...
```

```
Current configuration : 5545 bytes
!
version 16.3.2
service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname DSW-B1
!
!
no profinet
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
!
!
!
!
!
no ip cef
ip routing
!
no ipv6 cef
!
!
!
username admin privilege 15 secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
!
!
!
!
!
!
!
ip ssh version 2
ip domain-name ane-labs.net
ip name-server 10.10.60.6
!
!
spanning-tree mode rapid-pvst
spanning-tree vlan 30,90 priority 4096
spanning-tree vlan 40 priority 28672
!
!
!
!
!
!
interface Loopback0
ip address 7.7.7.7 255.255.255.255
ip ospf 1 area 0
!
```

```
interface Port-channel1
switchport trunk native vlan 100
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100
switchport mode trunk
!
interface GigabitEthernet1/0/1
switchport trunk native vlan 100
switchport trunk allowed vlan 30,40,70,80,90,100
switchport mode trunk
!
interface GigabitEthernet1/0/2
switchport trunk native vlan 100
switchport trunk allowed vlan 30,40,70,80,90,100
!
interface GigabitEthernet1/0/3
switchport trunk native vlan 100
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100
switchport mode trunk
channel-group 1 mode desirable
!
interface GigabitEthernet1/0/4
switchport trunk native vlan 100
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100
switchport mode trunk
channel-group 1 mode desirable
spanning-tree bpdufilter enable
!
interface GigabitEthernet1/0/5
no switchport
ip address 10.10.200.33 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
!
interface GigabitEthernet1/0/6
no switchport
ip address 10.10.200.37 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
!
interface GigabitEthernet1/0/7
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/8
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/9
```

```
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/10
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/11
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/12
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/13
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/14
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/15
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/16
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/17
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/18
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/19
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/20
switchport access vlan 99
```

```
switchport mode access
shutdown
!
interface GigabitEthernet1/0/21
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/22
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/23
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/24
no switchport
ip address 10.10.200.54 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
spanning-tree bpdufilter enable
!
interface GigabitEthernet1/1/1
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/1/2
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/1/3
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/1/4
switchport access vlan 99
switchport mode access
shutdown
!
interface Vlan1
no ip address
shutdown
!
interface Vlan30
mac-address 0001.64b9.5301
ip address 10.10.30.2 255.255.255.192
ip helper-address 10.10.60.5
```

```
ip helper-address 10.10.6.5
ip ospf 1 area 0
standby 30 ip 10.10.30.1
standby 30 priority 110
standby 30 preempt
!
interface Vlan40
mac-address 0001.64b9.5302
ip address 10.10.40.2 255.255.255.224
ip helper-address 10.10.60.5
ip helper-address 10.10.6.5
ip ospf 1 area 0
standby 40 ip 10.10.40.1
standby 40 preempt
!
interface Vlan70
mac-address 0001.64b9.5303
no ip address
shutdown
!
interface Vlan80
mac-address 0001.64b9.5304
no ip address
shutdown
!
interface Vlan90
mac-address 0001.64b9.5305
ip address 10.10.92.2 255.255.255.192
ip helper-address 10.10.6.5
ip ospf 1 area 0
standby 90 ip 10.10.92.1
standby 90 priority 110
standby 90 preempt
!
router ospf 1
log-adjacency-changes
passive-interface Loopback0
passive-interface Vlan30
passive-interface Vlan40
!
ip classless
!
ip flow-export version 9
!
!
ip access-list standard RESTRICT-SSH
remark *** Permit SSH access strictly from IT Staff Subnet ***
permit 10.10.50.0 0.0.0.31
deny any
!
banner motd ^C^C
!
!
!
!
logging trap debugging
```

```
logging 10.10.60.4
line con 0
exec-timeout 30 0
logging synchronous
login local
!
line aux 0
!
line vty 0 4
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
login local
transport input ssh
line vty 5 15
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
login local
transport input ssh
!
!
!
ntp server 10.10.60.7
!
end
```

DSW-B2

```
DSW-B2(config)#do sh run
Building configuration...

Current configuration : 5283 bytes
!
version 16.3.2
service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname DSW-B2
!
!
no profinet
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
!
!
!
!
!
no ip cef
ip routing
!
no ipv6 cef
```

```
!
!
!
username admin privilege 15 secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
!
!
!
!
!
!
!
ip ssh version 2
ip domain-name ane-labs.net
ip name-server 10.10.60.6
!
!
spanning-tree mode rapid-pvst
spanning-tree vlan 40 priority 4096
spanning-tree vlan 30,90 priority 28672
!
!
!
!
!
!
interface Loopback0
ip address 8.8.8.8 255.255.255.255
ip ospf 1 area 0
!
interface Port-channel1
switchport trunk native vlan 100
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100
switchport mode trunk
!
interface GigabitEthernet1/0/1
switchport trunk native vlan 100
switchport trunk allowed vlan 30,40,70,80,90,100
!
interface GigabitEthernet1/0/2
switchport trunk native vlan 100
switchport trunk allowed vlan 30,40,70,80,90,100
!
interface GigabitEthernet1/0/3
switchport trunk native vlan 100
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100
switchport mode trunk
channel-group 1 mode desirable
!
interface GigabitEthernet1/0/4
switchport trunk native vlan 100
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100
switchport mode trunk
channel-group 1 mode desirable
```

```
!
interface GigabitEthernet1/0/5
no switchport
ip address 10.10.200.41 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
!
interface GigabitEthernet1/0/6
no switchport
ip address 10.10.200.45 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
!
interface GigabitEthernet1/0/7
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/8
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/9
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/10
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/11
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/12
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/13
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/14
switchport access vlan 99
```

```
switchport mode access
shutdown
!
interface GigabitEthernet1/0/15
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/16
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/17
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/18
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/19
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/20
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/21
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/22
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/23
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/24
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/1/1
switchport access vlan 99
switchport mode access
```

```
shutdown
!
interface GigabitEthernet1/1/2
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/1/3
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/1/4
switchport access vlan 99
switchport mode access
shutdown
!
interface Vlan1
no ip address
shutdown
!
interface Vlan30
mac-address 00e0.b017.5401
ip address 10.10.30.3 255.255.255.192
ip helper-address 10.10.60.5
ip ospf 1 area 0
standby 30 ip 10.10.30.1
standby 30 preempt
!
interface Vlan40
mac-address 00e0.b017.5402
ip address 10.10.40.3 255.255.255.224
ip helper-address 10.10.60.5
ip ospf 1 area 0
standby 40 ip 10.10.40.1
standby 40 priority 110
standby 40 preempt
!
interface Vlan70
mac-address 00e0.b017.5403
no ip address
shutdown
!
interface Vlan80
mac-address 00e0.b017.5404
no ip address
shutdown
!
interface Vlan90
mac-address 00e0.b017.5405
ip address 10.10.92.3 255.255.255.192
ip helper-address 10.10.6.5
ip ospf 1 area 0
standby 90 ip 10.10.92.1
standby 90 preempt
!
```

```
router ospf 1
log-adjacency-changes
passive-interface Loopback0
passive-interface Vlan30
passive-interface Vlan40
!
ip classless
!
ip flow-export version 9
!
!
ip access-list standard RESTRICT-SSH
remark *** Permit SSH access strictly from IT Staff Subnet ***
permit 10.10.50.0 0.0.0.31
deny any
!
banner motd ^C^C
!
!
!
!
logging trap debugging
logging 10.10.60.4
line con 0
exec-timeout 30 0
logging synchronous
login local
!
line aux 0
!
line vty 0 4
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
login local
transport input ssh
line vty 5 15
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
login local
transport input ssh
!
!
!
ntp server 10.10.60.7
!
end
```

DSW-C1

```
DSW-C1(config)#do sh run
Building configuration...
```

```
Current configuration : 5859 bytes
!
version 16.3.2
service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname DSW-C1
!
!
no profinet
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
!
!
!
!
!
no ip cef
ip routing
!
no ipv6 cef
!
!
!
username admin privilege 15 secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
!
!
!
!
!
!
!
ip ssh version 2
ip domain-name ane-labs.net
ip name-server 10.10.60.6
!
!
spanning-tree mode rapid-pvst
spanning-tree vlan 50,70,90 priority 4096
spanning-tree vlan 60 priority 28672
!
!
!
!
!
!
interface Loopback0
ip address 9.9.9.9 255.255.255.255
ip ospf 1 area 0
!
interface Port-channel1
switchport trunk native vlan 100
```

```
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100
switchport mode trunk
!
interface GigabitEthernet1/0/1
switchport trunk native vlan 100
switchport trunk allowed vlan 50,60,70,80,90,100
switchport mode trunk
!
interface GigabitEthernet1/0/2
switchport trunk native vlan 100
switchport trunk allowed vlan 50,60,70,80,90,100
switchport mode trunk
!
interface GigabitEthernet1/0/3
switchport trunk native vlan 100
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100
switchport mode trunk
channel-group 1 mode desirable
!
interface GigabitEthernet1/0/4
switchport trunk native vlan 100
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100
switchport mode trunk
channel-group 1 mode desirable
!
interface GigabitEthernet1/0/5
no switchport
ip address 10.10.200.17 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
!
interface GigabitEthernet1/0/6
no switchport
ip address 10.10.200.21 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
!
interface GigabitEthernet1/0/7
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/8
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/9
switchport access vlan 99
switchport mode access
```

```
shutdown
!
interface GigabitEthernet1/0/10
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/11
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/12
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/13
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/14
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/15
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/16
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/17
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/18
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/19
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/20
switchport access vlan 99
switchport mode access
shutdown
```

```
!
interface GigabitEthernet1/0/21
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/22
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/23
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/24
no switchport
ip address 10.10.200.50 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
spanning-tree bpdufilter enable
!
interface GigabitEthernet1/1/1
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/1/2
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/1/3
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/1/4
switchport access vlan 99
switchport mode access
shutdown
!
interface Vlan1
no ip address
shutdown
!
interface Vlan50
mac-address 00d0.5853.1801
ip address 10.10.50.2 255.255.255.224
ip helper-address 10.10.60.5
ip ospf 1 area 0
standby 50 ip 10.10.50.1
```

```
standby 50 priority 110
standby 50 preempt
!
interface Vlan60
mac-address 00d0.5853.1802
ip address 10.10.60.2 255.255.255.240
ip ospf 1 area 0
standby 60 ip 10.10.60.1
standby 60 preempt
!
interface Vlan70
mac-address 00d0.5853.1803
ip address 10.10.70.2 255.255.255.128
ip helper-address 10.10.6.5
ip helper-address 10.10.60.5
ip ospf 1 area 0
standby 70 ip 10.10.70.1
standby 70 priority 110
standby 70 preempt
standby 0 preempt
!
interface Vlan80
mac-address 00d0.5853.1804
ip address 10.10.80.2 255.255.255.128
ip helper-address 10.10.6.5
ip helper-address 10.10.60.5
ip ospf 1 area 0
standby 80 ip 10.10.80.1
standby 80 preempt
!
interface Vlan90
mac-address 00d0.5853.1805
ip address 10.10.93.2 255.255.255.192
ip helper-address 10.10.6.5
ip ospf 1 area 0
standby 90 ip 10.10.93.1
standby 90 priority 110
standby 90 preempt
!
router ospf 1
log-adjacency-changes
passive-interface Loopback0
passive-interface Vlan50
passive-interface Vlan60
passive-interface Vlan70
passive-interface Vlan80
passive-interface Vlan90
!
router rip
!
ip classless
!
ip flow-export version 9
!
!
ip access-list standard RESTRICT-SSH
```

```
remark *** Permit SSH access strictly from IT Staff Subnet ***
permit 10.10.50.0 0.0.0.31
deny any
!
banner motd ^C^C
!
!
!
!
logging trap debugging
logging 10.10.60.4
line con 0
exec-timeout 30 0
logging synchronous
login local
!
line aux 0
!
line vty 0 4
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
login local
transport input ssh
line vty 5 15
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
login local
transport input ssh
!
!
!
ntp server 10.10.60.7
!
end
```

DSW-C2

```
DSW-C2(config)#do sh run
Building configuration...

Current configuration : 5760 bytes
!
version 16.3.2
service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname DSW-C2
!
!
```

```
no profinet
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
!
!
!
!
!
no ip cef
ip routing
!
no ipv6 cef
!
!
!
username admin privilege 15 secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
!
!
!
!
!
!
!
ip ssh version 2
ip domain-name ane-labs.net
ip name-server 10.10.60.6
!
!
spanning-tree mode rapid-pvst
spanning-tree vlan 60,80 priority 4096
spanning-tree vlan 50,90 priority 28672
!
!
!
!
!
!
interface Loopback0
ip address 10.10.10.10 255.255.255.255
ip ospf 1 area 0
!
interface Port-channel1
switchport trunk native vlan 100
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100
switchport mode trunk
!
interface GigabitEthernet1/0/1
switchport trunk native vlan 100
switchport trunk allowed vlan 50,60,70,80,90,100
switchport mode trunk
!
interface GigabitEthernet1/0/2
switchport trunk native vlan 100
```

```
switchport trunk allowed vlan 50,60,70,80,90,100
switchport mode trunk
!
interface GigabitEthernet1/0/3
switchport trunk native vlan 100
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100
switchport mode trunk
channel-group 1 mode desirable
!
interface GigabitEthernet1/0/4
switchport trunk native vlan 100
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100
switchport mode trunk
channel-group 1 mode desirable
!
interface GigabitEthernet1/0/5
no switchport
ip address 10.10.200.25 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
!
interface GigabitEthernet1/0/6
no switchport
ip address 10.10.200.29 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
!
interface GigabitEthernet1/0/7
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/8
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/9
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/10
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/11
switchport access vlan 99
switchport mode access
```

```
shutdown
!
interface GigabitEthernet1/0/12
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/13
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/14
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/15
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/16
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/17
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/18
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/19
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/20
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/21
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/22
switchport access vlan 99
switchport mode access
shutdown
```

```
!
interface GigabitEthernet1/0/23
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/0/24
no switchport
ip address 10.10.200.53 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
spanning-tree bpdufilter enable
!
interface GigabitEthernet1/1/1
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/1/2
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/1/3
switchport access vlan 99
switchport mode access
shutdown
!
interface GigabitEthernet1/1/4
switchport access vlan 99
switchport mode access
shutdown
!
interface Vlan1
no ip address
shutdown
!
interface Vlan50
mac-address 0050.0fa9.2001
ip address 10.10.50.3 255.255.255.224
ip helper-address 10.10.60.5
ip ospf 1 area 0
standby 50 ip 10.10.50.1
standby 50 preempt
!
interface Vlan60
mac-address 0050.0fa9.2002
ip address 10.10.60.3 255.255.255.240
ip ospf 1 area 0
standby 60 ip 10.10.60.1
standby 60 priority 110
standby 60 preempt
!
```

```
interface Vlan70
mac-address 0050.0fa9.2003
ip address 10.10.70.3 255.255.255.128
ip helper-address 10.10.60.5
ip ospf 1 area 0
standby 70 ip 10.10.70.1
standby 70 preempt
standby 0 preempt
!
interface Vlan80
mac-address 0050.0fa9.2004
ip address 10.10.80.3 255.255.255.128
ip helper-address 10.10.60.5
ip ospf 1 area 0
standby 80 ip 10.10.80.1
standby 80 priority 110
standby 80 preempt
standby 0 preempt
!
interface Vlan90
mac-address 0050.0fa9.2005
ip address 10.10.93.3 255.255.255.192
ip helper-address 10.10.6.5
ip ospf 1 area 0
standby 90 ip 10.10.93.1
standby 90 preempt
!
router ospf 1
log-adjacency-changes
passive-interface Loopback0
passive-interface Vlan50
passive-interface Vlan60
passive-interface Vlan70
passive-interface Vlan80
!
ip classless
!
ip flow-export version 9
!
!
ip access-list standard RESTRICT-SSH
remark *** Permit SSH access strictly from IT Staff Subnet ***
permit 10.10.50.0 0.0.0.31
deny any
!
banner motd ^C^C
!
!
!
!
logging trap debugging
logging 10.10.60.4
line con 0
exec-timeout 30 0
logging synchronous
login local
```

```
!
line aux 0
!
line vty 0 4
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
login local
transport input ssh
line vty 5 15
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
login local
transport input ssh
!
!
!
ntp server 10.10.60.7
!
end
```

Core Layer

CSW-1

```
CSW1(config)#do sh run
Building configuration...

Current configuration : 3575 bytes
!
version 16.3.2
service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname CSW1
!
!
no profinet
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
!
!
clock timezone AEST 10
!
!
!
!
no ip cef
ip routing
!
```

```
no ipv6 cef
!
!
!
username admin privilege 15 secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
!
!
!
!
!
!
!
ip ssh version 2
ip domain-name ane-labs.net
ip name-server 10.10.60.6
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface Loopback0
ip address 3.3.3.3 255.255.255.255
!
interface Port-channel1
no switchport
ip address 10.10.200.57 255.255.255.252
ip ospf 1 area 0
!
interface GigabitEthernet1/0/1
no switchport
ip address 10.10.200.62 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
!
interface GigabitEthernet1/0/2
no switchport
ip address 10.10.200.70 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
!
interface GigabitEthernet1/0/3
!
interface GigabitEthernet1/0/4
```

```
!
interface GigabitEthernet1/0/5
no switchport
no ip address
duplex auto
speed auto
!
interface GigabitEthernet1/0/6
no switchport
no ip address
duplex auto
speed auto
!
interface GigabitEthernet1/0/7
no switchport
ip address 10.10.200.2 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
!
interface GigabitEthernet1/0/8
no switchport
ip address 10.10.200.10 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
!
interface GigabitEthernet1/0/9
no switchport
ip address 10.10.200.18 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
!
interface GigabitEthernet1/0/10
no switchport
ip address 10.10.200.26 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
!
interface GigabitEthernet1/0/11
no switchport
ip address 10.10.200.34 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
```

```
speed auto
!
interface GigabitEthernet1/0/12
no switchport
ip address 10.10.200.42 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
!
interface GigabitEthernet1/0/13
!
interface GigabitEthernet1/0/14
!
interface GigabitEthernet1/0/15
!
interface GigabitEthernet1/0/16
!
interface GigabitEthernet1/0/17
!
interface GigabitEthernet1/0/18
!
interface GigabitEthernet1/0/19
!
interface GigabitEthernet1/0/20
!
interface GigabitEthernet1/0/21
!
interface GigabitEthernet1/0/22
!
interface GigabitEthernet1/0/23
!
interface GigabitEthernet1/0/24
!
interface GigabitEthernet1/1/1
!
interface GigabitEthernet1/1/2
!
interface GigabitEthernet1/1/3
!
interface GigabitEthernet1/1/4
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
log-adjacency-changes
!
ip classless
!
ip flow-export version 9
!
!
ip access-list standard RESTRICT-SSH
```

```
remark *** Permit SSH access strictly from IT Staff Subnet ***
permit 10.10.50.0 0.0.0.31
deny any
!
banner motd ^C^C
!
!
!
!
logging trap debugging
logging 10.10.60.4
line con 0
exec-timeout 30 0
logging synchronous
login local
!
line aux 0
!
line vty 0 4
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
login local
transport input ssh
line vty 5 15
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
login local
transport input ssh
!
!
!
ntp server 10.10.60.7
!
end
```

CSW-2

```
CSW2(config)#do sh run
Building configuration...

Current configuration : 3575 bytes
!
version 16.3.2
service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname CSW2
!
!
no profinet
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
```

```
!
!
!
clock timezone AEST 10
!
!
!
!
no ip cef
ip routing
!
no ipv6 cef
!
!
!
username admin privilege 15 secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
!
!
!
!
!
!
!
ip ssh version 2
ip domain-name ane-labs.net
ip name-server 10.10.60.6
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface Loopback0
ip address 4.4.4.4 255.255.255.255
!
interface Port-channel1
no switchport
ip address 10.10.200.58 255.255.255.252
ip ospf 1 area 0
!
interface GigabitEthernet1/0/1
no switchport
ip address 10.10.200.66 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
!
interface GigabitEthernet1/0/2
```

```
no switchport
ip address 10.10.200.74 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
!
interface GigabitEthernet1/0/3
!
interface GigabitEthernet1/0/4
!
interface GigabitEthernet1/0/5
no switchport
no ip address
duplex auto
speed auto
!
interface GigabitEthernet1/0/6
no switchport
no ip address
duplex auto
speed auto
!
interface GigabitEthernet1/0/7
no switchport
ip address 10.10.200.6 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
!
interface GigabitEthernet1/0/8
no switchport
ip address 10.10.200.14 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
!
interface GigabitEthernet1/0/9
no switchport
ip address 10.10.200.22 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
!
interface GigabitEthernet1/0/10
no switchport
ip address 10.10.200.30 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
```

```
ip ospf 1 area 0
duplex auto
speed auto
!
interface GigabitEthernet1/0/11
no switchport
ip address 10.10.200.38 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
!
interface GigabitEthernet1/0/12
no switchport
ip address 10.10.200.46 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
!
interface GigabitEthernet1/0/13
!
interface GigabitEthernet1/0/14
!
interface GigabitEthernet1/0/15
!
interface GigabitEthernet1/0/16
!
interface GigabitEthernet1/0/17
!
interface GigabitEthernet1/0/18
!
interface GigabitEthernet1/0/19
!
interface GigabitEthernet1/0/20
!
interface GigabitEthernet1/0/21
!
interface GigabitEthernet1/0/22
!
interface GigabitEthernet1/0/23
!
interface GigabitEthernet1/0/24
!
interface GigabitEthernet1/1/1
!
interface GigabitEthernet1/1/2
!
interface GigabitEthernet1/1/3
!
interface GigabitEthernet1/1/4
!
interface Vlan1
no ip address
```

```
shutdown
!
router ospf 1
log-adjacency-changes
!
ip classless
!
ip flow-export version 9
!
!
ip access-list standard RESTRICT-SSH
remark *** Permit SSH access strictly from IT Staff Subnet ***
permit 10.10.50.0 0.0.0.31
deny any
!
banner motd ^C^C
!
!
!
!
logging trap debugging
logging 10.10.60.4
line con 0
exec-timeout 30 0
logging synchronous
login local
!
line aux 0
!
line vty 0 4
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
login local
transport input ssh
line vty 5 15
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
login local
transport input ssh
!
!
!
ntp server 10.10.60.7
!
end
```

R1

```
R1(config)#do sh run
Building configuration...
```

```
Current configuration : 2370 bytes
!
version 15.1
service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname R1
!
!
!
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
username admin privilege 15 secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
license udi pid CISCO2911/K9 sn FTX1524VU32-
!
!
!
!
!
!
!
!
!
!
!
ip ssh version 2
ip domain-name ane-labs.net
ip name-server 10.10.60.6
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface Loopback0
ip address 1.1.1.1 255.255.255.255
ip ospf 1 area 0
!
```

```
interface GigabitEthernet0/0
ip address 203.0.113.2 255.255.255.248
ip nat outside
duplex auto
speed auto
!
interface GigabitEthernet0/1
ip address 10.10.200.61 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
!
interface GigabitEthernet0/2
ip address 10.10.200.65 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
log-adjacency-changes
default-information originate
!
ip nat inside source list 1 interface GigabitEthernet0/0 overload
ip classless
ip route 0.0.0.0 0.0.0.0 203.0.113.1
!
ip flow-export version 9
!
!
access-list 1 permit 10.10.0.0 0.0.255.255
ip access-list standard RESTRICT-SSH
remark *** Permit SSH access strictly from IT Staff Subnet ***
permit 10.10.50.0 0.0.0.31
deny any
ip access-list extended PERIMETER
remark *** 1. Permit Return Established TCP Traffic (Web/FTP) ***
permit tcp any 10.10.0.0 0.0.255.255 established
remark *** 2. Permit Explicit Cloud and FTP Server Web Responses ***
permit ip 198.51.100.0 0.0.0.255 10.10.0.0 0.0.255.255
remark *** 3. Block Spoofed Private RFC 1918 Ranges ***
deny ip 10.0.0.0 0.255.255.255 any
deny ip 172.16.0.0 0.15.255.255 any
deny ip 192.168.0.0 0.0.255.255 any
remark *** 4. Deny all other unsolicited traffic ***
deny ip any any
!
banner motd ^C^C
!
```

```
!
!
!
logging trap debugging
logging 10.10.60.4
line con 0
exec-timeout 30 0
logging synchronous
login local
!
line aux 0
!
line vty 0 4
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
login local
transport input ssh
line vty 5 15
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
login local
transport input ssh
!
!
ntp server 10.10.60.7
!
end
```

R2

```
R2(config)#do sh run
Building configuration...

Current configuration : 2370 bytes
!
version 15.1
service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname R1
!
!
!
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
!
!
!
!
```

```
!
ip cef
no ipv6 cef
!
!
!
username admin privilege 15 secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
license udi pid CISCO2911/K9 sn FTX1524NE5E-
!
!
!
!
!
!
!
!
!
!
!
ip ssh version 2
ip domain-name ane-labs.net
ip name-server 10.10.60.6
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface Loopback0
ip address 2.2.2.2 255.255.255.255
ip ospf 1 area 0
!
interface GigabitEthernet0/0
ip address 203.0.113.3 255.255.255.248
ip nat outside
duplex auto
speed auto
!
interface GigabitEthernet0/1
ip address 10.10.200.69 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
duplex auto
speed auto
!
interface GigabitEthernet0/2
ip address 10.10.200.73 255.255.255.252
ip ospf network point-to-point
ip ospf priority 1
ip ospf 1 area 0
```

```
duplex auto
speed auto
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
log-adjacency-changes
default-information originate
!
ip nat inside source list 1 interface GigabitEthernet0/0 overload
ip classless
ip route 0.0.0.0 0.0.0.0 203.0.113.1
!
ip flow-export version 9
!
!
access-list 1 permit 10.10.0.0 0.0.255.255
ip access-list standard RESTRICT-SSH
remark *** Permit SSH access strictly from IT Staff Subnet ***
permit 10.10.50.0 0.0.0.31
deny any
ip access-list extended PERIMETER
remark *** 1. Permit Return Established TCP Traffic (Web/FTP) ***
permit tcp any 10.10.0.0 0.0.255.255 established
remark *** 2. Permit Explicit Cloud and FTP Server Web Responses ***
permit ip 198.51.100.0 0.0.0.255 10.10.0.0 0.0.255.255
remark *** 3. Block Spoofed Private RFC 1918 Ranges ***
deny ip 10.0.0.0 0.255.255.255 any
deny ip 172.16.0.0 0.15.255.255 any
deny ip 192.168.0.0 0.0.255.255 any
remark *** 4. Deny all other unsolicited traffic ***
deny ip any any
!
banner motd ^C^C
!
!
!
!
logging trap debugging
logging 10.10.60.4
line con 0
exec-timeout 30 0
logging synchronous
login local
!
line aux 0
!
line vty 0 4
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
login local
transport input ssh
line vty 5 15
```

```
access-class RESTRICT-SSH in
exec-timeout 30 0
logging synchronous
login local
transport input ssh
!
!
ntp server 10.10.60.7
!
end
```

ISP-R

```
ISP-R#sh run
Building configuration...

Current configuration : 766 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname ISP-R
!
!
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO2911/K9 sn FTX1524D735-
!
!
!
!
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
```

```
!
!
!
!
!
!
interface GigabitEthernet0/0
ip address 203.0.113.1 255.255.255.248
duplex auto
speed auto
!
interface GigabitEthernet0/1
ip address 198.51.100.1 255.255.255.0
duplex auto
speed auto
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
shutdown
!
interface Vlan1
no ip address
shutdown
!
ip classless
ip route 0.0.0.0 0.0.0.0 GigabitEthernet0/0
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
end
```